

Virtuoso® Layout Pro: T7 Module Generator and Floorplanner

Course Version IC 25.1

Lecture Manual

Revision 1.0

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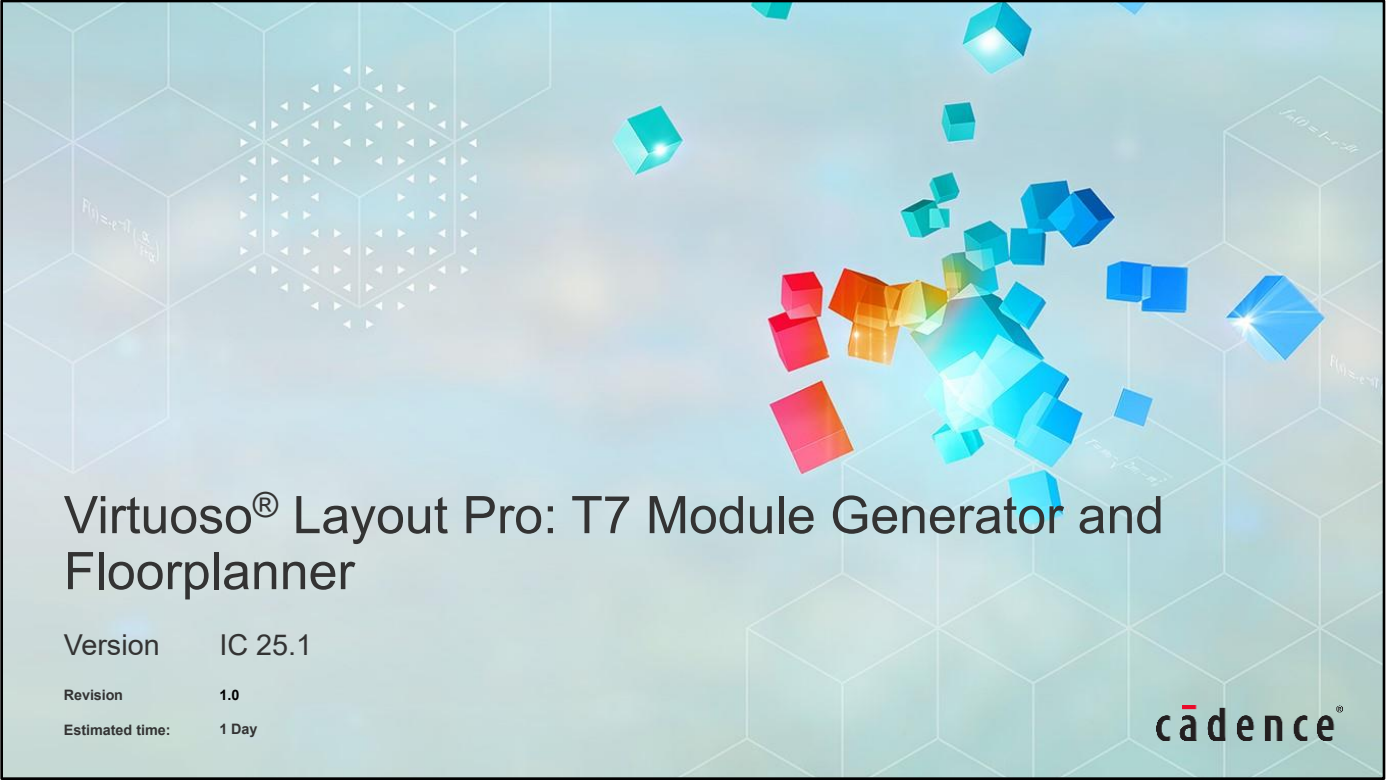
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Virtuoso® Layout Pro: T7 Module Generator and Floorplanner

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Revision 1.0

Estimated time: 1 Day

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have already completed the following courses:

- [Virtuoso® Layout Pro: T1 Environment and Basic Commands](#)
- [Virtuoso Layout Pro: T2 Create and Edit Commands](#)
- [Virtuoso Layout Pro: T3 Basic Commands](#)
- [Virtuoso Layout Pro: T4 Advanced Commands](#)
- [Virtuoso Layout Pro: T5 Interactive Routing](#)
- [Virtuoso Layout Pro: T6 Constraint-Driven Flow and Power Routing](#)

Or you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux



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Course Objectives

In this course, you

- Set up and run Module Generator
- Analyze Modgen Enhancements
- Set up and run Floorplanner
- Set up and run Top-Down Floorplanning



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Course Agenda

- Module Generator
 - Creating a Modgen from Schematic
 - Editing a Modgen
 - Routing a Modgen
- Modgen Enhancements
 - Using the Array Assistant
 - Working with Pattern Preset
 - Using the Modgen On-Canvas Commands
 - Using the Topology Editor
- Floorplanner
 - Defining the Cell Type Attribute
 - Using the Configure Physical Hierarchy Utility
 - Generating and Placing the Blocks with the Floorplanner
 - Editing the Level-1 Objects
 - Using the Pin Optimizer
 - Aligning the Soft Block Pins with the Top-Level Pins
 - Creating Feed Through Terminal Pins
- Top-Down Floorplanning
 - Creating a Top-Down Floorplan



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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/86121.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.



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How do I register to take the exam?

- Log in to our [Learning Management System](#), click on the course in your transcript, and go to the **Content** tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- [Credly](#) validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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Icons Used in This Class



Best Practice



Language Syntax



Concept/
Glossary



Frequently Asked
Questions /
Quiz



Error Message

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.



Problem & Solution



Quick Reference



Tool Command



How To



Lab List

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Module 2

Module Generator

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Module Objectives

In this module, you

- Analyze the various methods to create a Modgen
- Modify the default Modgen settings
- Route a Modgen using the Structured Router



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Useful Terms and Definitions



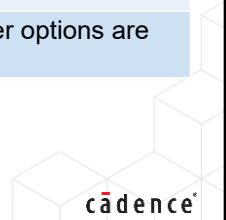
Virtuoso Studio Schematic Editor (VSE)	The Virtuoso® Studio Schematic Editor is a design entry tool that supports the work of logic and circuit design engineers, including drafters. Physical layout designers and printed circuit board designers can use the information as background material to support their work.
Virtuoso Studio Layout Suite (VLS)	The Virtuoso Studio Layout Suite is an editing tool that provides the basic creation and editing capabilities to work with layout designs. The layout editor lets you create custom layouts or edit any existing layouts. It lets you work with multipart paths and hierarchical designs. You can also use the layout editor to view the connected shapes in a design.
Module Generator (Modgen)	The Virtuoso Module Generator (Modgen) let you generate multiple Pcell instances into a complex, highly matched, structured array.
Device	A design element that has both a symbol and a layout.
Component	A pin or a device.



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Useful Terms and Definitions (continued)

Virtuoso Floorplanner (VFP)	Virtuoso Floorplanner is an IC physical block floorplanner from the Virtuoso Layout Suite.
Configure Physical Hierarchy (CPH)	CPH is a physical hierarchy configuration utility that lets you specify how a layout implementation is generated from a specific schematic design.
Generate Physical Hierarchy generates (GPH)	GPH generates the physical hierarchy based on the configuration defined in Configure Physical Hierarchy (CPH).
Array Assistant	The Array Assistant is a unified interface that allows you to quickly create and edit Modgens.
Plan menu	In Virtuoso Layout Suite EXL and higher tiers, the Floorplanner options are available under the Plan menu.



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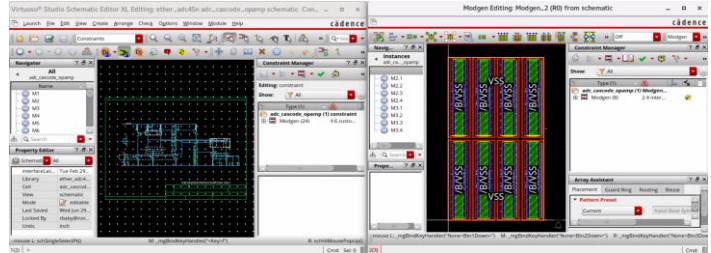
What Is the Module Generator?



The **Module Generator** provides a simple and fast method to generate multiple Pcell instances into a complex, highly matched, structured array based on defined Pcell leaf structures.

Use the Modgen tool to:

- Create one placeable object from multiple instances
- Specify the devices to be arrayed
- Specify an interdigitation pattern
- Abut and align instances
- Insert dummy devices, body contacts, and guard rings
- Backannotate dummies
- Control the routing style and generate internal routing geometry
- Create Modgen from either the schematic or layout design



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Understanding Module Generators

Module generators are designed to provide a way to generate multiple Pcell instances into a complex, highly matched, structured array. With the Modgen tool, you specify the devices to be arrayed, then specify an interdigitation pattern, insert dummy devices, body contacts, and guard rings. Finally, you control the routing style and generate internal routing geometry.

Module generators can be created from either the schematic or the layout. When you create a Modgen from the schematic, the tool creates a Modgen constraint on the schematic that will then be used when the layout is generated using *Generate All From Source* or *Generate Selected From Source*. When run from the schematic, by default, the Modgen creates a temporary physConfig. Use the *modgenPhysConfigs* environment variable to specify a different physconfig that already exists to be used by the Modgen.

Note: The Modgen router cannot route devices with horizontal terminals. If you want all devices in the Modgen to be horizontal, first route the Modgen with the terminals in a vertical orientation and then rotate the entire Modgen to accomplish the desired effect.

Licensing of the Module Generator



Licensing Requirements of the Module Generator:

- When launched from Layout XL, the Array Assistant checks out two Virtuoso Layout Suite GXL tokens (95321).
- When launched from higher tiers, no additional licenses are required.

Modgen features are available through scripting in the Command Interpreter Window (CIW) without a cockpit.



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Creating a Modgen



Use Modgen tool to create one placeable object from multiple instances.
There are several ways to create a Module Generator.

Modgens can be created from both the schematic and layout designs.

Different methods available are:

1. Using the Constraint Manager toolbar
2. Using the Place menu
3. Using Convert to Modgen
4. Using SKILL® functions

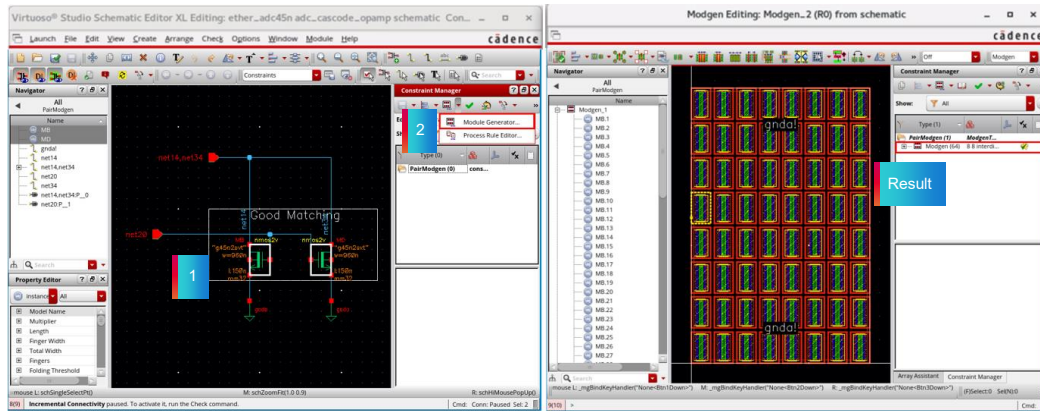


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1. Using the Constraint Manager Toolbar

1. Select the devices you want to include as one placeable object from the schematic or layout design.
2. Choose **Module Generator** from the Constraint Manager toolbar.

Result: A new *Modgen* constraint is created, and the *Modgen Editor* is invoked.

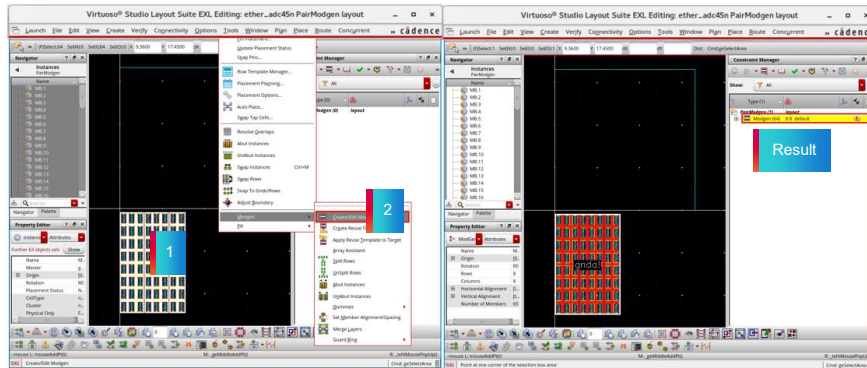


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2. Using the Place Menu

1. Select the instances, Modgens, figGroups, and mosaics you want to include as one placeable object from the schematic or layout design.
2. Choose **Place – Modgen – Create/Edit Modgen** or click the **Create/Edit Modgen** icon from the Placement toolbar. 

Result: A new *Modgen* constraint is created.



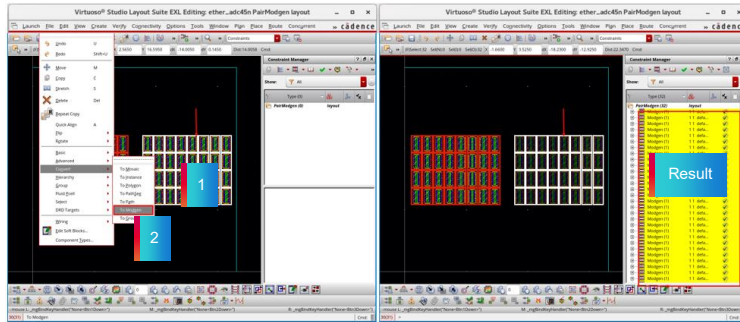
However, the **Modgen Editor** is not invoked. Use the **Modgen on-canvas** commands to edit the **Modgen** directly in the layout canvas.

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3. Using Convert to Modgen

- A mosaic is a compact array of instances that belong to the same mfactored instance and therefore have the same master and connectivity. You can convert a mosaic in the layout design to a Modgen. To do this:
 1. Select the mosaic to be converted into a Modgen.
 2. Choose **Edit – Convert – To Modgen** or right-click the mosaic in the canvas and choose **Convert to Modgen** from the *Mosaic* shortcut menu.

Result: A *Modgen* constraint is created.



However, the **Modgen Editor** is not invoked. Use the **Modgen on-canvas** commands to edit the **Modgen** directly in the layout canvas.

This page does not contain notes.

4. Using the SKILL Functions

Use one of the following SKILL functions:

- **ciConCreate**: Creates the Modgen constraint.
- **gpeCreateSandbox**: Creates a Modgen sandbox object that includes the specified instances.
- **mgCreateModgenAsLayout**: Creates a Modgen from the list of selected instances and keeps the instance locations inside the Modgen as close as possible to their original locations in the layout.
- **mgCreateModgenConstraintAsLayout**: Creates a Modgen constraint that uses the current layout to drive the initial Modgen constraint and row/column assignments.



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Setting the Default Workspace for the Modgen Editor Using the modgenWindowConfigFile



Use the **modgenWindowConfigFile** environment variable to specify the default workspace to be loaded when the Modgen is invoked.

To specify the default workspace for the Modgen Editing session to start with:

- In the CIW, enter the following SKILL command:

```
envSetVal("layoutXL" "modgenWindowConfigFile" 'string "modgenUserWorkspace")
```

Note: Here, *modgenUserWorkspace* is the required workspace that the Modgen needs to load during startup.

Use the *.cdsinit* file to specify the default workspace to be loaded automatically when Modgen Editor is invoked.



Setting the Default Workspace for Modgen Editor Using the modgenWindowConfigFile

Use the **modgenWindowConfigFile** environment variable to specify the default workspace to be loaded when Modgen is invoked.

To specify the default workspace for the modgen editing session to start with:

- In the CIW, enter the following SKILL command:
 - **envSetVal("layoutXL" "modgenWindowConfigFile" 'string "modgenUserWorkspace")**

Note: Here *modgenUserWorkspace* is the required workspace that the Modgen needs to load during startup.

Use the *.cdsinit* file to specify the default workspace to be loaded automatically when Modgen Editor is invoked.

The specified workspace, if available either in the local *.cadence* directory or in any directory in the CSF hierarchical search, is applied to the current Modgen session. If the workspace is invalid, then a warning message is displayed, and the default Modgen Edition workspace is loaded. You can also create a workspace for the current Modgen session.

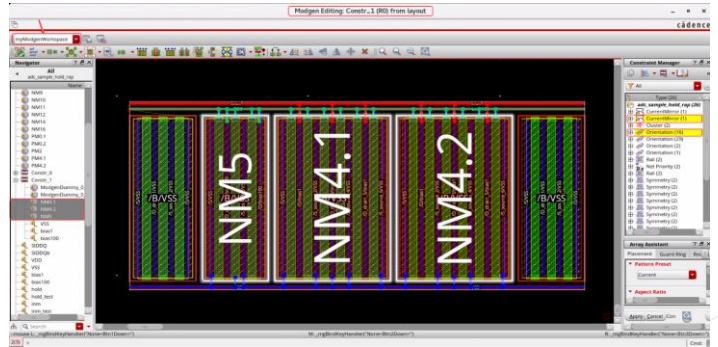
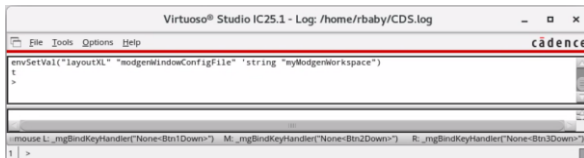
Example: Setting the Default Workspace for the Modgen Editor Using the modgenWindowConfigFile

In this example, to specify the default workspace to be loaded when the Modgen Editor is invoked:

- In the CIW, enter the following SKILL command and reinvoke the Modgen Editor.

```
envSetVal("layoutXL" "modgenWindowConfigFile" 'string "myModgenWorkspace")
```

Result: The Modgen Editor opens with *myModgenWorkspace* workspace in the **Workspace Configuration** menu.



Example: Setting the Default Workspace for Modgen Editor Using the modgenWindowConfigFile

In this example, to specify the default workspace to be loaded when the Modgen Editor is invoked:

- In the CIW, enter the following SKILL command and reinvoke the Modgen Editor.
 - `envSetVal("layoutXL" "modgenWindowConfigFile" 'string "myModgenWorkspace")`

Result: The Modgen Editor opens with *myModgenWorkspace* workspace in the **Workspace Configuration** menu.

Creating a Modgen from the Schematic



Use the Modgen tool to create one placeable object from multiple instances.

You can create a Modgen from the schematic in three different ways:

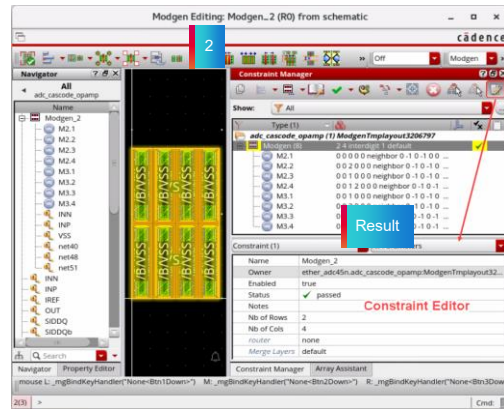
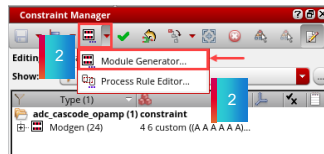
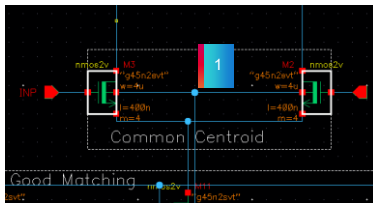
1. By selecting the instances in the Schematic Canvas
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
2. By selecting the instances in the Navigator Assistant
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
3. By using the Circuit Prospector Assistant for Special Configurations
 - Use the Rapid Analog Prototype (RAP) flow using the Circuit Prospector assistant to find special configurations



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1. Selecting the Instances In the Schematic Canvas

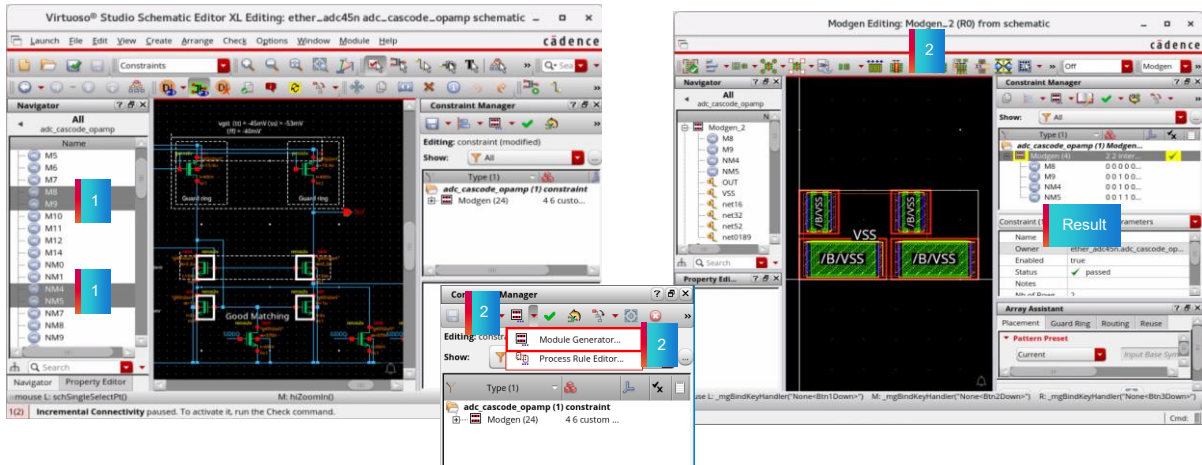
1. In the *Schematic* canvas, select the instances (use **SHIFT** with the left mouse button) you want to create as one placeable object.
2. Click the **Module Generator** icon in the *Modgen* constraint menu to invoke the *Modgen Editing* window and create the modgen.



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2. Selecting the Instances In the Navigator Assistant

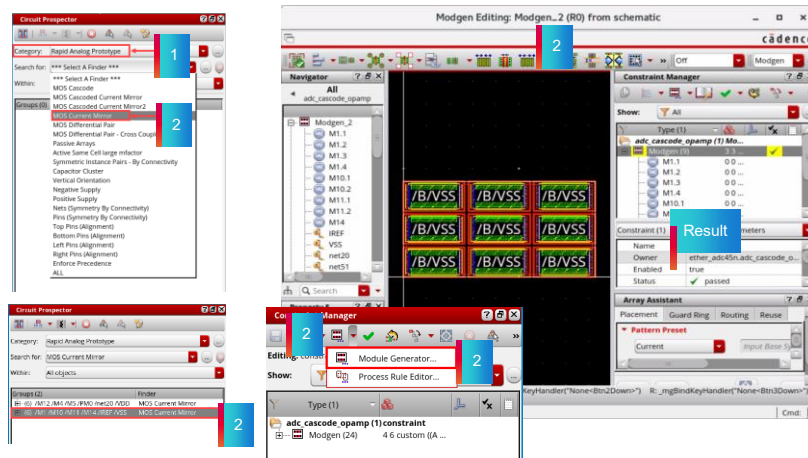
1. In the *Navigator* Assistant, select the instances (use **SHIFT** or **Ctrl** with the left mouse button).
2. Click the **Module Generator** icon in the *Modgen* constraint menu to invoke the *Modgen Editing* window and create the modgen.



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3. Using the Circuit Prospector Assistant for Special Configurations

1. In the *Circuit Prospector* assistant, choose the **Category** as *Rapid Analog Prototype*.
2. In the list of available *Finders*, search for **MOS Current Mirror** and create the Modgen for the **Groups**.



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Editing a Modgen



A Modgen is a placeable object that combines multiple device instances, dummy instances, Modgens, figGroups, or mosaics.

Once a modgen is created, you may need to edit a modgen for your design requirements.

Use one of the following methods to open and edit an existing Modgen in the Modgen Editor.

There are several options available for editing a Modgen:

1. **Double**-click the *Modgen*
2. **Double**-click the *Modgen* Constraint
3. Using the *Place* menu
4. Using the *Module Generator* icon
5. Using the *Edit Modgen* option

Each option is explained in detail in the following slides.

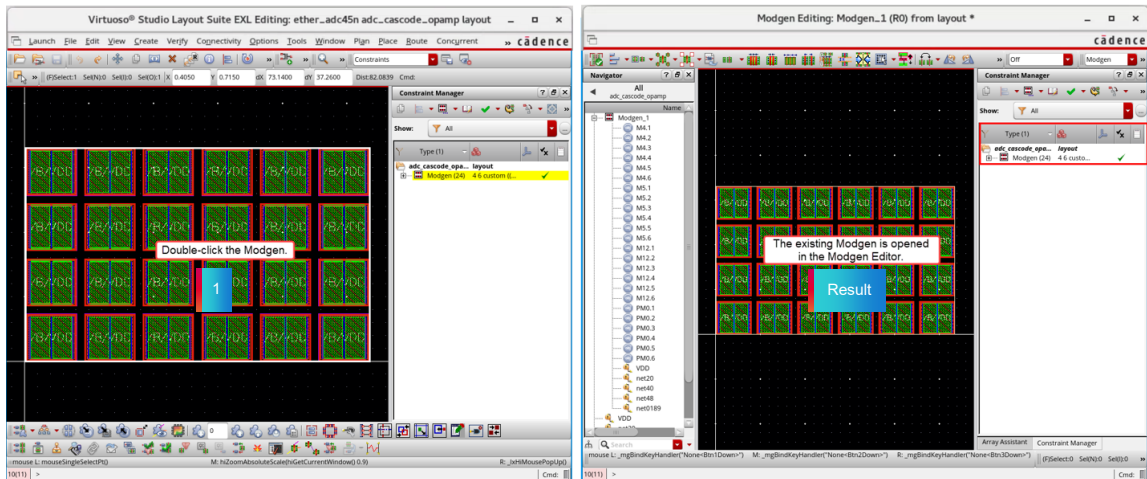


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1. By Double-Clicking the Modgen

1. Double-click the *Modgen* in the design which you want to edit.

Result: The existing *Modgen* is opened in the *Modgen Editor*.

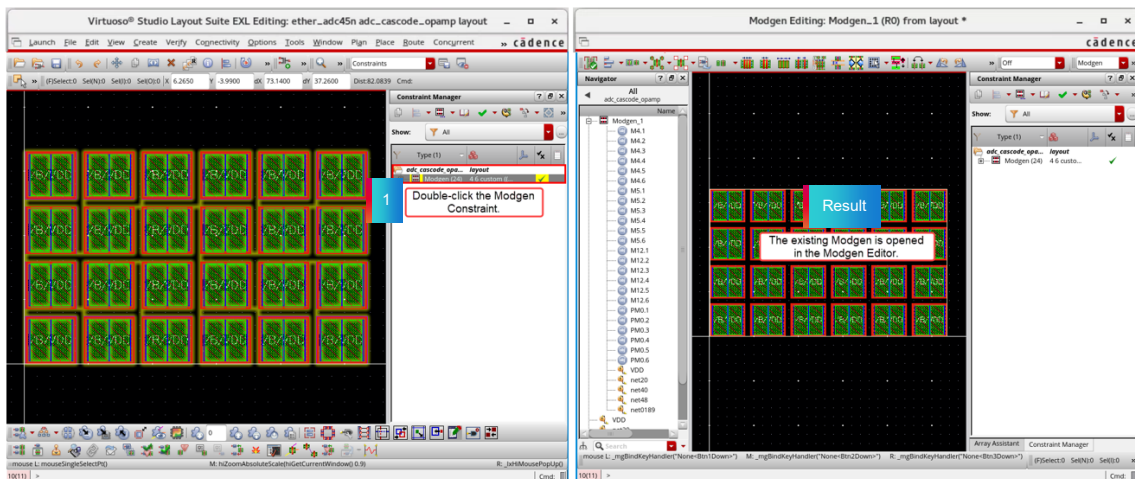


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2. By Double-Clicking the Modgen Constraint

1. Double-click the *Modgen* constraint in the Constraint Manager toolbar which you want to edit.

Result: The existing *Modgen* is opened in the *Modgen Editor*.

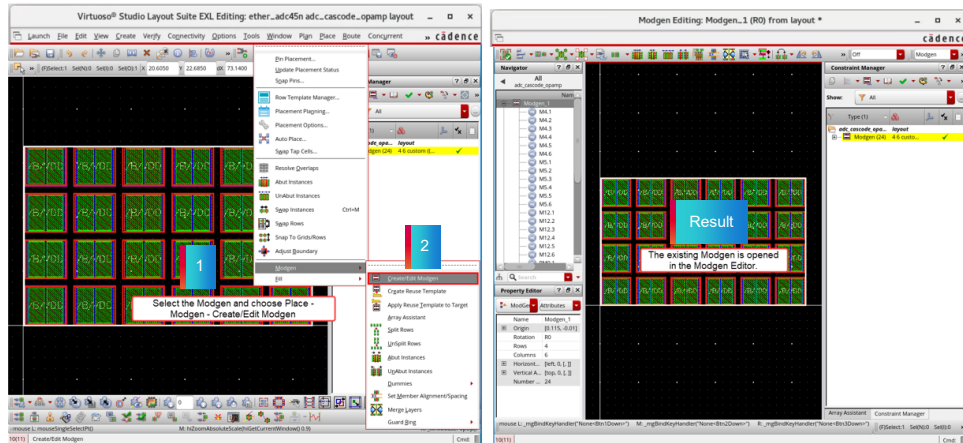


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3. Using the Place Menu

1. Select the *Modgen* which you want to edit in the layout canvas.
2. Choose **Place – Modgen – Create/Edit Modgen** or click the **Create/Edit Modgen** icon from the Placement toolbar.

Result: The existing *Modgen* is opened in the *Modgen Editor*.

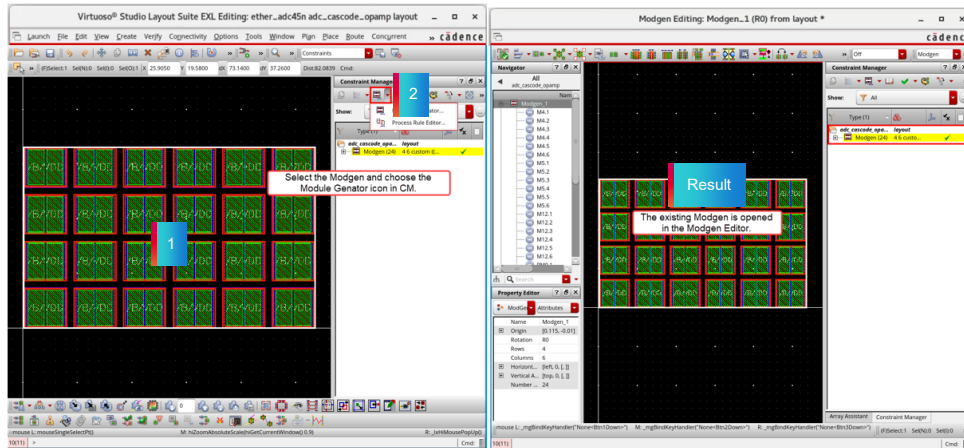


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4. Using the Module Generator Icon

1. Select the Modgen in the design which you want to edit.
2. Choose the **Module Generator** icon in the Constraint Manager assistant.

Result: The existing *Modgen* is opened in the *Modgen Editor*.

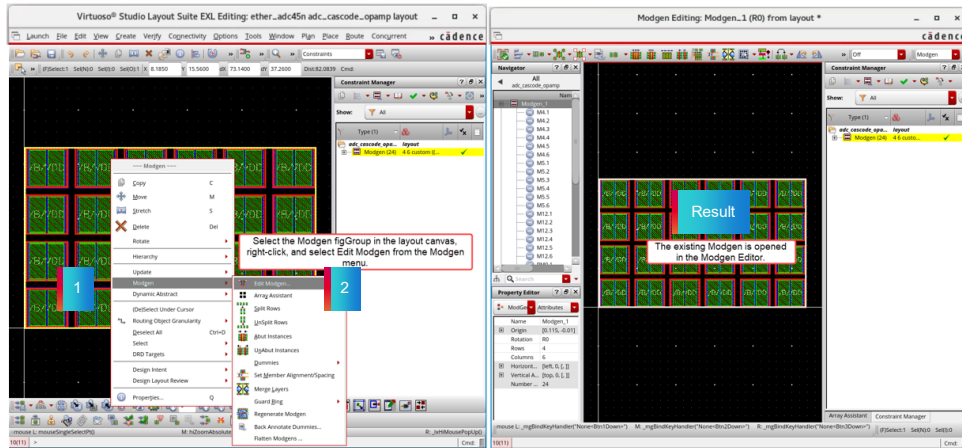


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5. Using the Edit Modgen Option

1. Select the Modgen figGroup which you want to edit in the layout canvas.
2. Right-click and select **Edit Modgen** from the **Modgen** menu.

Result: The existing *Modgen* is opened in the *Modgen Editor*.



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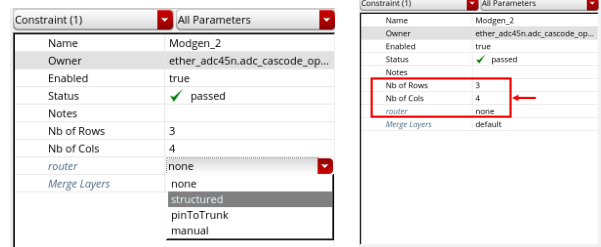
Specifying the Modgen Parameters



If you need to set up the Module Generator parameters, use the **Module Generator** panel.

In the Module Generator panel:

- Specify the number of rows for the array in the *Nb of Rows* field.
- Specify the number of columns for the array in the *Nb of Cols* field.
- In the *router* field, select the type of router.

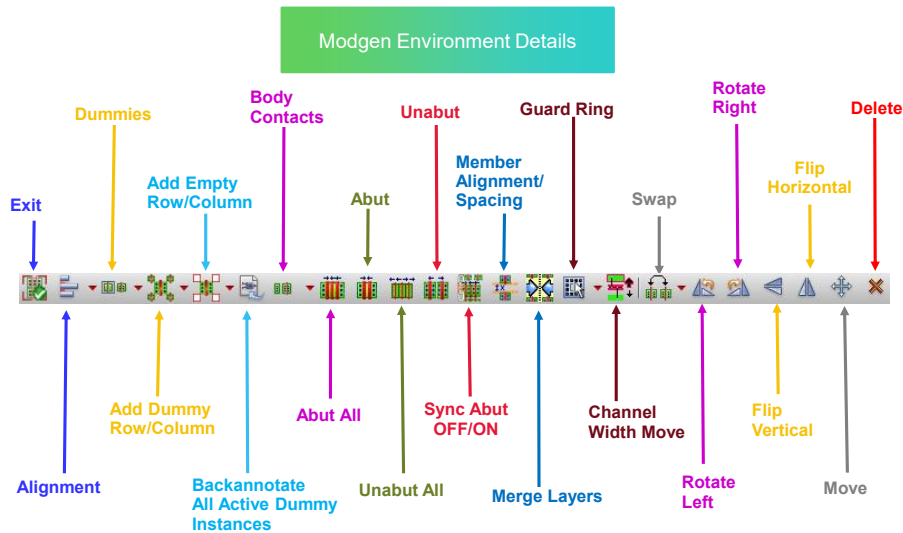


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Using the Modgen Placement Toolbar



If you need to edit the modgens, use the **Modgen Placement toolbar** in the **Modgen Editor**.



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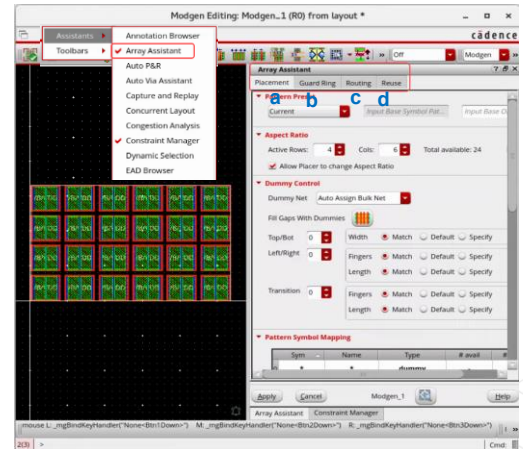
Invoking the Array Assistant



In the Modgen Editor, choose the **Window** → **Assistants** → **Array Assistant** to launch the *Array Assistant*.

Array Assistant has four tabs:

- **Define Modgen Placement Settings:** Use the **Placement** tab of the Array Assistant to specify the Modgen pattern settings.
- **Create Guard Rings:** Use the **Guard Ring** tab of the Array Assistant to create guard rings around the Modgens.
- **Define Modgen Topology Settings:** Use the **Routing** tab of the Array Assistant to define Modgen topology patterns and set routing preferences.
- **Reuse Modgen Templates:** Use the **Reuse** tab of the Array Assistant to load and save settings to Modgen template files.



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Invoking the Array Assistant

In the Modgen Editor, choose the **Window** → **Assistants** → **Array Assistant** to launch the *Array Assistant*.

Array Assistant has four tabs:

- Define Modgen Placement Settings: Use the *Placement tab* of the Array Assistant to specify the Modgen pattern settings.
- Create Guard Rings: Use the Guard Ring tab of the Array Assistant to create guard rings around the Modgens.
- Define Modgen Topology Settings: Use the *Routing tab* of the Array Assistant to define Modgen topology patterns and set routing preferences.
- Reuse Modgen Templates: Use the *Reuse tab* of the Array Assistant to load and save settings to Modgen template files.

Using the Placement Tab

The following describes the fields available on the **Placement** tab of the Array Assistant.

- **Pattern Preset:** Specifies the pattern in which the devices are to be placed within a Modgen.
- **Aspect Ratio:** Specifies the aspect ratio of the Modgen.
- **Dummy Control:** Specifies dummy parameters.
- **Pattern Symbol Mapping:** View and change the mapping of devices in the Pattern box.
- **Pattern:** Displays the Modgen pattern derived based on the specified preset, active rows, and dummies.
- **Save/Load CSV:** Save the current array settings to a comma-separated values (CSV) file or load array settings from a CSV file.
- **Spacing:** Specifies the spacing between devices in a Modgen.



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Using the Placement Tab

The following describes the fields available on the Placement tab of the Array Assistant.

- **Pattern Preset:** Specifies the pattern in which the devices are to be placed within a Modgen.
- **Aspect Ratio:** Specifies the aspect ratio of the Modgen.
- **Dummy Control:** Specifies dummy parameters.
- **Pattern Symbol Mapping:** View and change the mapping of devices in the Pattern box.
- **Pattern:** Displays the Modgen pattern derived based on the specified preset, active rows, and dummies.
- **Save/Load CSV:** save the current array settings to a comma-separated values (CSV) file or load array settings from a CSV file.
- **Spacing:** Specifies the spacing between devices in a Modgen.

Placement Tab: Pattern Preset

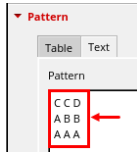
Pattern Preset	Description
Current	Uses the current Modgen pattern.
Clustered	Displays a pattern based on the bottom-up approach with row-based split.
Interdigitated	Applies an interdigitation of 1 to the devices.
Compact	Adjusts the pattern to achieve the maximum abutment of devices.
Custom	The repeat pattern of Custom patterns in each row starts from left to right.
Zigzag	Applies the base symbol pattern in the opposite horizontal direction for alternating the array rows, starting with left-to-right for row index 0.
Shift	Shifts the pattern by one position for each consecutive row, starting from the bottom-most row.
Common Centroid	Places devices in a grid following a common centroid pattern. Symbols are ordered in precedence, outward from the center.



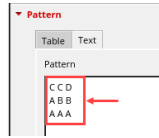
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Placement Tab: Pattern Preset (continued)

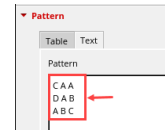
Clustered



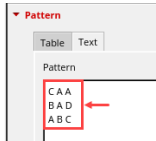
Compact



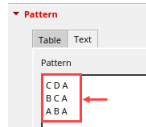
Custom



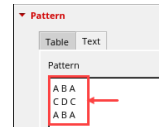
Zigzag



Shift



Common Centroid



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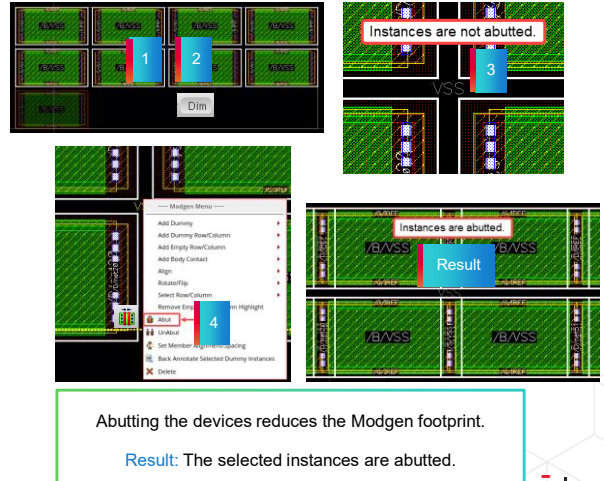
Abutting the Devices



Modgen supports the abutment of MOSFET member instances. When you abut a device, Modgen is aware of connectivity and may flip the device to ensure correct abutment.

To automatically abut the instances:

1. Select the cells which you want to abut using window select.
2. Add the **Dimming** option for better visibility of the selected cells. For this, **right-click** on the *Modgen Editing Window Banner*, add **Toolbars – Options**, and click on the icon. 
3. Zoom inside the array to view that the instances are not abutted.
4. Click the **Abut** icon in the *Modgen Placement* toolbar or **right-click** on the *Modgen Editing Window Banner* and click the **Abut** option from the *Modgen Menu* with the cells selected. 



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Abutting the Devices

For MOSFET member instances, Modgen supports abutment of devices. When you abut a device, Modgen remains aware of connectivity and may flip the device to correctly abut.

You can also abut dummy devices to other devices, including other dummies. Internally, the dummy device's net may be changed from the default specified in the Dummy Options form, but if the dummy device is removed from abutment, the net will revert to the default. When two instances are abutted, any body contacts between them are deleted.

Modgen abutment uses the Layout XL chaining engine. Therefore, the abutment results are subject to the settings of the Layout XL chaining environment variables. For more information about these environment variables, see *Layout XL Environment Variables* in the *Virtuoso Layout Suite XL User Guide*. Notable examples are *chainMirror* and *chainMirrorEquivOrients*. These environment variables are enabled by default and may therefore result in unintended changes to the orientation of instances. The devices to be abutted must be registered as component types.

Aligning the Devices



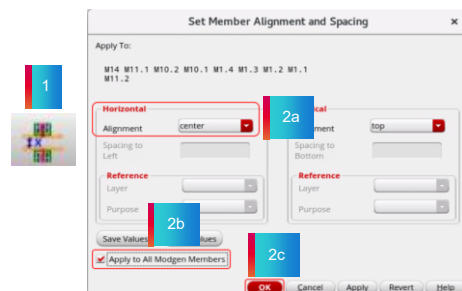
A Modgen module can include instances with different Pcell masters and dimensions. To arrange instances of varying heights, define the vertical bounding box alignment as top, center, or bottom.

Use the **Set Member Alignment and Spacing** form to set the alignment and spacing values for the Modgen devices.

To change the instance alignment in Modgen:

1. Click the **Member Alignment/Spacing** icon in the *Modgen Placement* toolbar or **right-click** on the *Modgen Editing Window Banner* and click the **Set Member Alignment/Spacing** option from the *Modgen* menu.
2. Select the following options in the Set Member Alignment and Spacing form:
 - a. Select **Alignment** as *center* in the **Horizontal** section.
 - b. Select **Apply to All Modgen Members** radio button.
 - c. Click **OK**.

Result: All members are aligned according to the *center* axis.



Instances are aligned to the center axis.



Specifying Device Alignment and Spacing

Since one Modgen module can contain instances with different Pcell masters, the instances can have different dimensions. To arrange instances with different heights within a module, you can specify the vertical bounding box alignment (top, center, or bottom).

When devices are aligned at the top, the top edges of the instances are aligned at the same Y-coordinate, while bottom edges are aligned at the same Y-coordinate for bottom alignment. For center alignment, the center of shorter instances are aligned with the center of the largest instance.

Aligning the Devices

Use the **Set Member Alignment and Spacing** form to specify the alignment and spacing values for the Modgen devices.

To change the instance alignment in Modgen:

- Click the **Member Alignment/Spacing** icon in the *Modgen Placement* toolbar or **right-click** on the *Modgen Editing Window Banner* and click the **Set Member Alignment/Spacing** option.
- Select the following options in the Set Member Alignment and Spacing form:
 - Select **Alignment** as *center* in the **Horizontal** section.
 - Select **Apply to All Modgen Members** radio button.
 - Click **OK**.

Result: All members are aligned according to the *center* axis.

Adding the Body Contacts



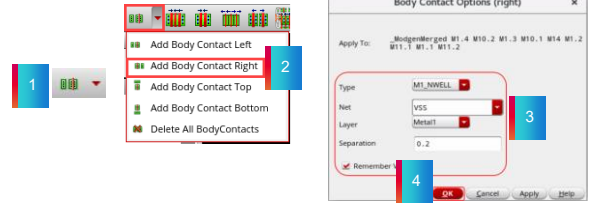
You can define properties for body contacts in the **Body Contact Options** form, then add body contacts to the array in the layout.

To add the body contacts at a specified location:

1. Select all Modgen instances and click the **Body Contacts** icon in the *Modgen Placement* toolbar. 
2. Click **Add Body Contact Right** in the pull-down menu to display the *Body Contact Options (right)* form.
3. Select the required options in the *Body Contact Options (right)* form.
4. Click **OK**.

Result: The Body Contacts are placed on the right of every member.

Body Contacts are placed on the right of every member, except the abutted ones.



Defining the Body Contact Properties

In the Body Contact Options form, you can specify the type, net, and separation distance for body contacts. Because the body contact created by the Modgen tool matches the height of the member instance, you also need to specify which reference layer the body contact matches. If you do not specify a reference layer, the Modgen tool will match the body contact to the height of the bounding box of the member instance. The Modgen tool also uses the reference layer to calculate the separation distance.

Adding the Body Contacts

You can define properties for body contacts in the **Body Contact Options** form, then add body contacts to the array in the layout.

To add the body contacts at a specified location:

- Select all Modgen instances and click the **Body Contacts** icon in the *Modgen Placement* toolbar.
- Click **Add Body Contact Right** in the pull-down menu to display the *Body Contact Options (right)* form.
- Select the required options in the *Body Contact Options (right)* form.
- Click **OK**.

Result: The Body Contacts are placed on the right of every member.

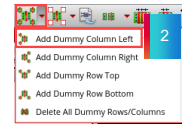
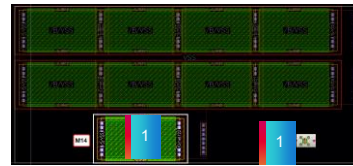
Note: Body Contacts are placed on the right of every member, except the abutted ones.

Adding the Dummy Devices



Dummy devices are created to counter electrical effects that are observed at small geometries. You can create dummies around devices in the Modgens.

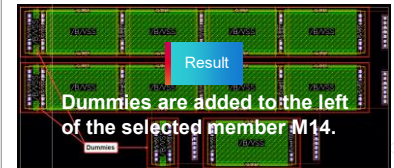
Use the **Dummy Options** form to specify the net to which dummy devices need to be attached and the type of dummy devices.



To add the dummies at a specified location:

1. Select the **bottom row** member **M14** and click the **Add Dummy Row/Column** icon in the *Modgen Placement* toolbar.
2. Click **Add Dummy Column Left** in the pull-down menu to open the *Dummy Options (left)* form.
3. Select the required options in the *Dummy Options (left)* form.
4. Click **OK**.

Result: Dummies are added to the left of the selected member.



Adding the Dummy Devices

Dummy devices are created to counter electrical effects that are observed at small geometries. You can create dummies around devices in Modgens.

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- Click **Add Dummy Column Left** in the pull-down menu to open the *Dummy Options (left)* form.
- Select the required options in the *Dummy Options (left)* form.
- Click **OK**.

Result: Dummies are added to the left of the selected member.

A device instance is considered a dummy if one or more of the following conditions are met:

The instance has one of the following properties set to *t*: *lxDummy*, *Ignore*, and *lvsIgnore*.

The instance has the *lxDummyOwner* property.

The instance is not bound to the schematic. This behavior is controlled by the *modgenCreateUnboundAsDummies* environment variable. The default value is *t*. In this state, the instance is considered a dummy. When set to *nil*, the instance is not considered a dummy.

The instance gate net is the same as that assigned to the environment variable *modgenDummyNet*.

Adding the Empty Row/Column

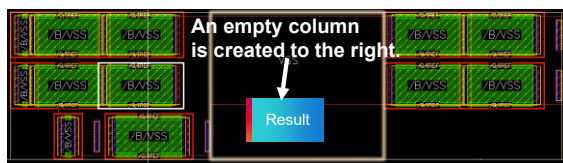
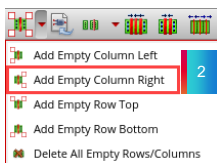


Use the **Add Empty Row/Column** icon in the *Modgen Placement* toolbar to add empty rows in your design.

To add an empty row/column in the design:

1. Select the member **M10.2** of the **middle row** and click the **Add Empty Row/Column** icon in the *Modgen Placement* toolbar or **right-click** on the canvas and access this command from the *Modgen Menu*. 
2. Click **Add Empty Column Right** in the pull-down menu that appears.

Result: An empty column is created to the right in the design.



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Adding the Empty Row/Column

Use the **Add Empty Row/Column** icon in the *Modgen Placement* toolbar to add empty rows in your design.

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- Select the member **M10.2** of the **middle row** and click the **Add Empty Row/Column** icon in the *Modgen Placement* toolbar or **right-click** on the canvas and access this command from the *Modgen Menu*.
- Click **Add Empty Column Right** in the pull-down menu that appears.

Result: An empty column is created to the right in the design.

Removing the Empty Row/Column

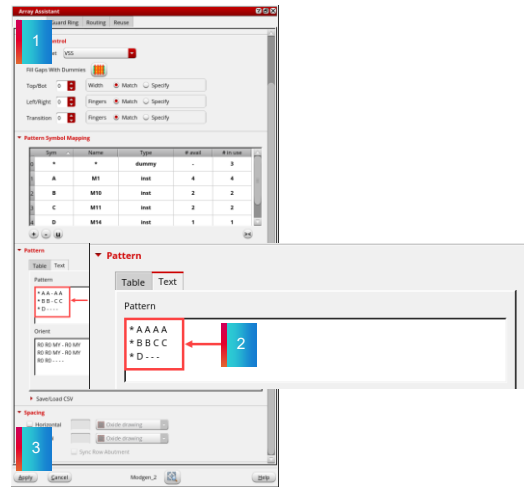
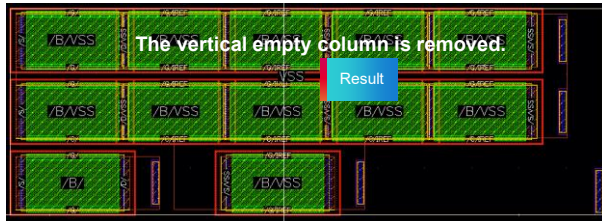


Use the **Array Assistant** to remove the empty rows/columns in your design.

To remove the empty row/column in the design:

1. Launch the **Array Assistant** from the *Window > Assistants > Array Assistant*.
2. Remove the added empty column manually by deleting "-" in the field in the **Array Assistant**.
3. Click the **Apply** button.

Result: The vertical empty column has been removed from the design.



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Adding the Guard Ring

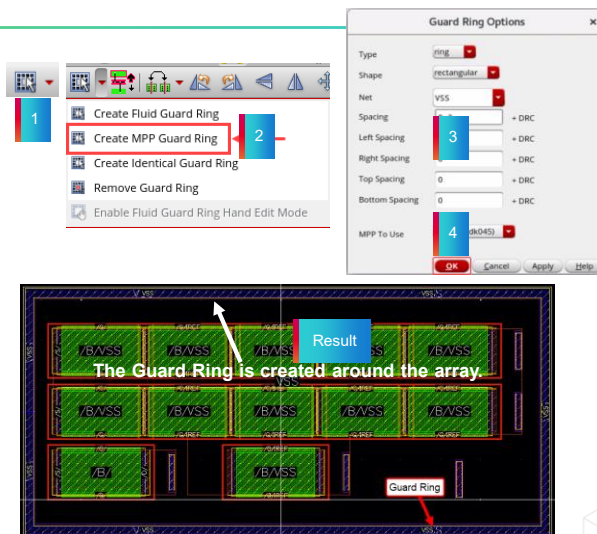


Guard rings are used to enclose one or more objects such as devices or device chains. Use the Modgen tool to create the multipart path (MPP) guard rings and fluid guard rings.

To add the guard ring in the design:

1. On the *Modgen Placement* toolbar, click the **Guard Ring** icon.
2. Click **Create MPP Guard Ring** in the pull-down menu to open the *Guard Ring Options* form.
3. Select the required options in the *Guard Ring Options* form.
4. Click **OK**.

Result: The guard ring is created around the array in the design.



Before creating the guard rings, ensure that they are defined in the technology file.

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Adding the Guard Ring

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- Select the required options in the *Guard Ring Options* form.
- Click **OK**.

Result: The Guard Ring is created around the array in the design.

Before creating the guard rings, ensure that they are defined in the technology file.

Backannotating the Dummy Devices



Modgen dummy devices in the layout can be backannotated to their corresponding schematic to keep the two views synchronized.

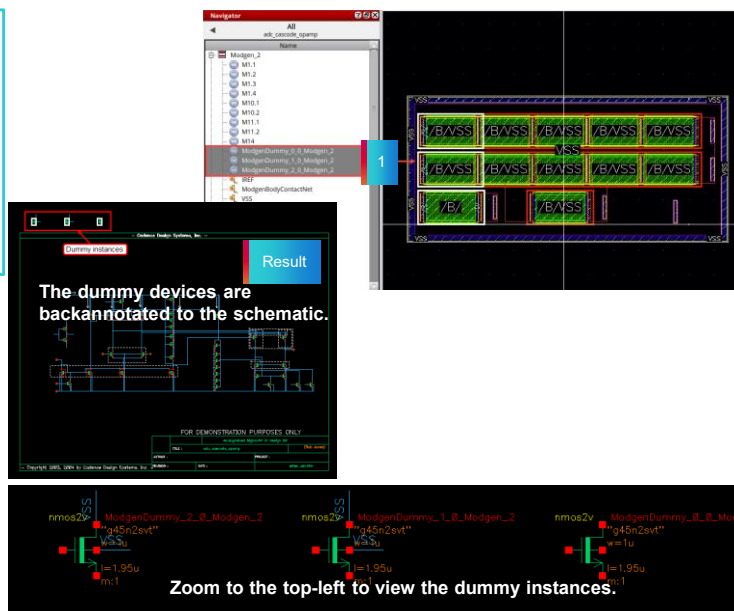
If the dummy layout instance is already bound to a symbol in the schematic, no backannotation is performed for that instance.

To backannotate the dummy instances to the schematic:

1. Select all the dummy members of the Modgen.
2. Click the **Backannotate All Active Dummy Instances** icon from the *Modgen Placement* toolbar.

Result: The dummy devices are backannotated to the schematic window.

You can backannotate dummies even when the layout view is open in read-only mode.



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Backannotating the Dummy Devices

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Routing a Modgen Using the Structured Router

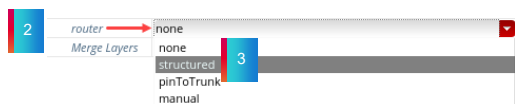


Use the **Structured Router** from the **Constraint Parameter Editor** to route a Modgen.

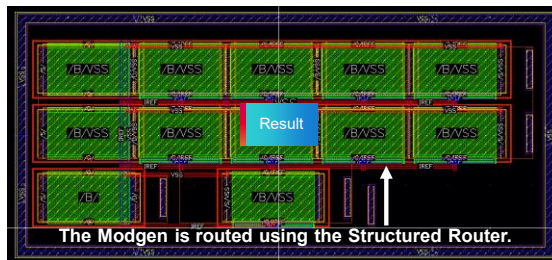
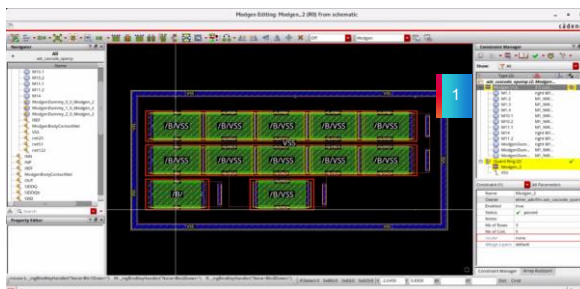
To route a Modgen using the Structured Router:

1. From the **Constraint Manager** assistant, select the *Modgen Constraint* to be routed.
2. Click the *router* parameter in the **Constraint Parameter Editor** to view the routing options: *structured*, *pinToTrunk*, and *manual*.
3. Click *structured* from the *router* parameter to route the Modgen.

Result: The Modgen is routed using the **Structured Router** in the design.



Click **none** to delete the routing.



Routing a Modgen Using the Structured Router

Use the **Structured Router** from the **Constraint Parameter Editor** to route a Modgen.

To route a Modgen using the Structured Router:

1. From the **Constraint Manager** assistant, select the *Modgen Constraint* to be routed.
2. Click the *router* parameter in the **Constraint Parameter Editor** to view the routing options: *structured*, *pinToTrunk*, and *manual*.
3. Click *structured* from the *router* parameter to route the Modgen.

Result: The Modgen is routed using the **Structured Router** in the design.

Modifying a Modgen



Once you have created a Modgen, you can modify the module's properties.

Use the following methods to modify a Modgen:

- Using the Module Generator Icon
- Using the Edit In Place Command



Modifying a Modgen

Once you have created a Modgen, you can modify the module's properties.

Use the following methods to modify a Modgen:

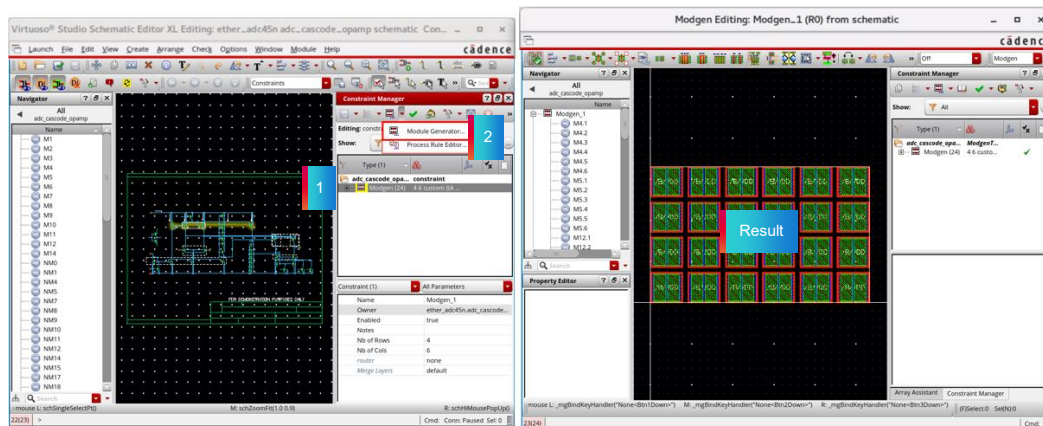
Using the Module Generator Icon and Using the Edit In Place Command.

1. Using the Module Generator Icon

To modify a module from the schematic or layout:

1. Select the *Modgen Constraint* from the Constraint Manager assistant.
2. Click the *Module Generator* icon from the Constraint Manager toolbar. 

Result: The Modgen Editor appears where you can modify the module's properties.



Step 1: Modifying a Modgen Using the Module Generator Icon

To modify a module, from the schematic or layout:

1. Select the *Modgen Constraint* from the Constraint Manager assistant.
2. Click the *Module Generator* icon from the Constraint Manager toolbar.

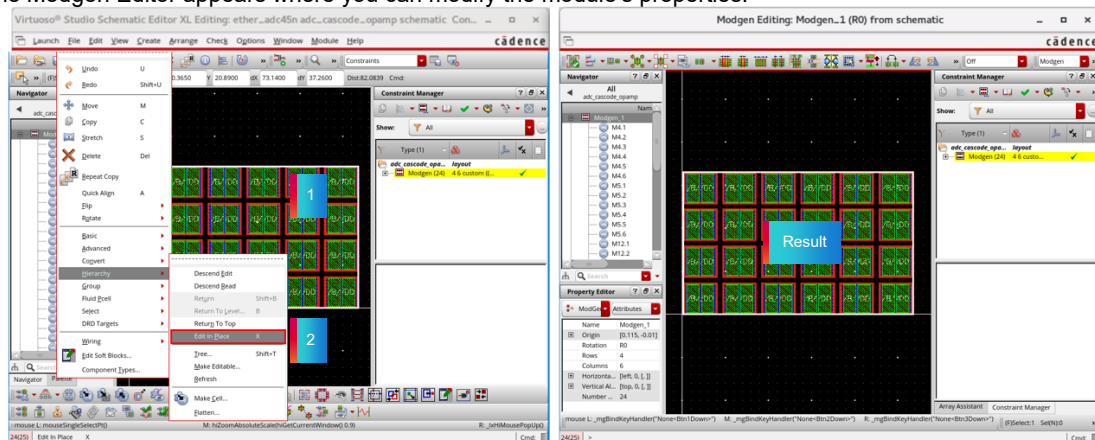
Result: The Modgen Editor appears where you can modify the module's properties.

2. Using the Edit In Place Command

To modify a module from the layout:

1. Select the *Modgen* in the layout.
2. In the layout, choose **Edit – Hierarchy – Edit In Place** command or use the bindkey *X*.

Result: The Modgen Editor appears where you can modify the module's properties.



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Step 2: Modifying a Modgen Using the Edit In Place Command

To modify a module, from the layout:

1. Select the *Modgen* in the layout.
2. In the layout, choose **Edit – Hierarchy – Edit In Place** command or use the bindkey *X*.

Result: The Modgen Editor appears where you can modify the module's properties.

Regenerating a Modgen



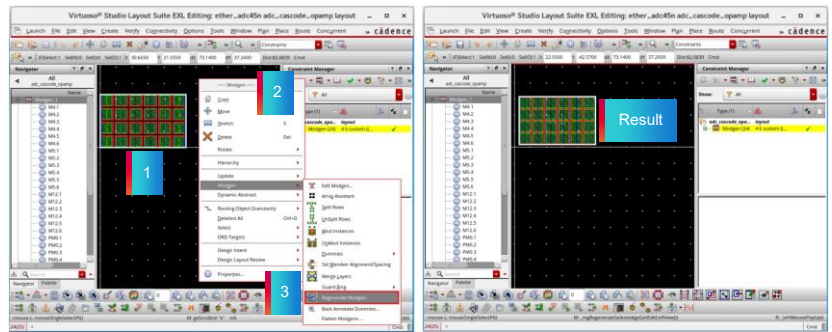
After performing certain tasks, such as moving a Modgen from one placement area to another, you can regenerate the Modgen.

Use the **Regenerate Modgen** option from the **Modgen** menu to regenerate the Modgen.

To regenerate the Modgen in the design:

1. In the layout canvas, select one or more Modgen-type FigGroups after performing certain tasks.
2. Then, **right-click** to display the *Modgen* shortcut menu.
3. Next, select **Modgen – Regenerate Modgen** option from the **Modgen** shortcut menu.

Result: The selected Modgen FigGroups are regenerated.



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Regenerating a Modgen

After performing certain tasks, such as moving a Modgen from one placement area to another, you can regenerate the Modgen.

Use the **Regenerate Modgen** option from the **Modgen** menu to regenerate the Modgen.

To regenerate the Modgen in the design:

1. In the layout canvas, select one or more Modgen-type FigGroups after performing certain tasks.
2. Then **right-click** to display the *Modgen* shortcut menu.
3. Next, select **Modgen – Regenerate Modgen** option from the **Modgen** shortcut menu.

Result: The selected Modgen FigGroups are regenerated.

Quitting the Module Generator Tool



When you are finished specifying your module, you can quit the Module Generator mode. There are several ways to quit the Modgen tool.

Use one of the following methods to quit the Modgen mode:

- Use the **Exit** icon in the Modgen Placement toolbar in the Modgen Editing window.
- Use the **Close (X)** button in the upper-right corner of the Modgen Editing window.
- Use the **Shift+B** key combination in the Modgen Editing window.
- **Right-click** to display the Modgen Menu and select **Exit Modgen** in the Modgen Editing window.



Quitting the Module Generator Tool

When you are finished specifying your module, you can quit the Module Generator mode.

There are several ways to quit the Modgen Tool. They are:

Using the Exit Icon, Using the Close (X) Button, Using the Shift+B Key, Using the Exit Modgen.

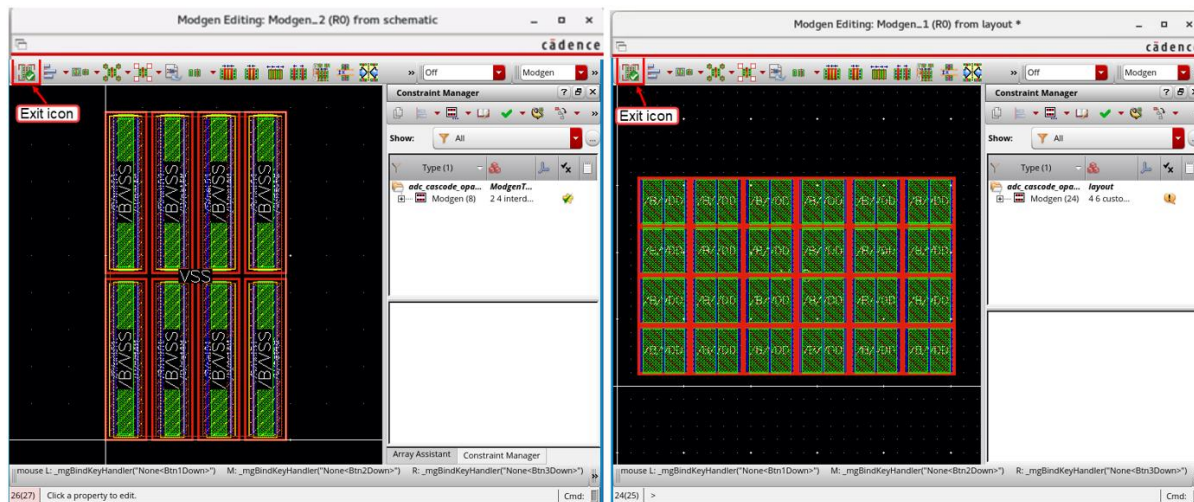
Use one of the following methods to quit the Modgen mode:

- Use the *Exit* icon in the *Modgen Placement* toolbar in the *Modgen Editing* window.
- Use the *Close* button in the *upper-right* corner of the *Modgen Editing* window.
- Use the *Shift+B* Key combination in the *Modgen Editing* window.
- *Right-click* to display the *Modgen Menu* and select *Exit Modgen* in the *Modgen Editing* window.

1. Using the Exit Icon

1. Use the **Exit** icon in the *Modgen Placement* toolbar in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.



Step 1: Quitting the Module Generator Tool Using the Exit Icon

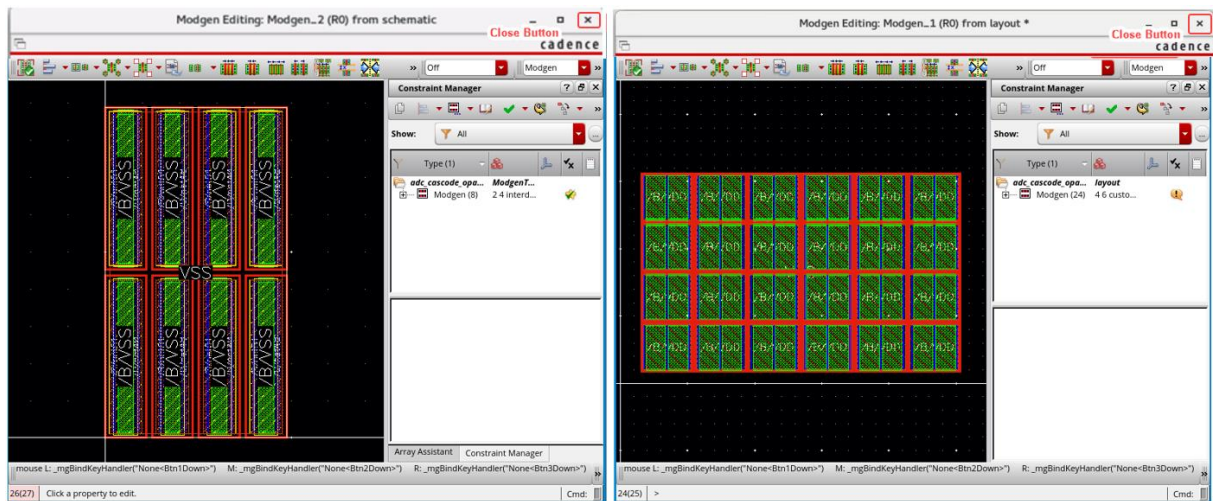
1. Use the *Exit* icon in the *Modgen Placement* toolbar in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.

2. Using the Close (X) Button

1. Use the **Close** button in the *upper-right* corner of the *Modgen Editing* window.

Result: Modgen Editor is dismissed.



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Step 2: Quitting the Module Generator Tool Using the Close (X) Button

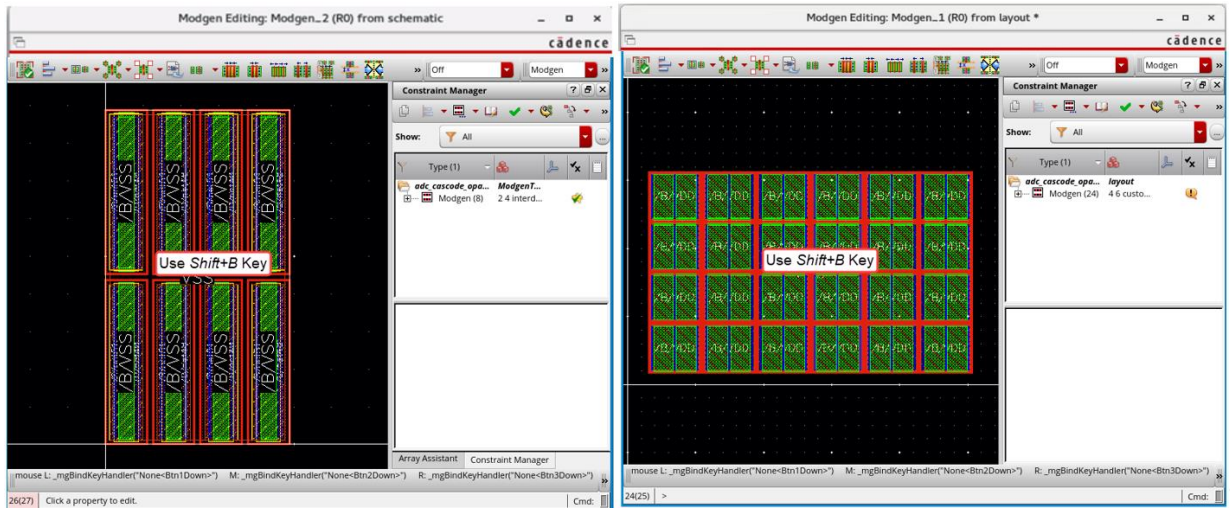
1. Use the *Close* button in the *upper-right* corner of the *Modgen Editing* window.

Result: Modgen Editor is dismissed.

3. Using the Shift+B Key

1. Use the **Shift+B** key combination in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.



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Step 3: Quitting the Module Generator Tool Using the Shift+B Key

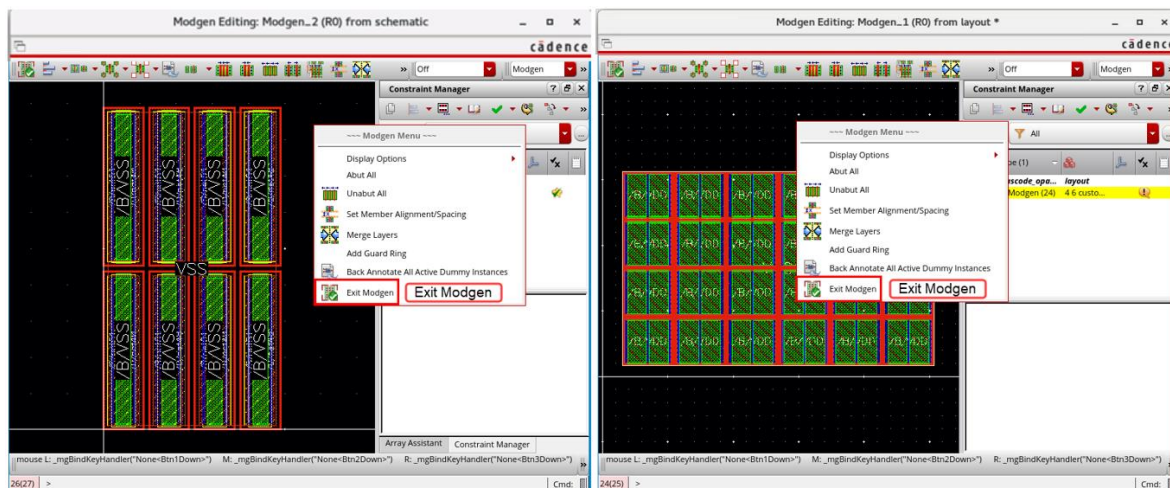
1. Use the *Shift+B* Key combination in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.

4. Using the Exit Modgen

1. Right-click to display the *Modgen Menu* and select **Exit Modgen** in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.



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Step 4: Quitting the Module Generator Tool Using the Exit Modgen

1. Right-click to display the *Modgen Menu* and select *Exit Modgen* in the *Modgen Editing* window.

Result: Modgen Editor is dismissed.

FAQ: Creating the Guard Ring



Where should you define the Multipart Path (MPP) guard rings before creating them in your design?

Before creating the Multipart Path (MPP) guard rings, ensure that they are defined in the technology file.



Where should you define the multipart path (MPP) guard rings before creating them in your design?
Before creating the multipart path (MPP) guard rings, ensure that they are defined in the technology file.

FAQ: Backannotating the Dummy Devices



Which option in the Modgen Placement toolbar can you use to backannotate the dummy instances to the schematic window?

You can use the Backannotate All Active Dummy Instances icon in the Modgen Placement toolbar to backannotate the dummy instances to the schematic window.



Which option in the Modgen tool can you use to backannotate the dummy instances to the schematic?

You can use the Backannotate All Active Dummy Instances icon from the Modgen toolbar to backannotate the dummy instances to the schematic.

FAQ: Quitting the Module Generator Tool



What are the different ways can you use to quit the Module Generator mode?

Use one of the following methods to quit the Modgen mode:

- Use the Exit icon in the Modgen Placement toolbar in the Modgen Editing window.
- Use the Close button in the upper-right corner of the Modgen Editing window.
- Use the Shift+B Key combination in the Modgen Editing window.
- **Right**-click to display the Modgen Menu and select Exit Modgen in the Modgen Editing window.



What are the different ways can you use to quit the Module Generator mode?

Use one of the following methods to quit the Modgen mode:

- Use the Exit icon in the Modgen Placement toolbar in the Modgen Editing window.
- Use the Close button in the upper-right corner of the Modgen Editing window.
- Use the Shift+B Key combination in the Modgen Editing window.
- Right-click to display the Modgen Menu and select Exit Modgen in the Modgen Editing window.

FAQ: Creating Placeable Object



Which tool can you use to create one placeable object from many instances?

You can use the ModGen tool to create one placeable object from many instances.



Which tool can you use to create one placeable object from many instances?

You can use the ModGen tool to create one placeable object from many instances.

FAQ: Adding the Body Contacts



Which form can you use to add body contacts to the array in the layout?

You can use the Body Contact Options form to add body contacts to the array in the layout.



Which form can you use to add body contacts to the array in the layout?

You can use the Body Contact Options form to add body contacts to the array in the layout.

Quiz: Creating a Modgen from Schematic



You can create a ModGen from schematic (select all that apply):

- A. By selecting the instances in the Schematic Canvas
- B. By selecting the instances in the Navigator Assistant
- C. By using the Circuit Prospector Assistant for Special Configurations
- D. By using the Palette Assistant
- E. All of the above

Answer: A, B, and C

You can create a Modgen from schematic:

- By selecting the instances in the Schematic Canvas
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
- By selecting the instances in the Navigator Assistant
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
- By using the Circuit Prospector Assistant for Special Configurations
 - Use the Rapid Analog Prototype (RAP) flow using the Circuit Prospector assistant to find special configurations

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Answer: A, B, and C

You can create a Modgen from schematic:

- By selecting the instances in the Schematic Canvas
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
- By selecting the instances in the Navigator Assistant
 - Use the ModGen icon in the Modgen constraint menu to invoke the Modgen Editing window
- By using the Circuit Prospector Assistant for Special Configurations
 - Use the Rapid Analog Prototype (RAP) flow using the Circuit Prospector assistant to find special configurations

Quiz: Creating Module Generators



Module Generators can be created from either the schematic or layout design.

- A. True
- B. False

Answer: A

Yes. Module generators can be created from either the schematic or layout design.



Answer: A

Yes. Module generators can be created from either the schematic or layout design.

Quiz: Launching the Circuit Prospector Assistant



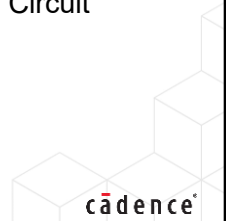
In the schematic window, which workspace can you choose to launch the Circuit Prospector assistant? Select all that apply.

- A. Constraints
- B. Basic
- C. Constraint-Helper
- D. Classic
- E. None of the above

Answer: C

In the schematic window, you can choose the Constraint-Helper workspace to launch the Circuit Prospector assistant.

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Answer: C

In the schematic window, you can choose the Constraint-Helper workspace to launch the Circuit Prospector assistant.

Quiz: Editing the Pattern Preset in the Array Assistant

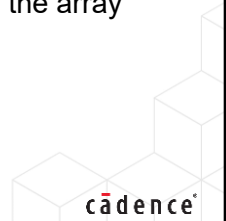


Which is the Pattern Preset that applies the base symbol pattern in the opposite horizontal direction, starting with left-to-right for row index 0? Select all that apply.

- A. Custom
- B. Compact
- C. Zigzag
- D. Shift
- E. None of the above

Answer: C

Zigzag applies the base symbol pattern in the opposite horizontal direction for alternating the array rows, starting with left-to-right for row index 0.



Answer: C

Zigzag applies the base symbol pattern in the opposite horizontal direction for alternating the array rows, starting with left-to-right for row index 0.

Quiz: Specifying the Alignment and Spacing Values



Which form can you use to specify the alignment and spacing values for instances in Modgens? Select all that apply.

- A. Set Member Alignment and Spacing
- B. Set Alignment and Spacing
- C. Specify Member Alignment and Spacing
- D. Specify Alignment and Spacing
- E. None of the above

Answer: A

You can use the Set Member Alignment and Spacing form to specify the alignment and spacing values for instances in Modgens.

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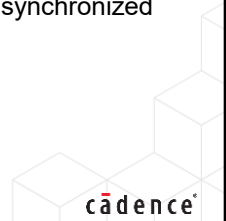
Answer: A

You can use the Set Member Alignment and Spacing form to specify the alignment and spacing values for instances in Modgens.

Module Summary

This module:

- Introduces the Virtuoso Module Generator (Modgen) as a constraint-driven tool to create one placeable, highly matched layout object from multiple Pcell instances.
- Examines multiple workflows to create Modgens from schematic or layout, including Constraint Manager, Place menu, Convert to Modgen, and SKILL functions.
- Demonstrates how to edit and configure Modgens using the Modgen Editor, Placement Toolbar, and Array Assistant.
- Covers array pattern control such as interdigitation, common-centroid placement, dummies, body contacts, empty rows/columns, and guard rings.
- Shows how to route Modgens using the Structured Router and manage routing through Modgen constraints.
- Highlights regeneration, backannotation, and modification flows to keep schematic and layout synchronized and design intent intact.



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Labs

Lab 2-1 Creating a Modgen from Schematic

Lab 2-2 Editing a Modgen

Lab 2-3 Routing a Modgen



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Module 3

Modgen Enhancements

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Module Objectives

In this module, you

- Use the Array Assistant
- Work with Pattern Preset
- Use the Modgen On-Canvas commands
- Use the Topology Editor



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Swapping the Instances



By using the **Swap** function, you can interchange the position of two instances, rows, or columns. This helps optimize routing.

You can modify the device placement directly on the **Pattern** section.

To swap the instances:

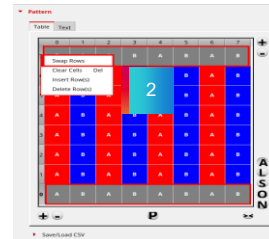
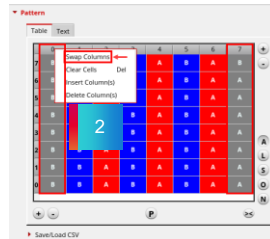
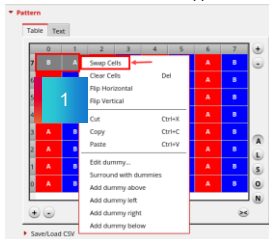
1. Select the two required instances to be swapped on the **Pattern** section.
 - a. Use LMB click to select one instance. Use **<Ctrl> + LMB** click to select multiple instances. **Right-click** to display a pop-up menu.
 - b. Choose **Swap Cells** to swap between the two selected cells.

Result: The two selected cells are swapped.

To swap an entire Column/Row:

2. Select an instance of each in the two columns/rows that you want to swap.
 - a. Use LMB click on a column/row number to select the entire column/row, and **<Ctrl> + LMB** click on another column/row number to select another column/row. RMB will display a pop-up menu.
 - b. Choose **Swap Columns/Swap Rows** to swap between the two selected columns/rows.

Result: The two selected Columns/Rows are swapped.



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Swapping Instances

By using the swap function, you can interchange the position of two instances, rows, or columns. This helps optimize routing.

You can modify the device placement tab directly on the **Pattern**.

To swap instances:

1. Select the two required instances in the Pattern.
 - a. LMB click selects one instance. Use **<Ctrl> + LMB** click to select multiple instances. **Right-click** to display a pop-up menu.
 - b. Choose **Swap Cells** to swap between the two selected cells.

To swap an entire row/column:

1. Select an instance of each in the two columns/rows that you want to swap.
 - a. LMB click on a column/row number to select the entire column/row, and **<Ctrl> + LMB** click on another column/row number to select another column/row. RMB will display a pop-up menu.
 - b. Choose **Swap Columns/Rows** to swap between two selected columns/rows.

Assigning Device(s) to Empty Slot(s) Using the Pattern Symbol Mapping Section



Use the *Pattern Symbol Mapping* region of the Array Assistant to view and change the mapping of instances and dummies.

The graphic explains the columns in the *Pattern Symbol Mapping* table.

Pattern Symbol Mapping

Sym	Name	Type	# avail	# In use
*	*	dummy	-	2
A	MB	inst	32	32
	MD	inst	32	32

Annotations:

- Dummies:** Points to the row with Symbol '*' and Name '*'.
- Symbol A represents device MB:** Points to the row with Symbol 'A' and Name 'MB'.
- There are 32 available instances of the device:** Points to the '# avail' column for the row with Symbol 'A'.
- All 32 instances are placed on the grid:** Points to the '# In use' column for the row with Symbol 'A'.

Assigning Device(s) to Empty Slot(s) Using the Pattern Symbol Mapping Assistant

Use the *Pattern Symbol Mapping* region of the Array Assistant to view and change the mapping of instances and dummies.

The following figure explains the columns in the *Pattern Symbol Mapping* table.

Using the Paste on Grid Option

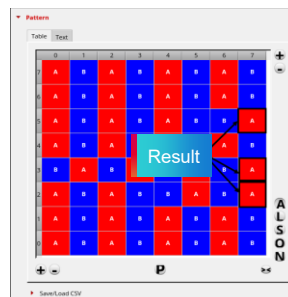
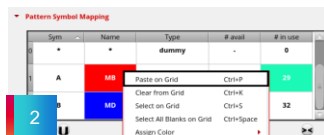
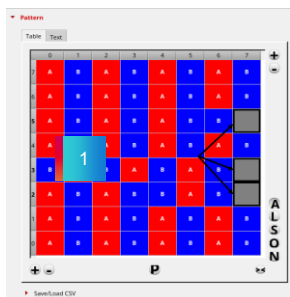


Select multiple devices in the **Pattern** section and then run this command. The selected devices will be replaced with the device that is highlighted in the Pattern Symbol Mapping.

In this example:

1. LMB to select an empty slot in the array in the Pattern.
2. Click the LMB on the unplaced device row and choose the **Paste on Grid** in the Pattern Symbol Mapping section.

Result: Unplaced devices are now placed in the selected empty slot of the Array Assistant.



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Paste on Grid

Select multiple devices in the Pattern and then run this command. The selected devices will be replaced with the device that is highlighted in the Pattern Symbol Mapping.

In the following example,

1. LMB to select an empty slot in the array in the Pattern.
2. Click the LMB on the unplaced device row and choose the paste on grid in the Pattern Symbol Mapping.
3. Unplaced devices are now placed in the selected empty slot of the array assistant.

Creating Guard Rings

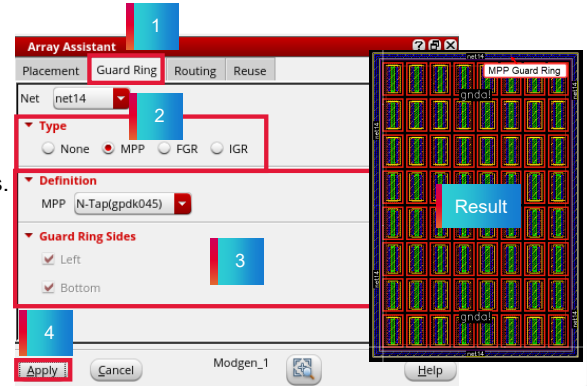


A ring-shaped conductor is used around one or more sensitive devices or circuit sections. It helps guard against spurious or unwanted signals or leakage to or from that device or circuit section.

Creating guard rings using the Array Assistant:

1. Open the **Guard Ring** tab.
2. Select the *Type* of guard ring.
 - a. **None**:(Default).
 - b. **MPP**: Displays options to create multipart path (MPP) guard rings.
 - c. **FGR**: Displays options to create Fluid guard rings (FGR).
 - d. **IGR**: Displays options to create Identical guard rings (IGR).
3. Specify the guard ring parameters.
4. Click **Apply**.

Result: Guard Ring is created as per the specified values.



Before creating guard rings, ensure that they are defined in the technology file.

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Creating the Multipart Path (MPP) Guard Ring



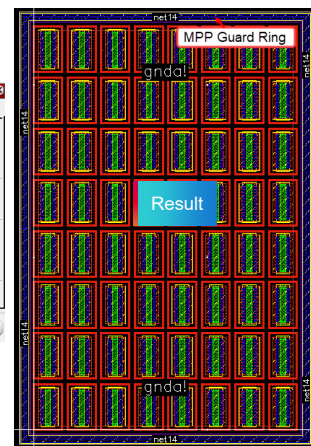
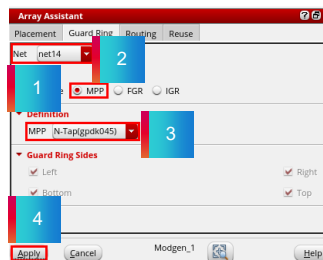
A multipart path is a named path that has relative object design (ROD) information associated with it. You can create simple one-part paths or complex paths containing several parts, such as a guard ring, transistor, bus, or shielded path. A path that consists of more than one part is called a multipart path.

Multipart Path (MPP): The tool supports only surrounding MPP guard rings. Although the sides are listed, you cannot control the sides along which the guard ring is added.

Creating MPP Guard Ring:

1. In the *Type* field, select **MPP**.
2. Select Net as net14.
3. In the Definition field, choose an MPP for the guard ring.
 - a. Here, you can choose N-Tap(gpdk045).
4. Click **Apply**.

Result: MPP guard ring is created.



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Creating the Fluid Guard Ring (FGR)

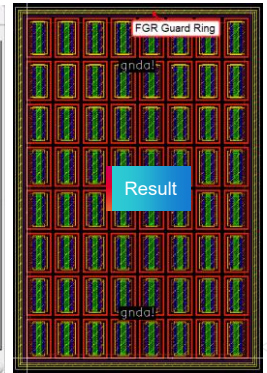
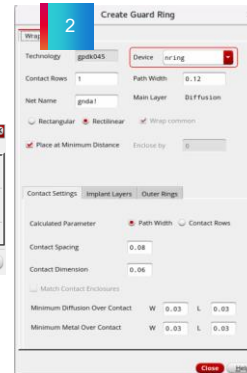


In addition to multipart path guard rings, you can use the Modgen Editor to create fluid guard ring (FGR) around objects. FGR are a type of fluid Pcells that can be used to enclose one or more objects such as devices or device chains.

Creating FGR Guard Ring:

1. Select Create/Replace FGR.
2. The Create Guard Ring form is displayed.
 - a. In the Wrap mode, an FGR is created around the objects you select. The FGR parameters are stored in the Modgen constraint.
 - b. Select nring in the device.
 - c. Remaining with the default setting, click on the **Close** button.
3. Click **Apply** in the **Guard Ring** tab.

Result: FGR guard ring is created.



FGRs can be created by drawing a path, a rectangle, or a polygon, or by using the wrap mode.

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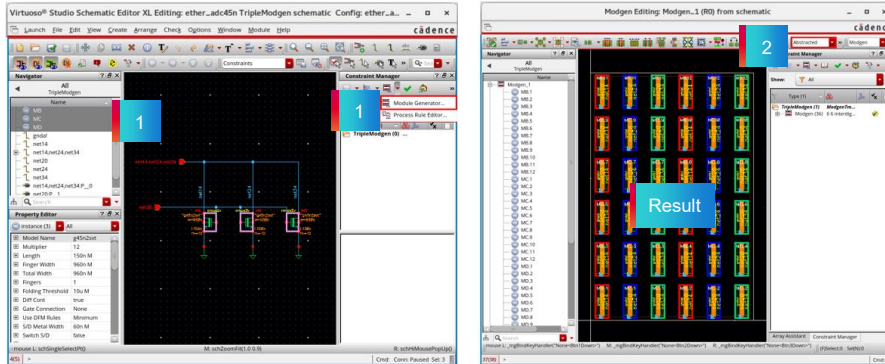
Using the Pattern Preset



The **Pattern Preset** specifies the arrangement of devices within a Modgen. You can choose a pattern preset from a drop-down list, specify the base symbol pattern, or specify the base orientation pattern.

1. Select the devices MB, MC, and MD and invoke the Modgen Editor by clicking on the Module Generator in the Constraint Manager.
2. Set the Smart Display Configuration to *Abstracted* in the Modgen Editor window.

Result: Modgen Editor displays in *Abstracted* mode.



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Input Base Symbol Pattern

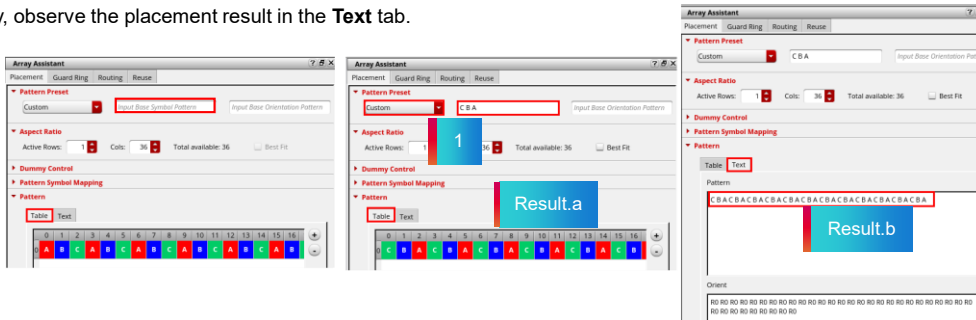


Specifies the base symbol pattern.

1. In the Placement tab, enter CBA in the Input Base Symbol Pattern field. Set Pattern Preset to **Custom** and click **Apply**.

Result: In the Pattern,

- a. In the **Table** tab, observe the Grid result.
- b. Similarly, observe the placement result in the **Text** tab.



The **Table** tab displays the grid pattern in a tabular format.
The **Text** tab displays a textual pattern for the grid.

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Input Base Orientation Pattern

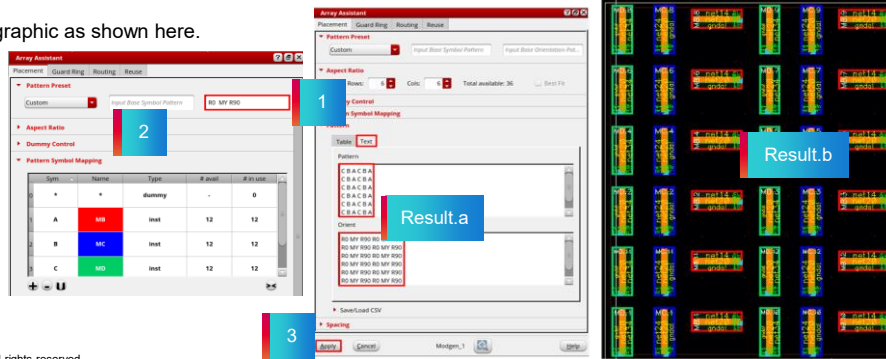


Specifies the base orientation pattern.

1. In the **Placement** tab, enter R0 MY R90 in the Input Base Orientation Pattern field.
2. Select **Custom** in Pattern Preset.
3. Click **Apply**.

Result: In the Pattern,

- Observe the placement result in the **Text** tab.
- Observe the graphic as shown here.



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Common Centroid

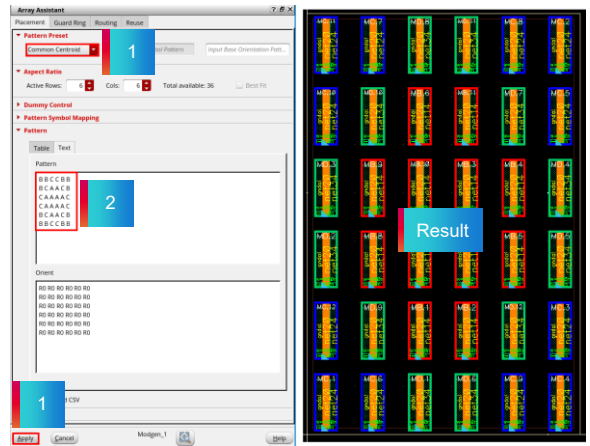


Devices are placed in a grid using a centroid pattern. Symbols are arranged in order of precedence starting from the center. Priority is determined by the number of instances, in ascending order, and then alphabetically if the numbers are the same.

In the **Placement** tab,

1. Select *Common Centroid* in the Pattern Preset field and click on **Apply**.
2. It automatically updates in the Pattern field.

Result: Generated pattern in the layout canvas.



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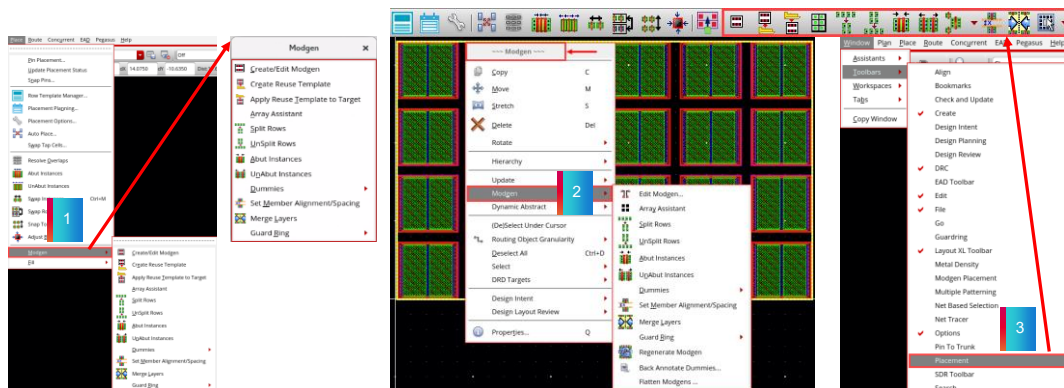
Accessing the Modgen On-Canvas Commands



Virtuoso® provides the Modgen on-canvas commands that can be used to perform certain tasks without opening the Modgen Editor. In the Modgen menu, you will find all the operations that can be made on a Modgen directly from the layout canvas.

To access the Modgen on-canvas commands:

1. In the layout canvas, choose **Place – Modgen** to access the Modgen menu commands.
2. Select the *Modgen* constraint in the layout canvas and choose **Modgen** from the *Modgen* shortcut menu to access the Modgen menu commands.
3. In the layout canvas, choose **Window – Toolbars – Placement** to access the *Placement* toolbar, which provides the same set of operations.



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Accessing the Modgen On-Canvas Commands

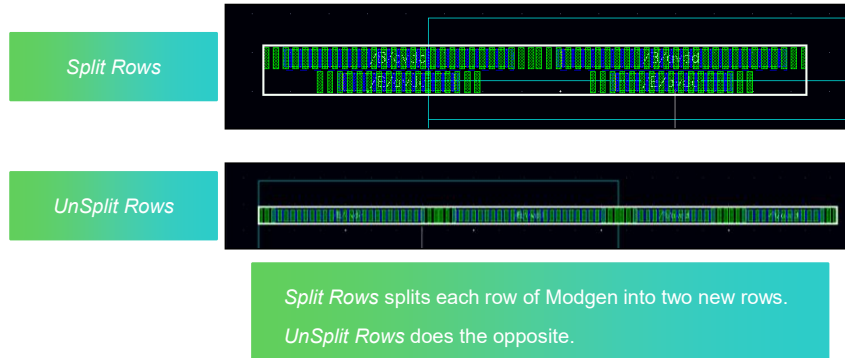
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- In the layout canvas, choose **Window – Toolbars – Placement** to access the *Placement* toolbar which provides the same set of operations.

Example: Split Rows and UnSplit Rows

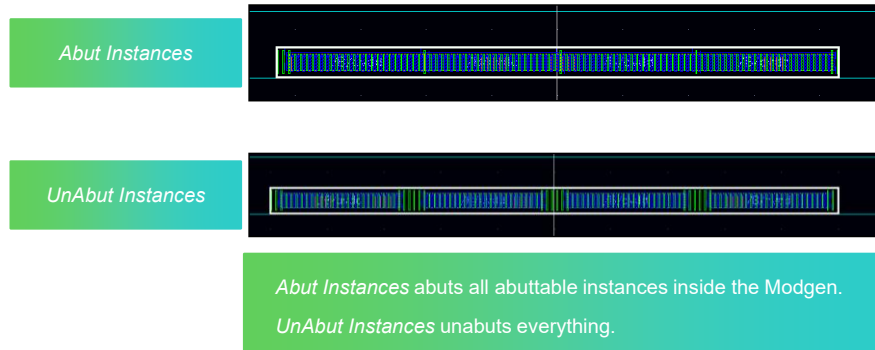


Example: Split Rows and UnSplit Rows

Split Rows splits each row of Modgen into two new rows.

UnSplit Rows does the opposite.

Example: Abut Instances and UnAbut Instances

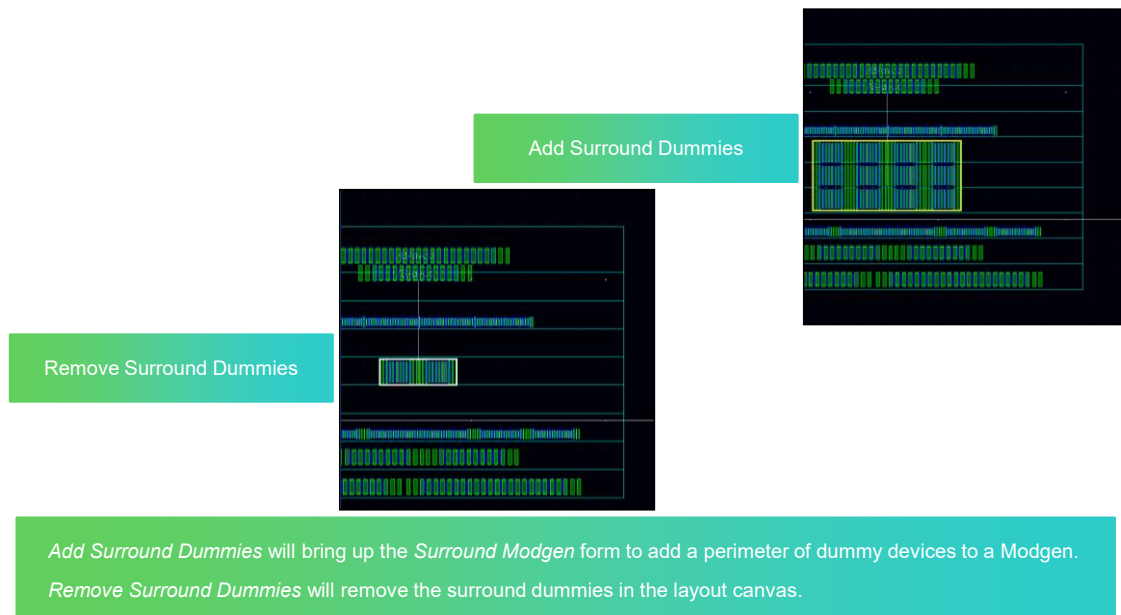


Example: Abut Instances and UnAbut Instances

Abut Instances abuts all abutable instances inside the Modgen.

UnAbut Instances unabuts everything.

Example: Add Surround Dummies and Remove Surround Dummies



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Example: Add Surround Dummies and Remove Surround Dummies

Add Surround Dummies will bring up the *Surround Modgen* form to add a perimeter of dummy devices to a Modgen.

Remove Surround Dummies will remove the surround dummies in the layout canvas.

Creating Generic Modgen Constraints

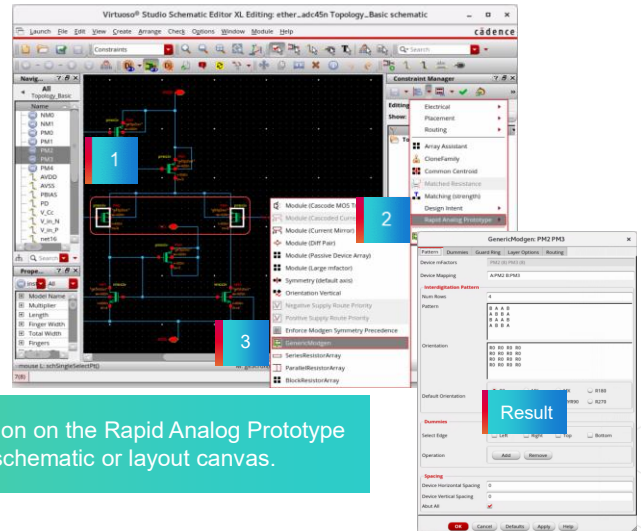


The GenericModgen constraint generator enables you to create generic modgen constraints on one or more selected devices.

To run the Generic Modgen:

1. Select the instances PM2 and PM3 in the schematic.
2. Go to the Rapid Analog Prototype submenu in the Constraint Creation menu of the Constraint Manager assistant.
3. Click on the GenericModgen option.

Result: GenericModgen form opens.



To enable the GenericModgen constraint generator option on the Rapid Analog Prototype sub-menu, ensure to select at least one device on the schematic or layout canvas.

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GenericModgen Form

The GenericModgen form appears with the following tabs: Pattern, Dummies, Guard Ring, Layer Options, and Routing.

In the **Pattern** tab:

1. In the Device Mapping field, specify the symbols to be used for creating the device interdigitation pattern. By default, the field shows the device mapping of the currently selected devices.
2. Set the number of rows in the Num Rows field to adjust the arrangement of device symbols in the interdigitation pattern. The Num Rows field is linked to the Pattern and Orientation fields, and changing any of these fields will also adjust the other two field values.
3. Specify the device interdigitation Pattern. In this field, each device symbol must appear as many times as the mFactor of the corresponding device. The dummies are specified by an asterisk (*) and gaps by a hyphen (-).

You can undo the changes made using the Ctrl+Z shortcut key, which works only in these three fields.

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GenericModgen Form (continued)

4. Specify the device Orientation. Each orientation must appear as many times as the mFactor of the corresponding device.
5. Select the Default Orientation from the given radio buttons: R0, MX, MY, R180, R90, MXR90, MYR90, or R270. These radio buttons are useful for defining the default orientation of the dummies, if any are added using the Dummies group box.
6. Select the required check box adjacent to the Select Edge field in the Dummies group box. This specifies the side on which the dummies should be added in the pattern, that is, Left, Right, Top, and Bottom.
7. Click the required button, Add or Remove, to specify the Operation to be performed to add new or remove existing dummies.
8. Enter the Device Horizontal Spacing in the Spacing group box.
9. Enter the Device Vertical Spacing.
10. Select the Abut All checkbox if the abutment of devices is needed.

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What Is a Topology Pattern?



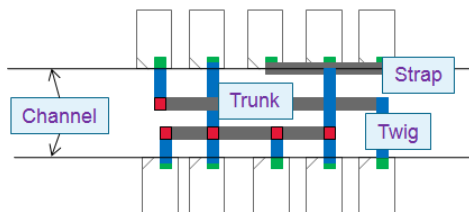
A **Topology Pattern** is a routing plan for a design that is used by the Pin-to-Trunk Router while creating routes.

Use the topology options in Modgen to visualize, configure, and store the routing information.

Topology patterns comprise the following topology objects:

- **Trunks:** Specify the location of the routes, including their anchor points and offsets and the routing extents.
- **Twigs:** Define the connections between the trunks and the devices.
- **Channels:** Define the routing areas in a design.
- **Straps:** Define the connections between multiple objects. Straps are used for specific purposes, depending on the constraints.

Use of *topology objects* in a *topology pattern*.



What Is Topology Pattern?

A **Topology Pattern** is a routing plan for a design which is used by the Pin-to-Trunk Router while creating routes.

Use the topology options in Modgen to visualize, configure, and store the routing information.

Topology patterns comprise the following topology objects:

- **Trunks:** Specify the location of the routes, including their anchor points and offsets, and the routing extents.
- **Twigs:** Define the connections between the trunks and the devices.
- **Channels:** Define the routing areas in a design.
- **Straps:** Define the connections between multiple objects. Straps are used for specific purposes, depending on the constraints.

Modgen Topology and Pin-to-Trunk (P2T) Routing



Overview

- Pin-to-Trunk Router will be the router for Modgen.
- Topology can be created both interactively and procedurally.
- Topology can be created in either the schematic preview editor or in the layout canvas (RMB edit modgen).
- Topology is used to store the layer attributes, route information, and directives used by the Pin-to-Trunk Router.
- Toolbars and menu items to access the features and commands.
- Constraint Template Flows for topology and P2T are supported from the schematic and can be customized by the user.
- SKILL® APIs are available for topology creation, editing, querying, and setting of topology layer attributes, constraints, and router directives.
- Modgen Pin-to-Trunk router uses Virtuoso-defined constraint look-up precedence.



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Creating Topology Patterns



- Topology options in the Modgen Editor allow for visualization, configuring, and storing of layer and advanced routing attributes and constraints.
 - Create all topologies and twigs in all channels.
 - Create a set of topologies in selected channels or over devices.
 - Add individual topologies in channels or over devices.
 - Define straps horizontally across multiple device instTerms.
 - Example gate strapping
 - Define straps vertically across channels for aligned instTerms.
- Set routing attributes on Trunks, Twigs, and Straps.
 - At creation, through editing, and through SKILL APIs.
- Copy Trunks Topologies from one channel to another.
 - Allows users to create, modify, and define topology in a single channel and copy to another, saving significant effort and time.



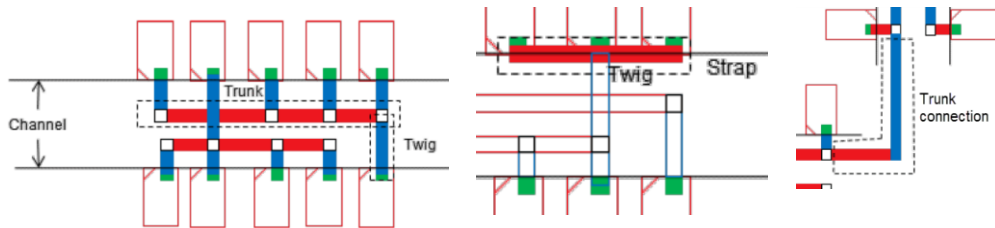
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Definitions of Topology Patterns

Topology Pattern

- Contains the set of rules for the Pin-to-Trunk Router.
- The different topology patterns specify specific routing types.
 - Trunk – Straight line interconnect used to connect instTerms.
 - Twig – Straight line interconnect between an instTerm and a Trunk.
 - Strap – The connection between a number of instTerms. A strap is connected to a trunk using a twig.
 - Trunk connection – The connection between more than one trunk.



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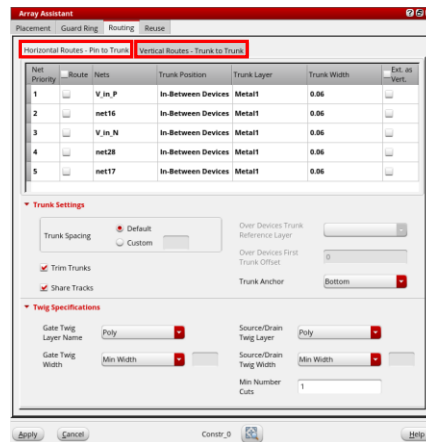
Defining Modgen Topology Settings Using the Array Assistant



The **Routing** tab of the Array Assistant provides options to use the Pin to Trunk router to define Modgen topology patterns. This involves specifying routing preferences for horizontal nets and vertical nets. Modgen topology patterns help you to visualize, configure, and store routing information.

To define the Modgen topology pattern and routing options:

- Use the **Routing** tab of the Array Assistant.
 - **Horizontal Routes – Pin to Trunk** tab: Specify the routing preferences for horizontal nets.
 - **Vertical Routes – Trunk to Trunk** tab: Specify the routing preferences for vertical nets.



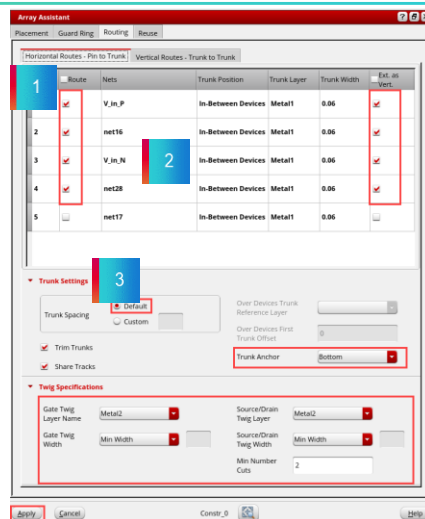
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Creating Horizontal Routes - Pin to Trunk



Pin to Trunk routing allows you to add connections from the power pins of pad or macro instances to existing rings and rails. A macro instance is defined as any instance whose area is greater than 90 percent of the entire cell view area.

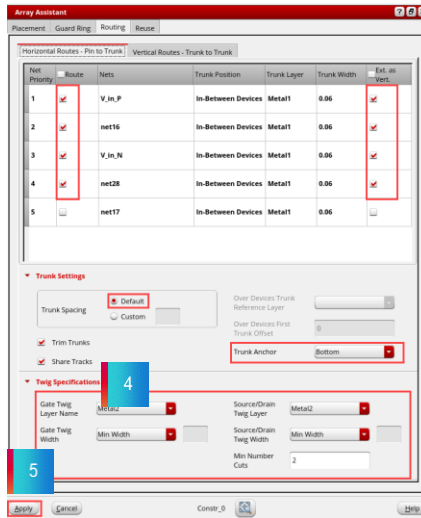
1. Open the **Routing** tab of the Array Assistant. The **Horizontal Routes - Pin to Trunk** tab is displayed by default.
2. Customize the settings in the net table, which lists all the channel nets in the current Modgen.
3. Set the required options in the Trunk Settings section. These settings are applied only to the nets selected for routing.



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Creating Horizontal Routes - Pin to Trunk (continued)

- Set the required options in the **Twig Specifications** section. These settings are applied only to the nets selected for routing.
- Click **Apply** to apply the routing settings to create the Pin to Trunk routing.



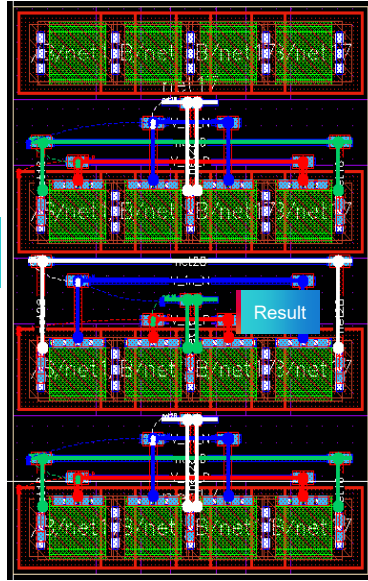
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Result: Creating Horizontal Routes – Pin to Trunk

Result: Pin to Trunk routing is created.



Select *Ext. as Vert.* to specify the net is to be listed on the *Vertical Routes – Trunk to Trunk* tab.

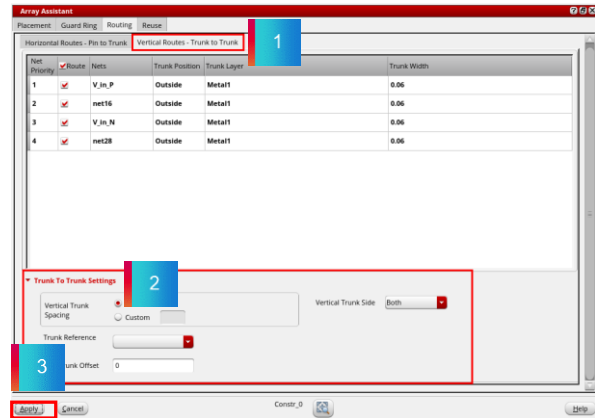
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Creating Vertical Routes – Trunk to Trunk



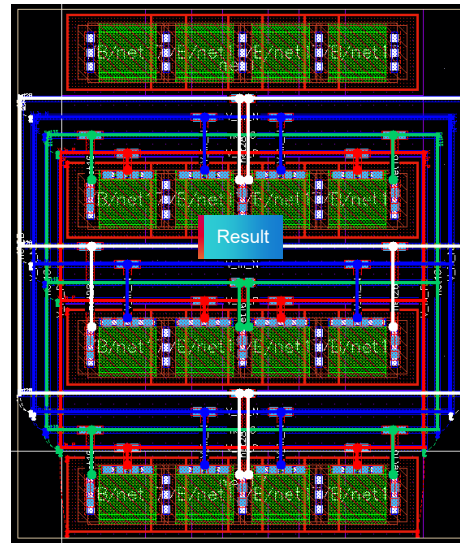
Trunk to Trunk tab specifies routing preferences for vertical nets.

1. Switch to the **Vertical Routes – Trunk to Trunk** tab.
2. Specify the required options in the Trunk to Trunk section.
3. Click **Apply** to apply the routing settings to create the Trunk to Trunk routing.



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Result: Creating Vertical Routes – Trunk to Trunk



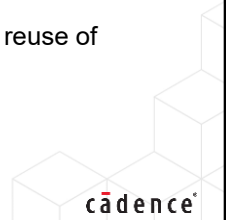
Result: Trunk to Trunk routing is created.

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Module Summary

This module:

- Introduces advanced use of the Array Assistant to control placement, guard rings, routing, and reuse settings for Modgens.
- Demonstrates enhanced pattern manipulation, including swapping instances/rows/columns and assigning devices to empty slots using Pattern Symbol Mapping.
- Examines Pattern Preset enhancements, including custom base symbol patterns, orientation patterns, and common-centroid placement.
- Shows how to use Modgen On-Canvas commands to perform common operations directly in the layout without opening the Modgen Editor.
- Covers creation and configuration of guard rings, including MPP, Fluid, and identical guard rings using the Array Assistant.
- Introduces the Topology Editor and Pin-to-Trunk routing, enabling definition, visualization, and reuse of routing topology patterns for Modgen designs.



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Labs



Lab 3-1 Using the Array Assistant

Lab 3-2 Working with Pattern Preset

Lab 3-3 Using the Modgen On-Canvas Commands

Lab 3-4 Using the Topology Editor

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Module 4

Floorplanner

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Module Objectives

In this module, you

- Define the Cell Type attribute
- Use the Configure Physical Hierarchy utility
- Generate and place the blocks with the floorplanner
- Edit the Level-1 objects
- Use the Pin Optimizer
- Align the soft block pins with the top-level pins
- Create Feed Through Terminal pins



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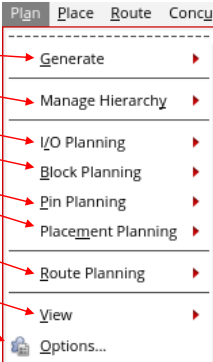
What Is the Floorplanner?

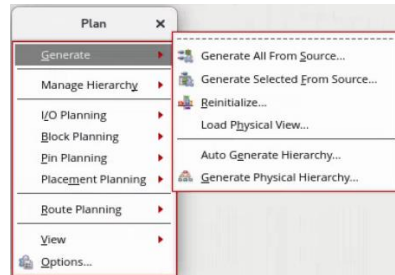


The **Floorplanner** enables you to estimate the area and to work with the hierarchy early in the design cycle, thus enabling you to adopt a top-down approach to design, a bottom-up approach, or a mix of both methods.

- The Virtuoso® Floorplanner contains automatic and interactive floorplanning tools and technologies that help the mixed-signal designers develop a layout from a schematic in a hierarchical manner.
- Use the *Plan* menu in the Virtuoso Studio Layout Suite for your floorplanning tasks.

The commands in the *Plan* menu are:

- Generate
 - Manage Hierarchy
 - I/O Planning
 - Block Planning
 - Pin Planning
 - Placement Planning
 - Route Planning
 - View
 - Options
- 



Example: In the *Plan* menu, the *Generate* sub-menu commands are shown here.

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Licensing of the Floorplanner



In IC 25.1, the floorplanning commands require one of the following licenses:

- A Virtuoso_Layout_Suite_XL license combined with 4 GXL flexible license tokens
- A Virtuoso_Layout_Suite_EXL license

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Licensing of the Floorplanner

Virtuoso Floorplanner requires the following licenses:

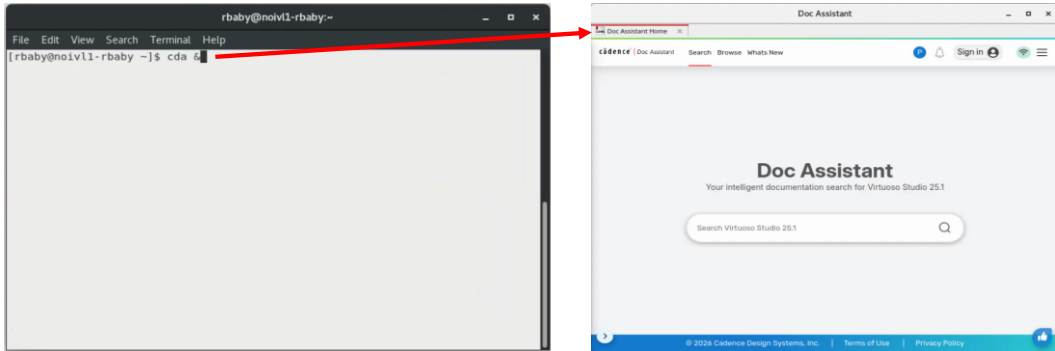
In IC 25.1, the floorplanning commands require one of the following licenses:

- A Virtuoso_Layout_Suite_XL license combined with 4 GXL flexible license tokens
- A Virtuoso_Layout_Suite_EXL license

Using the Cadence Doc Assistant



In the terminal window, to start **Cadence® Doc Assistant**, enter:
Unix>>cda &



You can access, view, and search the doc assistant for all the Cadence products installed on your machine.



Using the Cadence Doc Assistant

In the terminal window, to start **Cadence Doc Assistant**, enter:

Unix>>cda &

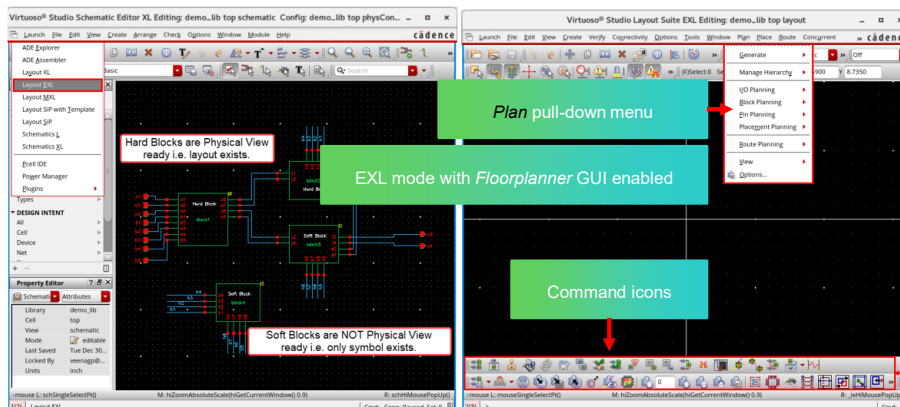
You can access, view, and search the doc assistant for all Cadence products installed on your machine.

Starting the Virtuoso Floorplanner



In the schematic window, choose **Launch – Layout EXL**. The layout view opens in EXL mode with the *Floorplanner* GUI enabled.

The Virtuoso Layout Suite EXL includes the *Plan* menu that enables you to develop a layout from a schematic in a hierarchical manner.



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Floorplanning: An Overview

The Virtuoso Layout Suite EXL includes the *Plan* menu that enables you to develop a layout from a schematic in a hierarchical manner. Floorplanner enables you to estimate the area and to work with the hierarchy early in the design cycle, thus enabling you to adopt a top-down approach to design, a bottom-up approach, or a mix of both methods.

Using the Floorplan Commands

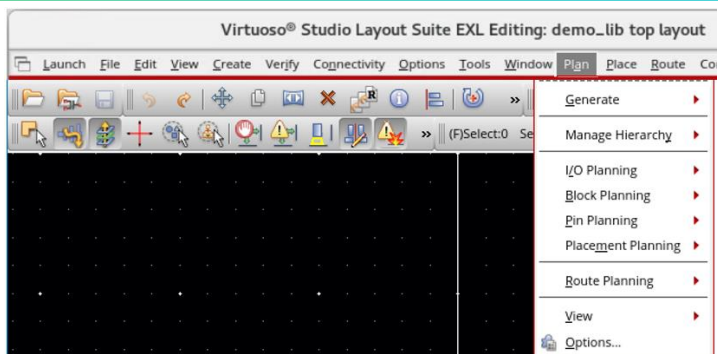


From the **Layout** → **Plan** menu, you can access the *Floorplan* commands.

The *Floorplan* commands are available in the Virtuoso Layout Suite EXL and MXL Suites in the *Plan* menu.

The available commands in the plan menu are:

- Generate
- Manage Hierarchy
- I/O Planning
- Block Planning
- Pin Planning
- Placement Planning
- Route Planning
- View
- Options



The *Plan* menu contains commands that manipulate the hierarchy and define floor plans.

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Using the Floorplan Commands

The *Floorplan* commands are available in Virtuoso Studio Layout Suite in the *plan* menu. The *Floorplan* menu is available by default in the layout window. To view the Floorplan commands, click the *plan* menu in the layout window.

The *plan* menu provides commands to manipulate hierarchy and define floorplans. Using the Floorplan commands, you can create a rough topology of the design rather than the final layout solution. You can use these commands to perform floorplanning tasks such as reinitializing a layout, placing I/Os and macros, optimizing pins, and update labels. The *plan* menu also includes additional commands to block planning, pin planning, placement planning, route planning

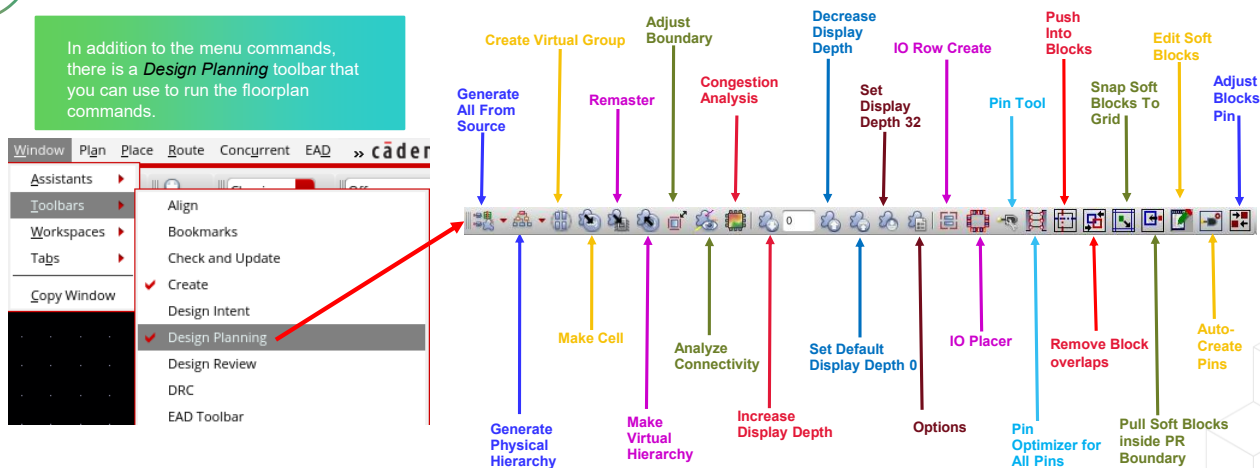
Using the Design Planning Toolbar



Use the *Design Planning* toolbar to run the Floorplan commands.

It can be accessed using the **Window – Toolbars – Design Planning** menu command.

In addition to the menu commands, there is a *Design Planning* toolbar that you can use to run the floorplan commands.



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Using the Floorplan Toolbar

In addition to the menu commands, there is a Design planning toolbar that you can use to run the floorplan commands. The Design planning toolbar is available by default.

Refer to *Virtuoso Floorplanner User Guide Product Version IC23.1* for more details.

Using the Design Planning Workspace

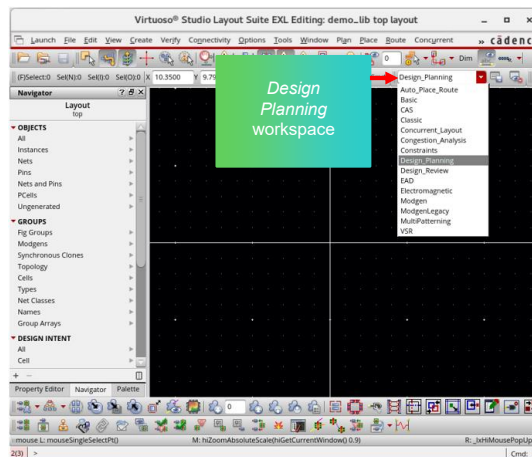


Floorplanner has its own workspace *Design Planning* in Layout EXL and MXL.

In the layout, choose **Workspace – Design Planning** to bring up the Design Planning workspace.

The *Design Planning* workspace shows:

- The Layout Canvas
- The Toolbars
- The Navigator Assistant
- The Property Editor Assistant
- The Palette Assistant



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Using the Design Planning Workspace

Floorplanner has its own workspace *design Planning* in Layout EXL and MXL.

In the layout, choose **Workspace – Design Planning** to bring up the Design Planning workspace.

The *Design Planning* workspace shows the layout canvas, the toolbars, the Navigator assistant, the Property Editor assistant, and the Palette assistant.

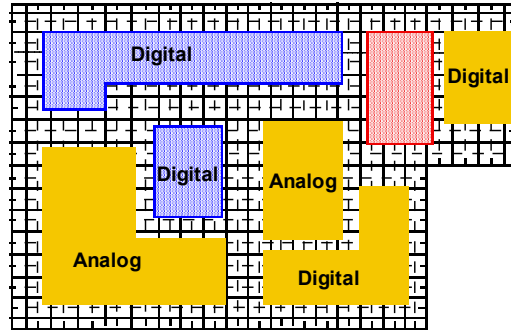
Virtuoso Floorplanner Overview



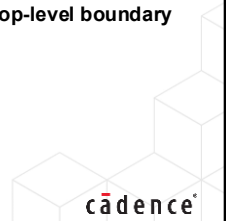
The Virtuoso Floorplanner is a mixed-signal custom floorplanner (non-timing-driven).

It provides:

- Analog and digital block support:
 - Soft block support
 - Rectilinear soft and hard block support
 - Rectilinear top-level support
 - Blockages



- Placement Grid and Routing Grid
- Manufacturing Grid
- Hard Macro
- Soft Block
- Blockages (routing and placement)
- Top-level boundary



Virtuoso Floorplanner Overview

Virtuoso Floorplanner is a mixed-signal custom floorplanner (non-timing-driven).

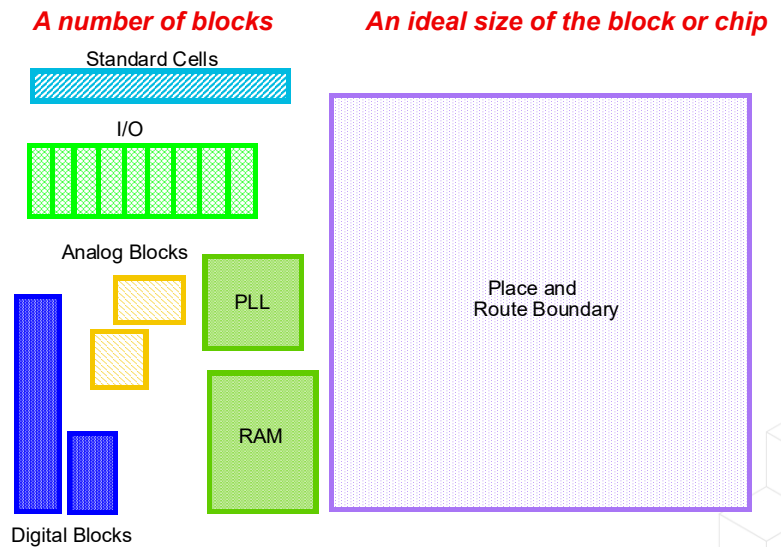
It provides:

- Analog and digital block support:
 - Soft block support
 - Rectilinear soft and hard block support
 - Rectilinear top-level support
 - Blockages

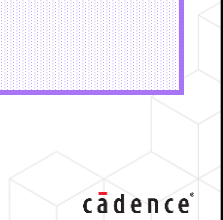
Reviewing the Block or Top-Level Design

- Existing and proposed abstract blocks.
- When generating the Physical Hierarchy, the following are created:
 - The place-and-route boundary.
 - A number of blocks like:
 - Standard Cells
 - I/O
 - PLL
 - RAM
 - Analog Blocks
 - Digital Blocks

– They are generated and placed around the prBoundary.



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Block or Top Level

- Existing and proposed abstract blocks.
 - When generating the Physical Hierarchy, the following are created:
 - The place-and-route boundary.
 - A number of blocks like:
 - Standard Cells
 - I/O
 - PLL
 - RAM
 - Analog Blocks
 - Digital Blocks
- They are generated and placed around the prBoundary.

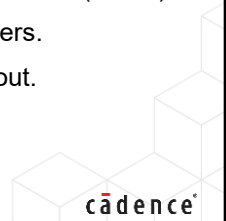


Virtuoso Floorplanner Key Features

Review the Floorplanner Key Features:

- Configure the hierarchy to specify the cells for placement.
- Categorize various blocks to match their Floorplanner purpose:
 - I/O pads
 - Standard cells
 - Abstracts
 - Cells
 - Editable cells (soft blocks)
- Use connectivity-driven and constraint-driven information:
 - Constraint group lookup sequence
- Support for rectilinear blocks to create and edit the prBoundary at the top level (level0) and the block (level1).
- Create and edit various blockages that will be respected by the Pin Optimizer, placer, and routers.
- Load instance and pin locations, constraints, and place-and-route objects from an existing layout.

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Floorplanning Approaches



Refer to the information about *Top-Down* and *Bottom-Up* approaches.

Top-Down Approach

- Allows planning higher-level blocks (or chip level) without waiting for sub-blocks to finalize.
- Gives designers preliminary block sizes (soft blocks) to design the block's layout (derived constraints from top level to subordinate levels).
- Pins on soft blocks can be optimized to adjacent blocks.

Bottom-Up Approach

- Create low-level cells to establish the area.
- Use abstracts to represent unfinished layout views.
- Re-run the floorplan as blocks finalize and mature.
- Propagate constraints for creating cell size estimates.

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Setting the Cell Type from the CIW



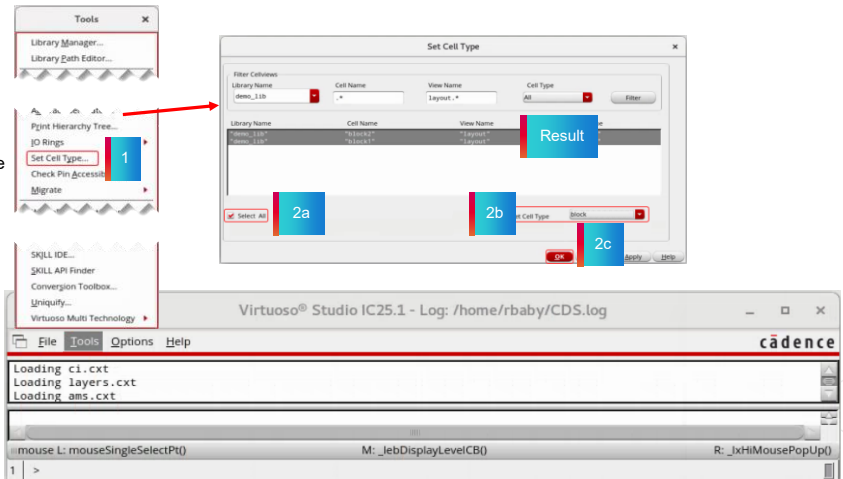
The Virtuoso Floorplanner requires information about the Cell Type to distinguish the cells at the floorplanning level.

To set the Cell Type from CIW in your design:

1. In the CIW window, choose **Tools – Set Cell Type** to open the Set Cell Type form.
2. In the Set Cell Type form, select the options as following:
 - a. Click **Select All** to select the required blocks in the design.
 - b. Change **Set Cell Type** to *block*.
 - c. Click **OK** in the Set Cell Type form.

Result: This will enable the Virtuoso Floorplanner to treat both the blocks, *block1* and *block2*, as hard blocks.

You can set the *Cell Type* for multiple instances.



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Setting the Cell Type

Floorplanner requires information about the celltype to distinguish the cells at the floorplanning level. Floorplanner behaves differently depending on celltype. For example, IO Placer will work on pads only and not on other cells.

Setting the Cell Type from the CIW

You can set the cell type for multiple instances using the *Tools – Set Cell Type* command from the CIW menu bar.

- From the CIW, choose *Tools – Set Cell Type*. The Set Cell Type form is displayed.
- To find the cells you want to set the cell type for:
 - Select a *Library Name* from the cyclic list.
 - Type the strings you want to filter by in the *Cell Name* and *View Name* fields and select a *Cell Type* from the drop-down list.

Note: For performance reasons, the *View Name* field does not accept regular expressions.

- Click *Filter* to filter the list to contain only those cells that match the options settings.
- Select the cells you want from the list and choose the cell type you want to set from the *Set Cell Type* drop-down list.
- Click *Apply* or *OK*. The specified cell type is set for the selected cells.

Launching the Configure Physical Hierarchy (CPH) Utility

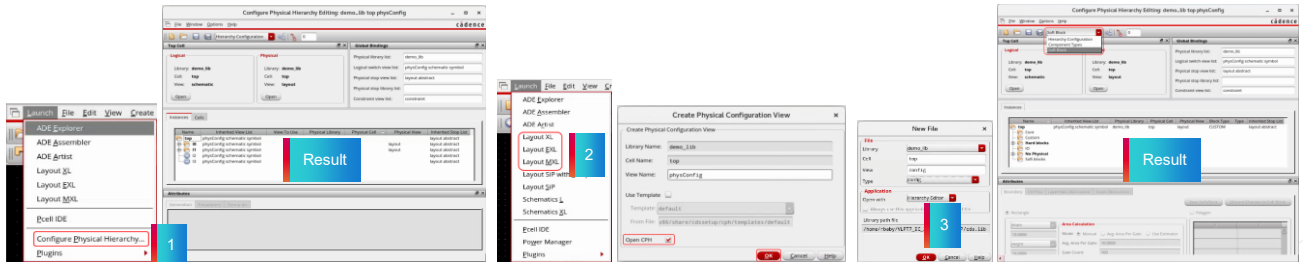


Configure Physical Hierarchy (CPH) is a physical hierarchy configuration utility that lets you specify how a layout implementation is generated from a specific schematic design.

To open the CPH in your design:

1. From the Layout XL/EXL/MXL window, choose **Launch – Configure Physical Hierarchy** to open the CPH form.
2. When launching Layout XL/EXL/MXL from a schematic, check the **Open CPH** option in either the **Create Physical Configuration View** or **Open Physical Configuration View** dialog to open the CPH form.
3. From the CIW, choose **File – New** or **File – Open** and create or open a cellview of type *physConfig* to open the CPH form.

Result: You can open the Configure Physical Hierarchy window by using any one of these steps.



CPH has 3 modes: *Hierarchy Configuration*, *Component Types*, and *Soft Block*.

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Configuring the Physical Hierarchy

Configure Physical Hierarchy (CPH) is a physical hierarchy configuration utility that lets you specify how a layout implementation is generated from a specific schematic design.

The introduction of this functionality means that the layout designer no longer needs to have “write” access to the schematic design to add or modify the properties that drive layout generation. As these properties are now controlled in the *Configure Physical Hierarchy* window, a layout designer can define different physical configurations and view different physical implementation possibilities without changing the schematic view.

Launching the Configure Physical Hierarchy (CPH) Utility

To open the *Configure Physical Hierarchy* window, do one of the following.

- From the Layout XL (EXL or MXL) window, choose *Launch – Configure Physical Hierarchy*.
- When launching Layout EXL (XL or MXL) from a schematic, check the *Open CPH* option in either the *Create Physical Configuration View* or *Open Physical Configuration View* dialog.
- From the CIW, choose *File – New* or *File – Open* and create or open a cellview of type *physConfig*.

Note: The system checks out a Layout XL license if there is not one currently checked out.

The CPH window has three modes of operation: *Hierarchy Configuration*, *Component Types*, and *Soft Block*.

Configuring the Physical Hierarchy: **Hierarchy Configuration Mode**

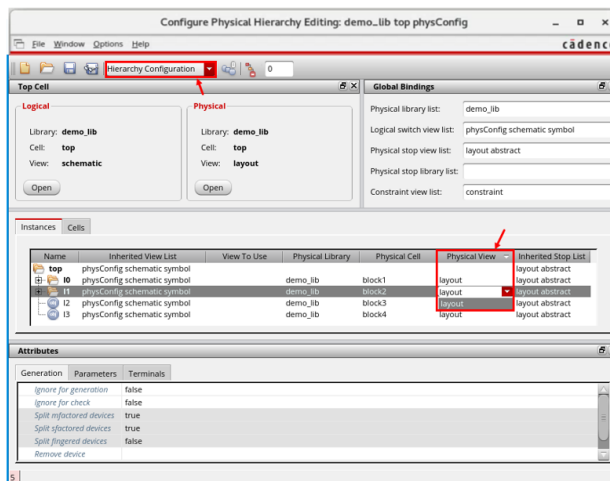


Select **Configure Physical Hierarchy** → **Hierarchy Configuration** to enable the *Hierarchy Configuration* mode.

You will get *Hierarchy Configuration* mode by default in CPH.

Hierarchy Configuration mode controls:

- How the physical hierarchy is generated from your logical design.
- Which logical components are to be ignored in the physical implementation.
- Which physical views are used to implement the logical components.
- Use the *Physical View* field to specify the individual cellviews to place.



Configuring the Physical Hierarchy: **Hierarchy Configuration Mode**

Hierarchy Configuration mode controls how the physical hierarchy is generated from your logical design, including which logical components are to be ignored in the physical implementation and which physical views are used to implement the logical components.

Use the *Physical View* field to specify the individual cellviews to place.

Use Configure Physical Hierarchy in the Hierarchy Configuration mode:

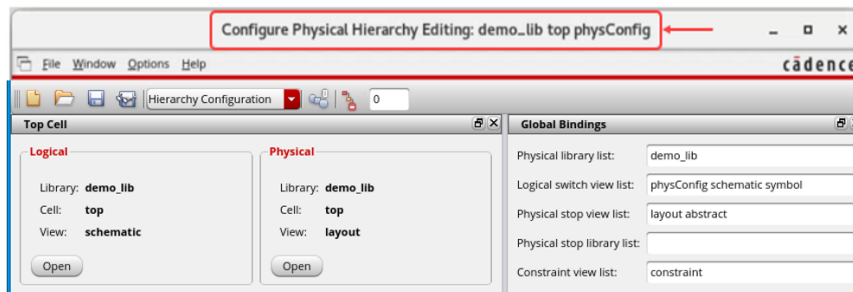
- To define how the layout is generated from the schematic (or from a Verilog netlist for the digital parts of a mixed-signal design that uses the Virtuoso® Schematic and Verilog-Driven Mixed-Signal flow and has the *Virtuoso_MixedSignalOpt_Layout* license checked out)
- To make any required changes before you update the layout from the schematic
- To produce alternative physical implementations of an existing design by using a different logical to physical configuration as a “what-if” experiment

In each case, the main role of the physical hierarchy configuration is to specify the mappings between logical and physical views. When you change a setting, you can see the effects of that change immediately in the layout view by choosing either the *Generate All From Source* or *Update Components And Nets* command. The layout view is regenerated (or updated) based on the latest settings in the *Configure Physical Hierarchy* window.

Configuring the Physical Hierarchy: *physConfig* View

Physical Configuration View (*physConfig*)

- This view stores design-specific overrides, including schematic expansion rules that drive layout generation, design component types, and cell and instance mapping rules.
- Use CPH to create and edit the *physConfig* view.



The *Configure Physical Hierarchy* window is the interface used to create and edit the *physConfig* view.

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Configuring the Physical Hierarchy: *physConfig* View

For layout design data, there is a *physical configuration* (*physConfig*) view which comprises a number of files that store design-specific overrides, including the schematic expansion rules that drive layout generation, design component types, and cell and instance mapping rules. Every Layout XL session operates in the context of a physical configuration view, whether generated automatically or user specified.

When you launch Layout XL from a schematic window, the schematic view is re-opened in the context of the physical configuration view being used for the session, which might be different from the configuration the schematic was using previously. The schematic window banner updates to indicate the name of the physical configuration view currently being used.

One effect of this is that you might need to re-extract the schematic view to take into account the settings in the physical configuration view. The system will inform you if this extraction is required.

If you close the Layout XL session by closing the layout window only, the schematic view is re-opened with no physical configuration context.

The *Configure Physical Hierarchy* window is the interface used to create and edit this view.

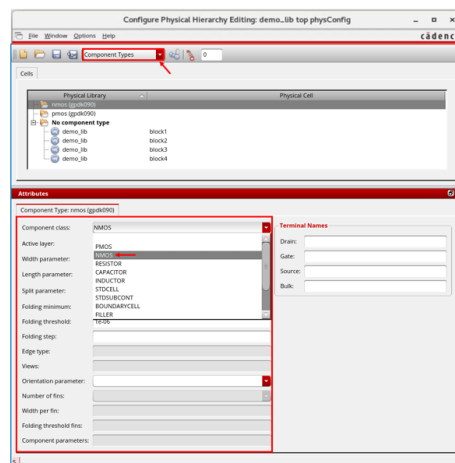
Configuring the Physical Hierarchy: **Component Types Mode**



Select **Configure Physical Hierarchy** → **Component Types** to enable the *Component Types* mode.

- *Component Types* mode lets you view the library-level component types.
- This mode lets you create, edit, and remove the design-level component types.
 - Identifies the *NMOS* and *PMOS* transistor cells in the design.
 - Set the parameters for device chaining and folding.
- Use this mode to designate common characteristics for device generation, including the following:
 - *PMOS*
 - *NMOS*
 - *RESISTOR*
 - *CAPACITOR*
 - *INDUCTOR*
 - *STDCELL*
 - *STDSUBCONT*
 - *BOUNDARYCELL*
 - *FILLER*
- Use this mode for folding, chaining, and placing in rows.

Select *Component Types* from CPH.



Component Types mode and *Cell Type* are different.

Configuring the Physical Hierarchy: **Component Types Mode**

Component Types mode lets you view the library-level component types. In addition, this mode lets you create, edit, and remove the design-level component types, which identify the *NMOS* and *PMOS* transistor cells in the design and set the parameters for device chaining and folding.

Component Types Mode

A component type is the name given to that classification of layout cells which share common characteristics.

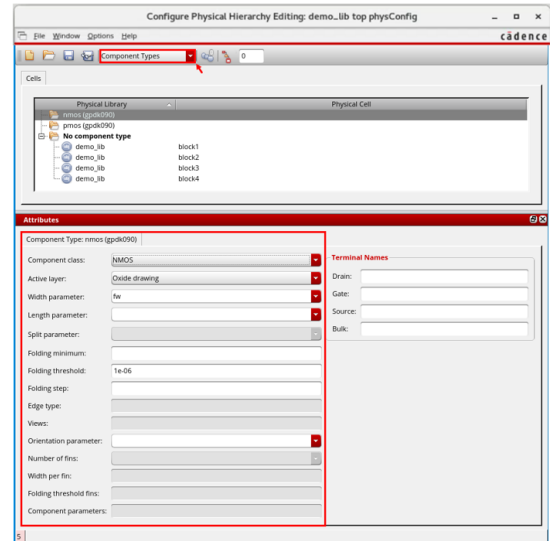
- *Component class* (one of *PMOS*, *NMOS*, *RESISTOR*, *CAPACITOR*, *INDUCTOR*, *STDCELL*, *STDSUBCONT*, *BOUNDARYCELL*, or *FILLER*)
- *Width parameter* name, *Length parameter* name, *Folding threshold*, *Active layer* and *View* names, and other associated component attributes
- Names defined for *Source*, *Gate*, *Drain*, and *Bulk* terminals

Component Types Attributes in CPH: **Attributes for NMOS**

The *Attributes* pane lets you edit the attributes for the selected component type.

Select and show the attributes for the NMOS device.

- Attributes: Cyclic
 - Component class
 - Active layer
 - Width parameter
 - Length parameter
 - Split parameter
 - Folding minimum
 - Folding threshold
 - Folding step
 - Edge type
 - Views
 - Orientation parameter
 - Number of fins
 - Width per fin
 - Folding threshold fins
 - Component parameters
 - Terminal names



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Component Types Attributes in CPH: **Attributes for NMOS**

The *Attributes* pane lets you edit the attributes for the selected component type.

Select and show the attributes for the NMOS device. Attributes: Cyclic

- Component class
- Active layer
- Width parameter
- Length parameter
- Split parameter
- Folding minimum
- Folding threshold
- Folding step
- Edge type
- Views
- Orientation parameter
- Number of fins
- Width per fin
- Folding threshold fins
- Component parameters
- Terminal names

Configuring the Physical Hierarchy: **Soft Block Mode**



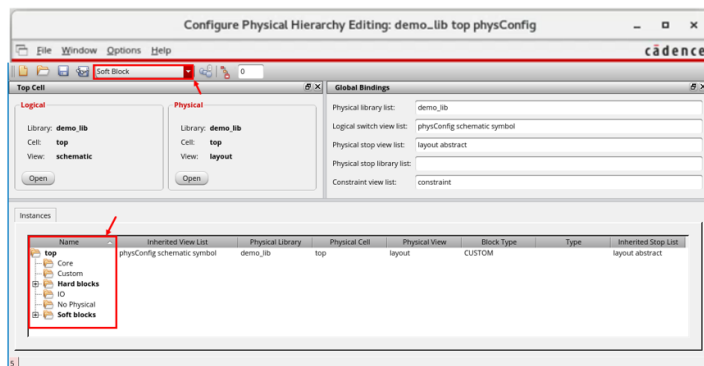
Select **Configure Physical Hierarchy** → **Soft Block** to enable the *Soft Block* mode.

Soft Block mode lets you configure and specify bindings for the soft blocks that will be created by the *Floorplan – Generate Physical Hierarchy* command.

Using the *Soft Block* mode, you can:

- Partition your logical design by grouping into bins.
 - Core
 - Custom
 - Hard blocks
 - IO
 - No Physical
 - Soft blocks
- Edit parameters for the soft blocks that are created when you generate the physical hierarchy.

Select *Soft Block* from CPH.



Configuring the Physical Hierarchy: **Soft Block Mode**

Soft Block mode lets you configure and specify bindings for the soft blocks that will be created by the *Plan - Generate – Generate Physical Hierarchy* command. You select the instances to be included and define the boundary, pin, and blockage parameters for the block you want to create. You can also remove the parameters from an existing soft block to prevent it from being created when the physical hierarchy is generated.

Note: To edit an existing soft block that has already been generated in the layout view, use the *Edit Soft Blocks* command. For more information, see *Editing Soft Blocks in Virtuoso Layout Suite EXL User Guide*.

Using the *Soft Block* mode, you can:

- Partition your logical design by grouping into bins: *Core*, *Custom*, *Hard blocks*, *IO*, *No Physical*, and *Soft blocks*.
- Edit parameters for the soft blocks that are created when you generate the physical hierarchy.

Configuring the Physical Hierarchy: **Hard Blocks Section**

Hard blocks section of *Instances* tab.

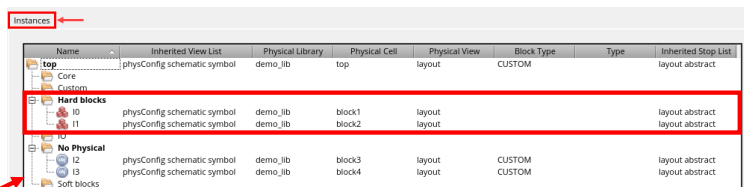
- Cells organized by Cell Type:

- Hard blocks
- IO
- No Physical
- Soft blocks

Hard blocks section of the *Instances* tab.

- Bins used by the Floorplanner.

- If a soft block has already been generated, it will be listed under *Soft blocks*. If it does not exist yet, it is displayed under *No Physical*.



Name	Inherited View List	Physical Library	Physical Cell	Physical View	Block Type	Type	Inherited Stop List
top	physConfig schematic symbol	demo_lib	top	layout	CUSTOM	layout abstract	layout abstract
Core							
Custom							
Hard blocks							
IO	physConfig schematic symbol	demo_lib	block1	layout		layout abstract	layout abstract
IO	physConfig schematic symbol	demo_lib	block2	layout		layout abstract	layout abstract
No Physical							
IO	physConfig schematic symbol	demo_lib	block3	layout	CUSTOM	layout abstract	layout abstract
IO	physConfig schematic symbol	demo_lib	block4	layout	CUSTOM	layout abstract	layout abstract
Soft blocks							

Configuring the Physical Hierarchy: **Hard blocks Section**

Hard blocks section of *Instances* tab:

- Cells organized by Cell Type:

- Hard blocks
- IO
- No Physical
- Soft blocks

- Bins used by the Floorplanner.

- If a soft block has already been generated, it will be listed under *Soft blocks*. If it does not exist yet, it is displayed under *No Physical*.

Configuring the Physical Hierarchy: Soft Blocks Section

Soft blocks section of *Instances* tab.

- Cells organized by Cell Type:

- Hard blocks
- IO
- No Physical
- Soft blocks

- Bins used by the Floorplanner.

- If a soft block has already been generated, it will be listed under *Soft blocks*. If it does not exist yet, it is displayed under *No Physical*.

Soft blocks section of the *Instances* tab.

Name	Inherited View List	Physical Library	Physical Cell	Physical View	Block Type	Type	Inherited Step List
top	physConfig schematic symbol	demo_lib	top	layout	CUSTOM		layout abstract
Core							
Custom							
Hard blocks							
IO							
No Physical							
Soft blocks							



Configuring the Physical Hierarchy: Soft blocks Section

Soft blocks section of *Instances* tab:

- Cells organized by Cell Type:

- Hard blocks
- IO
- No Physical
- Soft blocks

- Bins used by the Floorplanner.

- If a soft block has already been generated, it will be listed under *Soft blocks*. If it does not exist yet, it is displayed under *No Physical*.

Defining/Editing the Soft Block Parameters in CPH



Use the **Define Soft Block Parameters** command to define the boundary, pins, and blockage parameters for a selected instance in the schematic design.

These parameters are used by the **Generate Physical Hierarchy** command to create soft blocks for these instances in the physical implementation.

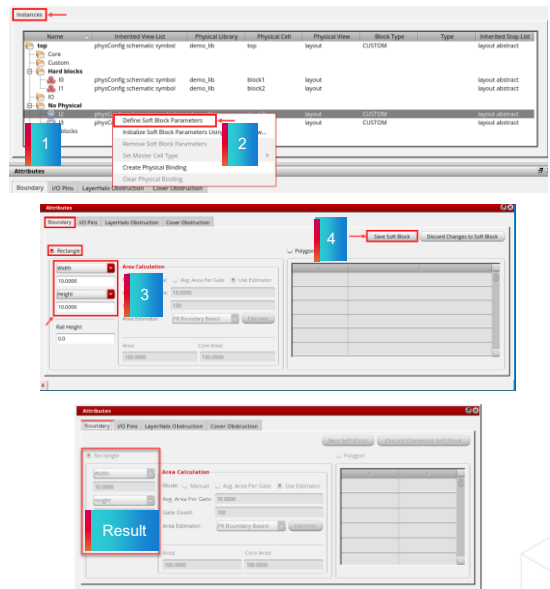
Use the **Instances** tab in the **Soft Block** mode of the CPH form to define or edit the soft block parameters.

To define/edit soft block parameters in CPH:

1. Select a **Soft Block** or **No Physical** Instance in the **Instances** tab.
2. Right-click the instance and choose **Define Soft Block Parameters**. If the selected component has soft block parameters defined already, the command is labeled as **Edit Soft Block Parameters**. The **Instances** tab is disabled, and the **Attributes** pane is enabled.
3. Define or edit the soft block parameters you require for your soft block.
4. To save the parameters for the selected component, click **Save Soft Block** in the **Attributes** pane. The **Attributes** pane is disabled, and the **Instances** tab is re-enabled.

Result: The Soft Block Attributes are saved for the selected component.

Discard the changes by clicking **Discard Changes to Soft Block**.



Defining Soft Block Parameters

Use the *Define Soft Block Parameters* command to define the boundary, pins, and blockage parameters for a selected instance in the schematic design. These parameters are used by the *Generate Physical Hierarchy* command to create soft blocks for these instances in the physical implementation. Use the *Instances* tab in the *Soft Block* mode of the CPH form to define or edit the soft block parameters.

When the *Define Soft Block Parameters* command is running, only the controls in the *Attributes* pane are enabled, ensuring you do not change the instance selection while you are editing the parameters. All other controls in the *Configure Physical Hierarchy* window remain disabled until you either save or discard the changes you have made to the soft block parameters. Similarly, cross-selection from other Virtuoso windows or assistants is delayed until you save or discard your changes.

Note: To make changes to existing soft blocks that do not require the physical hierarchy to be regenerated, use the *Edit Soft Blocks* command instead.

Editing Soft Blocks

The *Edit Soft Blocks* command lets you modify the soft blocks in your design. Use this command if you need to modify the attributes of individual soft blocks to take account of placement and connectivity requirements in your design.

Note: This command is recommended if you want to make changes that do not require the physical hierarchy to be regenerated. If you need to make more radical changes (by, for example, generating new soft blocks or flattening others that you no longer require), use the *Launch – Configure Physical Hierarchy* command in *Soft Block* mode.

Accessing the Soft Block Attributes in CPH



In the **Configure Physical Hierarchy** form, select **Soft Block → Attributes** to access the *Soft Block Attributes* in CPH.

The Soft Block Attributes in CPH are:

- Boundary
- I/O Pins
- LayerHalo Obstruction
- Cover Obstruction



Accessing the Soft Block Attributes in CPH

In the **Configure Physical Hierarchy** form, select **Soft Block → Attributes** to access the *Soft Block Attributes* in CPH.

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- Cover Obstruction

Soft Block Attributes in CPH: **Boundary**

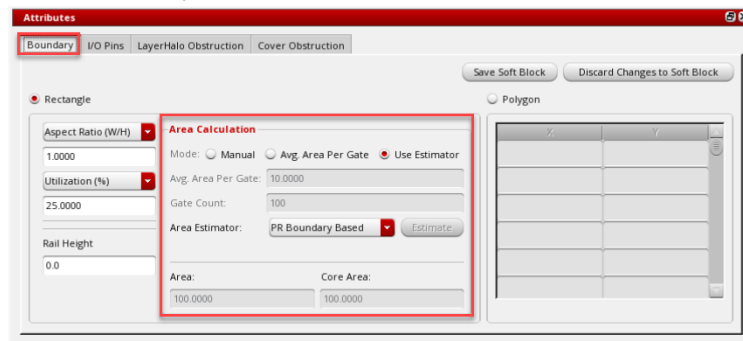
The Soft Block Attributes in CPH are:

- **Boundary**
- I/O Pins
- LayerHalo Obstruction
- Cover Obstruction

The **Boundary** attributes let you specify the shape and size of the boundary that encloses the soft block.

Methods for estimating the Boundary Area Calculation:

- **Manual:** Enter the value into the *Core Area* field
- **Avg. Area Per Gate:** Specify the *Avg. Area Per Gate* and *Gate Count*
- **Use Estimator:** Select *PR Boundary Based* or *BBox Based* for the Area Estimator



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Specifying the Boundary

The *Boundary* tab lets you specify the shape and size of the boundary that is created when the soft block is generated.

Boundary Area Calculation

Manual lets you directly type the value you want into the *Core Area* field.

Avg. Area Per Gate lets you specify the gate count and the average area per gate. The system calculates the area by multiplying the two values.

Use Estimator lets you choose between *PR Boundary Based* and *Bbox Based* area estimator. The PR Boundary-based estimation sums up the polygonal area of the individual instances to derive the overall area estimation. On the other hand, the Bbox-based estimation sums up the minimum bounding box area of the individual instances to derive the overall area estimation.

You can also register your own area estimation functions and use them as appropriate. For more information, see *Creating and Registering a User Defined Area Estimation Function* in *Virtuoso Layout Suite XL User Guide*.

Soft Block Attributes in CPH: I/O Pins

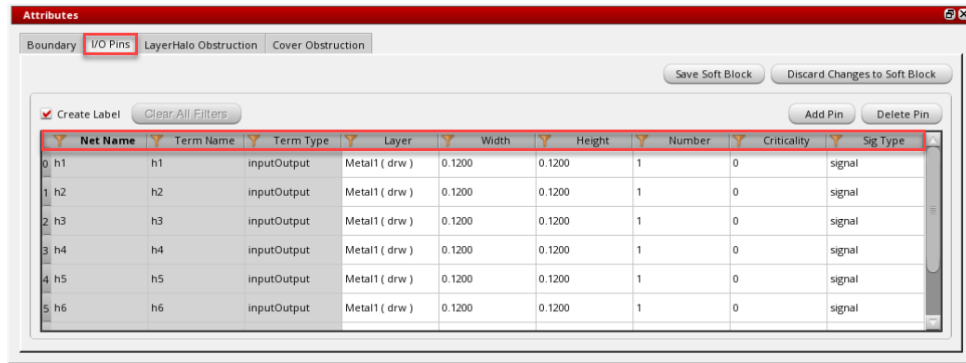
The Soft Block Attributes in CPH are:

- Boundary
- I/O Pins
- LayerHalo Obstruction
- Cover Obstruction

The **I/O Pins** attributes pane lists the I/O pins in the design.

Soft Block I/O Pins:

- Pin information from the schematic
- Edit: *Layer, Width, Height, Number, Criticality, and Sig Type*



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Specifying I/O Pins

In addition to defining layer-based rules, you can now define layer-purpose based rules while defining the soft block parameters in the *I/O Pins* tab.

Soft Block Attributes in CPH: LayerHalo Obstruction

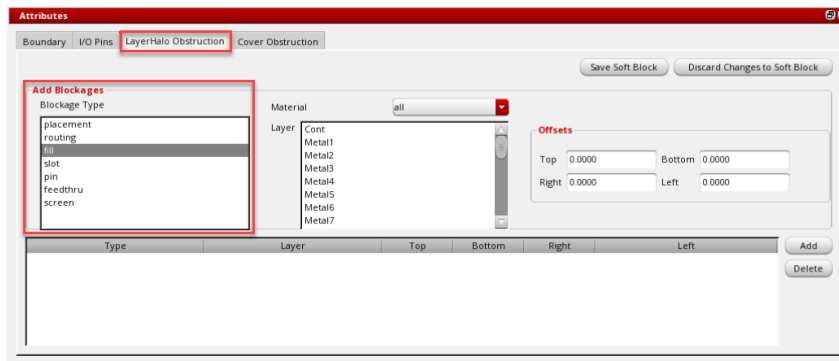
The Soft Block Attributes in CPH are:

- Boundary
- I/O Pins
- LayerHalo Obstruction
- Cover Obstruction

The **LayerHalo Obstruction** attributes pane lets you create the routing and placement obstructions.

Obstructions

- **routing, fill, slot, pin, feedthru, and screen:** Set the *Blockage Type*, and specify the *Layer*, and *Offsets*
- **placement:** Specify the *Offsets* only



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Specifying LayerHalo Obstructions

LayerHalo obstructions are defined at the child-level, and are effective at the parent-level. Alternatively, you can define cover obstructions to block routing on the specified layer at the top level. The selected layer and all lower layers are automatically obstructed. For more information about the differences between the two types of obstructions, see *Comparison between LayerHalo and Cover Obstructions* in *Virtuoso Layout Suite XL User Guide*.

A soft block obstruction is implemented using a halo object instead of a blockage object, because there is no need to manually remove the halo prior to routing.

To define a *routing, fill, slot, pin, feedthru, or screen* obstruction, you need to set the *Blockage Type*, and specify the *Layer* and *Offsets*. There is also a *Material* filter which you can use to select layers based on the material defined for that layer in the technology file.

The list of extractable layers is retrieved from the constraint group set for the current layout cellview. If there is no constraint group set for the current layout cellview, the software retrieves the layers from the constraint group specified by the *setupConstraintGroup* environment variable. The CPH soft block reads the environment variable and looks for the constraint group carrying the required information in the associated technology file. The technology file may be an Incremental Technology Database file (ITDB) or a design techfile.

If there are no layers defined in either location, it issues a message and disables the soft block definition.

To define a *placement* obstruction, you need to set the *Offsets* only. You can define only one placement obstruction per block.

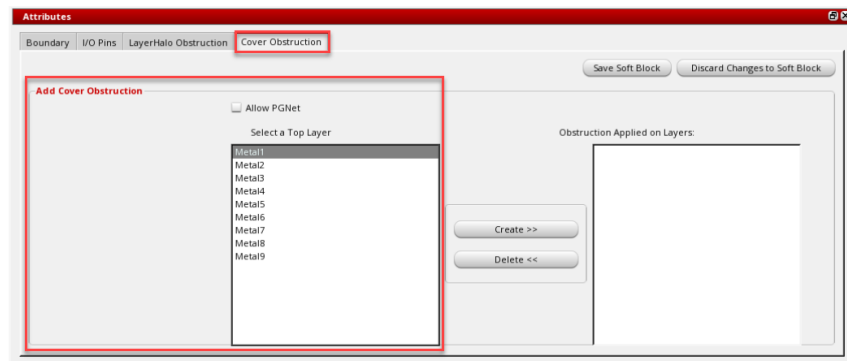
Soft Block Attributes in CPH: **Cover Obstruction**

The Soft Block Attributes in CPH are:

- Boundary
- I/O Pins
- LayerHalo Obstruction
- **Cover Obstruction**

The **Cover Obstruction** attributes pane lets you create a cover obstruction on the selected layer and lower layers.

- Use the **Cover Obstruction** tab to create cover obstructions in your soft block.
- Cover obstructions are defined at the layer level.
- When you create a cover obstruction, the selected layer and all lower layers are automatically obstructed.



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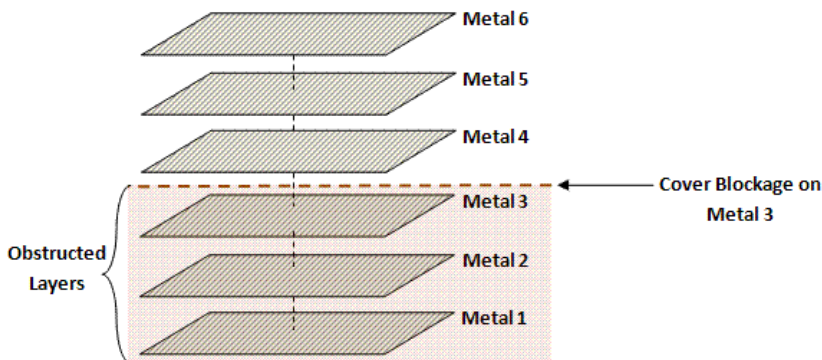
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Specifying Cover Obstructions

You use the *Cover Obstruction* tab to create cover obstructions in your soft block. Cover obstructions are defined at the layer level. When you create a cover obstruction, the selected layer and all lower layers are automatically obstructed.

Example: Soft Block Cover Obstruction

- Use the **Cover Obstruction** tab to create cover obstructions in your soft block.
- Use the layers where you want to prevent over-the-cell routing to improve performance or to avoid electrical effects between tracks.
- An example of cover obstruction is shown here.



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Example: Soft Block Cover Obstruction

Use the *Cover Obstruction* tab to create cover obstructions in your soft block.

An example cover obstruction is given here. In this example, *Metal1* is the lowest layer and *Metal6* is the top-most layer. Adding a cover obstruction on *Metal3* will generate obstructions on layers *Metal3*, *Metal2*, and *Metal1*. As a result, routing cannot be done on these layers.

This type of obstruction is typically generated for layers where you want to prevent over-the-cell routing to improve performance or to avoid electrical effects between tracks.

A soft block can be associated with only one cover obstruction. So, any new cover obstruction that is created on a soft block replaces the existing cover obstruction (if any) on it.

Use LayerHalo obstructions to block routing at the child-level. These obstructions are effective at the parent-level. For more information about the differences between the two types of obstructions, see *Comparison between LayerHalo and Cover Obstructions* in *Virtuoso Layout Suite XL User Guide*.

Saving/Loading the CPH Information



In the Configure Physical Hierarchy (CPH) form, choose **File – Save / Save Floorplan File / Load Floorplan File** to save/load the CPH information.

To save the current physical configuration view:

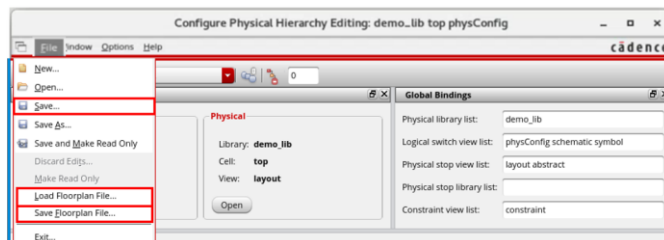
- **File – Save:** Saves changes to the physical configuration of the design.

To save the current soft block parameters:

- **File – Save Floorplan File:** Saves the current soft block parameters to a floorplan file, which you can then re-use in a different Virtuoso session.

To restore the *physConfig* view from the Floorplan file values:

- **File – Load Floorplan File:** Loads soft block parameters from a floorplan file.



Both the *physConfig* view and Floorplan file values complete the information available for configuring the physical hierarchy.

Saving the Physical Configuration View

From the Configure Physical Hierarchy menu bar, choose **File – Save** to save the physical configuration view.

Saving Soft Block Parameters to a Floorplan File

You can save the current block parameters to a floorplan file, which you can then re-use in a different Virtuoso session. To do this,

1. Define the soft block parameters you require using the procedures described in *Defining Soft Block Parameters*. Refer to the *Virtuoso Layout Suite XL User Guide* for more details.
2. From the Configure Physical Hierarchy menu bar, choose *File – Save Floorplan File*.
3. Specify a name for the file and click *OK*.

The floorplanning parameters are saved to the specified file.

Loading Soft Block Parameters from a Floorplan File

To load parameters from a floorplan file,

1. From the Layout XL window, choose *Launch – Configure Physical Hierarchy*.

The Configure Physical Hierarchy window appears.

2. Change to Soft Block mode using the drop down list in the toolbar.
3. From the Configure Physical Hierarchy menu bar, choose *File – Load Floorplan File*.

The Load Floorplan File dialog appears.

4. Specify the file you want to load and click *OK*.

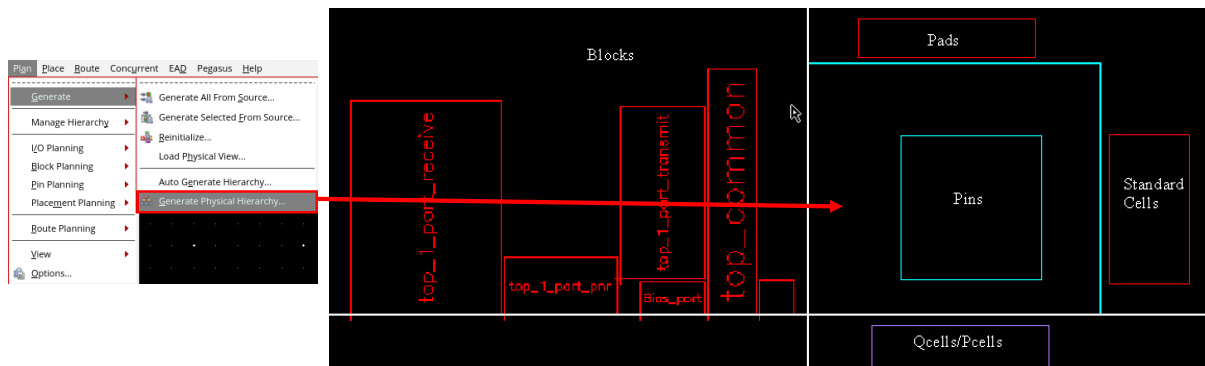
The floorplanning parameters defined in the file are loaded into the current configuration view.

Using the Generate Physical Hierarchy Command for Data Generation



In the layout canvas, choose **Plan – Generate – Generate Physical Hierarchy(GPH)** or click the **Generate Physical Hierarchy** icon  to open the *Generate Physical Hierarchy* form.

- GPH command creates the *prBoundary* and all the design elements of the cellview.
- Elements are positioned around the *prBoundary* based on the cell type.



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Physical Hierarchy Generation

Generate Physical Hierarchy generates the physical hierarchy based on the configuration defined in Configure Physical Hierarchy (CPH). It allows you to generate a physical hierarchy from the logical hierarchy in a schematic. It also creates soft layouts and soft abstracts in the layout canvas corresponding to the configuration view generated by CPH. Generate Physical Hierarchy uses two main engines which are Generate All From Source and Generate Abstract From Symbol to generate the existing blocks,

Generating Soft Layout and Soft Abstract

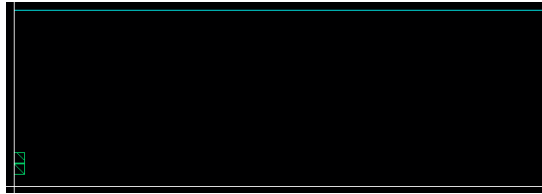
Generate Physical Hierarchy runs **Generate All From Source** for soft layouts and also generates soft abstracts in batch mode. 

Soft Layout

- All the views generated by the Virtuoso® Floorplanner are layout views. The tool adds the *softMacro* cell type to make the layout *soft*. After the soft layout is created, only the *prBoundary* and pins are visible. All the other components and their connectivity are hidden at the top level. You can descend in the layout view and make the other components visible.

Soft Abstract

- The tool adds the *blockBlackBox* cell type to make it *soft*. The *soft abstract* is a layout view with only the *prBoundary* and pins but no connectivity and components.



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Generating the Physical Hierarchy

Use the *Generate Physical Hierarchy* command to generate the components in your design while maintaining the hierarchy levels you have defined in the Soft Block mode.

Generating the Physical Hierarchy

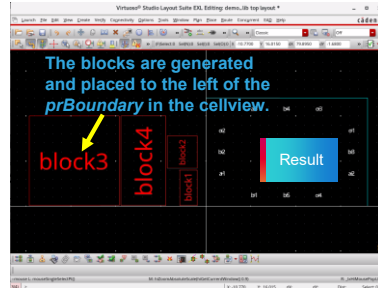
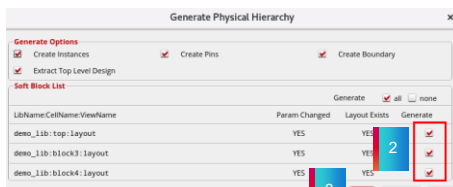
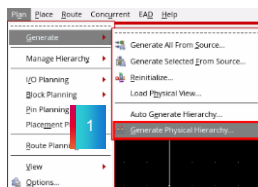


The **Generate Physical Hierarchy (GPH)** command generates the physical hierarchy based on the configuration defined in **Configure Physical Hierarchy (CPH)**.

To generate the physical hierarchy of your design:

1. In the layout window, choose **Plan – Generate – Generate Physical Hierarchy** or click the **Generate Physical Hierarchy** icon to open the GPH window.
2. Enable the **Generate** button for each of the cellviews in the list.
3. Click **OK**.

Result: The blocks are generated and placed to the left of the *prBoundary* in the cellview.



Observe the CIW window for the **Generate Physical Hierarchy (GPH)** results.

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Setting the Design Planning and Analysis Options

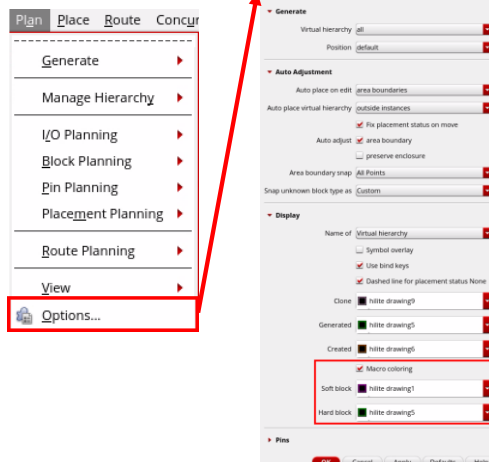


In the layout canvas, choose **Plan – Options** or click the **Options** icon to display the Design Planning and Analysis Options form.

Use this form to customize the layout design environment.

Use the *Design Planning and Analysis Options* form to do these tasks:

- Control the display of virtual hierarchies and blocks in a design
- Specifies whether to generate all or specific virtual hierarchies
- Specifies the options for automatically adjusting a virtual hierarchy
- Specifies the display options for a virtual hierarchy
- Assign highlight colors for hard and soft blocks
- Specifies the routing, pin connection, and update options



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Color Coding the Hard and Soft Blocks

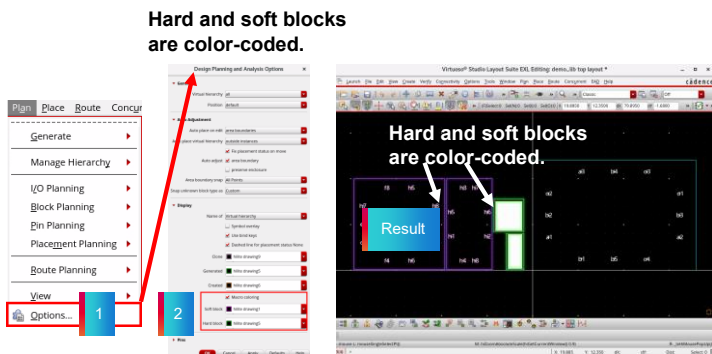


In the **Design Planning and Analysis Options** form, use the color-coding options in the Display section by enabling **Macro coloring** to assign different highlight colors for the hard and soft blocks so that they are visually distinguishable in the layout canvas.

To highlight the hard and soft blocks in different colors in your design:

1. In the layout window, choose **Plan – Options** or click the **Options** icon  to open the Design Planning and Analysis Options form.
2. Select the options as follows in the form.
 - i. Select the **Macro coloring** option.
 - ii. Choose the desired **hilite** color for the soft and hard blocks.
 - iii. Click **OK**.

Result: The hard and soft blocks are highlighted in different colors.



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Using the Block Placer

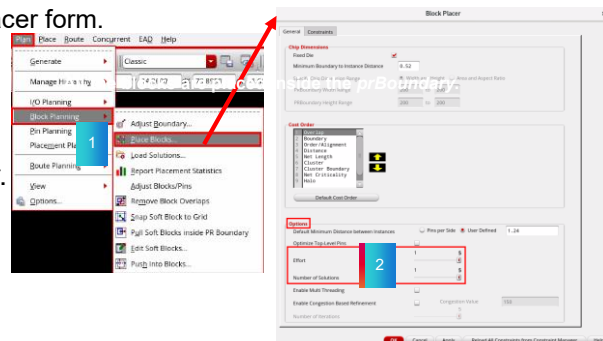


Use the **Block Placer** command to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area.

To place the blocks in the design:

1. In the layout window, Choose **Plan – Block Planning – Place Blocks** to open the Block Placer form.
2. Set the following in the **Options** section on the Block Placer form.
 - i. Effort as **5**.
 - ii. Number of Solutions as **5**.
 - iii. Click **OK**.

Result: All the blocks are placed inside the *prBoundary*.



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Block Placer

The *Place Blocks* command enables you to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area. Block placer runs on the entire design. Before you run block placer, ensure that the design has a *PRBoundary* and at least one soft or hard block. The command can adjust soft blocks within the specified width and height range to maximize chip density. Block placer supports both top-level and block-level rectilinear design boundaries.

Place Blocks

To invoke block placer, click The *Plan – Block Planning – Place Blocks* . The *Block Placer* form is displayed.

The *Block Placer* form has the following tabs: *General* and *Constraints*.

The *General* tab has the following sections that you can use to place blocks: *Chip Dimensions*, *Cost Order*, and *Options*.

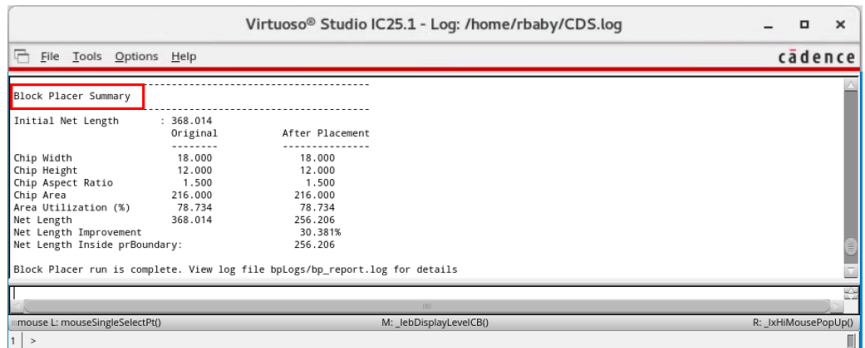
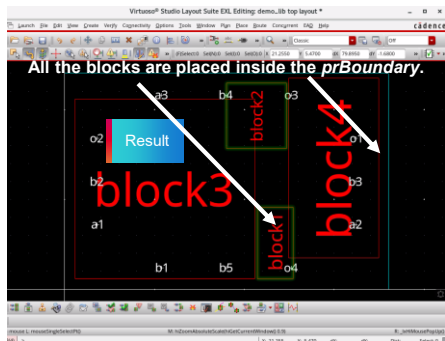
Constraints: You use the *Constraints* tab of the *Block Placer* form to specify the constraints that you want block placer to use to converge on a solution. The initial values in the *Constraints* tab are populated from the values specified in the Constraint Manager. You use the *Constraints* tab to modify the displayed values. However, the values you specify must not conflict with the constraint values in the Constraint Manager. In addition, you can add constraints in this tab.

The *Constraints* tab has the following sub-tabs: *Soft Block Constraints*, *Instance Constraints*, and *Relative Constraints*.

Block Placer Summary in the CIW

Observe the CIW window for the *Block Placer* run details.

View log file *bpLogs/bp_report.log* for the *Block Placer* run details.

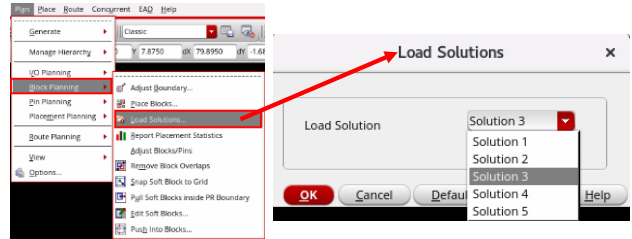


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Loading the Solutions

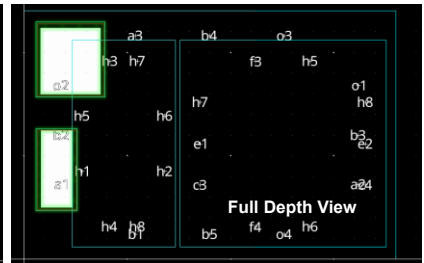
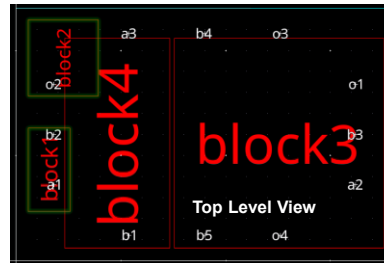


In the layout canvas, choose **Plan – Block Planning – Load Solutions** to display the *Load Solutions* form.



Choice of possible solutions:

- Varies by constraints chosen.
- Cells selected by CPH.
- The Floorplanner generates solutions based on constraints that you have previously set.
- One example of the placement is shown here.



After the solution is loaded, the design automatically fits in the workspace.

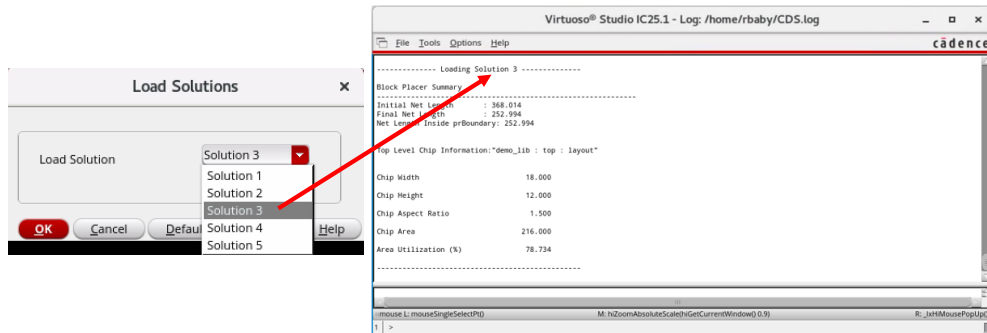
Load Solutions

By default, block placer loads the best solution among the solutions it generates. You can also load the other solutions that block placer generates. To load a solution, click **Plan – Block Planning – Load Solutions** command. The *Load Solutions* form is displayed.

The load solution command reads the number of solutions present for a design in the directory specified by BPDIR variable (or *bpLogs* directory within the testcase directory) and lists the number of solutions in the *Load Solution Number* field in the form. You can select a solution number in the field and load it. You can undo the changes made by the new solution by using the *Undo* command. When you undo the changes, block placer restores the previous state.

Summary Report: Loading the Solutions

After loading the solution, observe the CIW window for the summary report of loading the solution.



You can choose from **1** through **5** possible solutions.



Summary Report: Loading the Solutions

After loading the solution, observe the CIW window for the summary report of loading the solution.

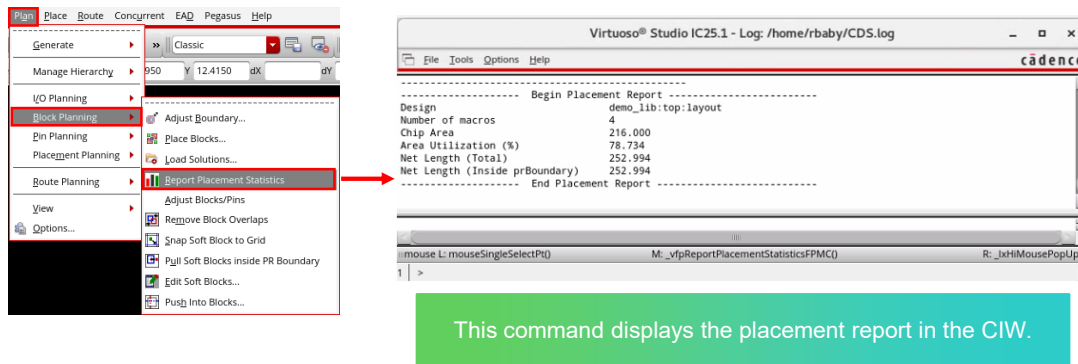
You can choose from 1 through 5 possible solutions.

Reporting the Placement Statistics



In the layout canvas, choose **Plan – Block Planning – Report Placement Statistics** to display the placement report in the CIW.

You can use the placement report to review the placement statistics and analyze the results.



Report Placement Statistics

Use the **Plan – Block Planning – Report Placement Statistics** option to view the placement statistics report at any time after block placement.

Typically, after running the block placer, you may want to make minor manual adjustments or automated adjustments such as using the Soft Block Update commands to remove overlaps or pulling blocks inside boundary. After making the adjustments, use the Plan – Block Planning – Report Placement Statistics option to display the placement report in the CIW. You can use the placement report to review the placement statistics and analyze the results.

Snapping the Soft Pins

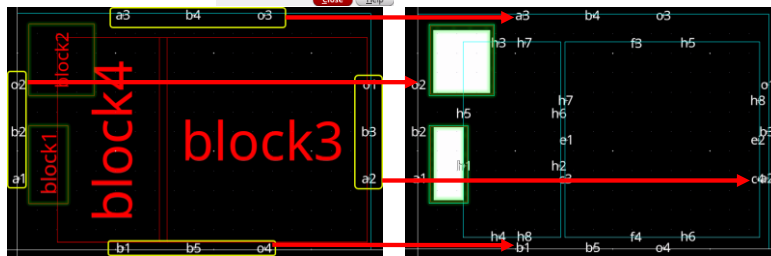
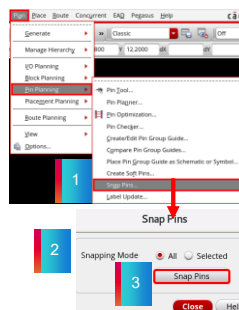


Use the **Snap Pins** command to snap both the top-level and Level-1 pins to the grid appropriate to the block type.

To snap all the soft block pins to the appropriate grid:

1. In the layout canvas, choose **Plan – Pin Planning – Snap Pins** to open the Snap Pins form. You can also access this command from **Connectivity – Pins – Snap Pins**.
2. Choose **Snapping Mode – All**.
3. Click **Snap Pins**.

Result: All the top level and Level-1 pins are snapped to the appropriate grid.



The pins with *Placement Status* as *Fixed* are also snapped, and the status is retained as *Fixed*.

Snap Pins

The *Snap Pins* command is used to snap both top-level and level-1 pins to the grid appropriate to the block type. To snap all soft block pins, select *Plan – Pin Planning – Snap Pins*. You can also access this command from *Connectivity – Pins – Snap Pins*. For more information on snapping soft block pins, refer to the *Virtuoso Layout Suite EXL User Guide*.

Snapping the Soft Pins

To snap all the soft block pins to the appropriate grid:

1. In the layout canvas, choose **Plan – Pin Planning – Snap Pins**. You can also access this command from **Connectivity – Pins – Snap Pins**.
2. Choose **Snapping Mode – All**.
3. Click **Snap Pins**.

Result: All the top-level and level-1 pins are snapped to the appropriate grid.

The pins with *Placement Status* as *Fixed* are also snapped and the status is retained as *Fixed*.

Analyzing the Connectivity

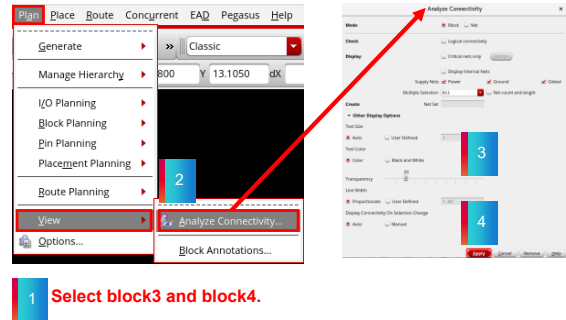


Use the **Analyze Connectivity** command to calculate the number of connections between each pair of selected instances, draw flight lines between each connected pair, and indicate the number of connections by displaying a numerical value on the flight lines.

To display the number of connections and analyze the connectivity between preselected blocks:

1. Select **block3** and **block4**.
2. In the layout canvas, choose **Plan – View – Analyze Connectivity** or click the **Analyze Connectivity** icon to open the Analyze Connectivity form.
3. Keep the required values in the Analyze Connectivity form.
4. Click **Apply** in the form.

Result: You get the number of connections between the selected blocks, **block3** and **block4**.



For *Analyze Connectivity* command, only the *pre-selection* mode is supported.

Analyze Connectivity

Use the *Analyze* command to calculate the number of connections between each pair of selected instances, draw flight lines between each connected pair, and indicate the number of connections by displaying a numerical value on the flight lines. This is a command that generates information about connectivity and other useful design information. It is useful to find out the connectivity (the number of connections) of a block with other blocks, so that during manual floorplanning, you can keep the blocks that have more connectivity between them closer to each other.

For *Analyze* command, only the pre-selection mode is supported. You can select multiple blocks whose connectivity you want to examine and can analyze the connectivity. You can also access this command from *Connectivity – Analyze*.

Analyzing the Connectivity

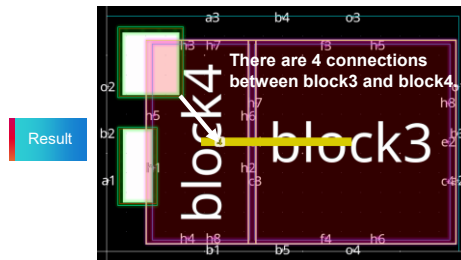
To display the number of connections and analyze the connectivity between preselected blocks:

1. Select **block3** and **block4**.
2. In the layout canvas, choose **Plan – View – Analyze Connectivity** or click the **Analyze Connectivity** icon to open the Analyze Connectivity form.
3. Keep the required values in the Analyze Connectivity form.
4. Click **Apply** in the form.

Result: You get the number of connections between the selected blocks, **block3** and **block4**.

Result: Analyzing the Connectivity

Result: The **numeric 4** represents the number of connections between the selected blocks.



A highlight line is placed between the selected blocks showing the number of connections between the blocks.

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Enabling Level-1 Editing



In *Level-1 Editing* mode, you can edit the objects inside a soft block from the top level. In this mode, you can edit the soft block without using the *Descend* or *Edit In Place* commands. You can enable the *Level-1 Editing* using: The *Display Options* Form and the *Objects* Assistant.

Different methods available are:

1. Enabling Level-1 Editing: Using the *Display Options* Form
 2. Enabling Level-1 Editing: Using the *Objects* Assistant
- After setting the *Level-1 Editing*, you are only able to access the *prBoundary* and the pins



Enabling Level-1 Editing

In *Level-1 Editing* mode, you can edit the objects inside a soft block from the top level.

In this mode, you can edit the soft block without using the *Descend* or *Edit In Place* commands.

You can enable the *Level-1 Editing* using: The *Display Options* Form and the *Objects* Assistant.

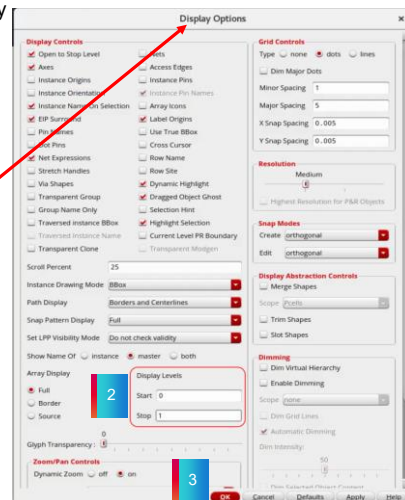
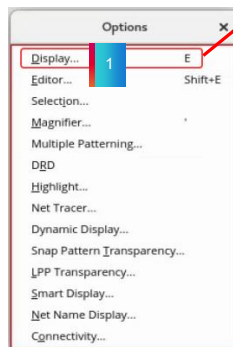
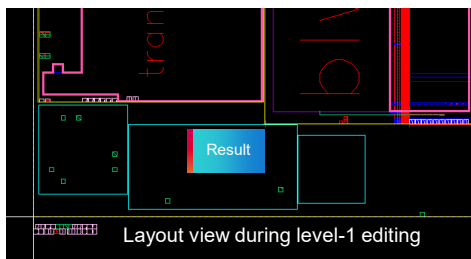
- Enabling Level-1 Editing: In the *Display Options* Form
- Enabling Level-1 Editing: In the *Objects* Assistant
- After setting the *Level-1 Editing*, you are only able to access the *prBoundary* and the pins

1. Enabling Level-1 Editing: Using the Display Options Form

To enable the Level-1 editing in your design:

1. In the layout canvas, choose **Options – Display** or press the bindkey **E** to bring up the Display Options form.
2. In the Display Options form, set the *Display Levels Stop* field to **1**.
3. Click **OK** in the form.

Result: Setting it to **1** gives you the cleanest view of Level-1 shapes.



Step 1: Enabling Level-1 Editing: Using the Display Options Form

To enable the Level-1 editing in your design:

- In the layout canvas, choose **Options – Display** or press the bindkey **E** to bring up the Display Options form.
- In the Display Options form, set the *Display Levels Stop* field to **1**.
- Click **OK** in the form.

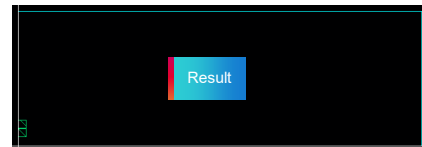
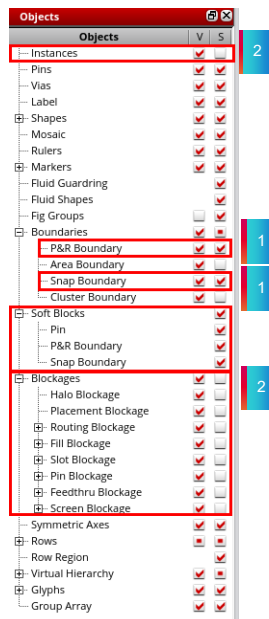
Result: Setting it to **1** gives you the cleanest view of Level-1 shapes.

2. Enabling Level-1 Editing: Using the Objects Assistant

To enable the Level-1 editing in your design:

1. In the **Objects** assistant, turn **ON** the Selectability for the following:
 - Boundaries: P&R Boundary and Snap Boundary
 - Soft Blocks: Pin, P&R Boundary, and Snap Boundary
2. In the **Objects** assistant, turn **OFF** the Selectability for the following:
 - Instances and Blockages

Result: You can access only the *prBoundary* and the pins.



Step 2: Enabling Level-1 Editing: Using the Objects Assistant

To enable the level-1 editing in your design:

1. In the **Objects** assistant, turn **ON** the Selectability for the following:
 - Boundaries: P&R Boundary and Snap Boundary
 - Soft Blocks: Pin, P&R Boundary, and Snap Boundary
2. In the **Objects** assistant, turn **OFF** the Selectability for the following:
 - Instances and Blockages

Result: You can access only the *prBoundary* and the pins.

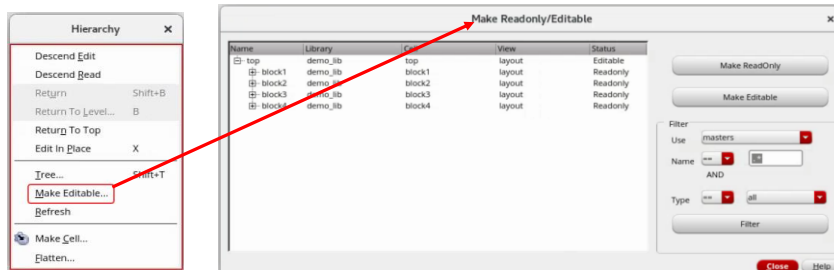
Controlling Which Blocks Are Editable



In the layout canvas, choose **Edit – Hierarchy – Make Editable** to display the Make Readonly/Editable form.

Use the **Make Editable** command to switch to the edit mode for updating the hierarchy.

- In the **Make Readonly/Editable** form:
 - Filter to search for specific cell names and types.
 - Cells are changed to **Editable** status during Level-1 editing.



You can select multiple master cellviews and change their mode.

Making Hierarchy Editable

Use the Make Editable command to switch to the edit mode for updating the hierarchy.

To make a hierarchy editable,

1. Choose Edit – Hierarchy – Make Editable.

The Make Readonly/Editable form opens. This form lists master cellviews of all the instances existing in the current cellview.

2. To list only specific master cellviews, use the Filter options to specify a search criteria.

- You can display the master cellviews of all or only the instances selected in the cellview
- You can specify a Name and Type criteria for filtering the master cellviews to display in the left pane.

3. Select a Readonly master cellview in the left pane and click Make Editable. Alternatively, select an Editable master cellview in the left pane and click Make Readonly. You can select multiple master cellviews and change their mode.

Level-1 Editing: Place-and-Route Boundary and Pin Snapping



Block Type specifies that:

- The type of block to be generated, either **CUSTOM** or **DIGITAL**
- Influences how the place-and-route boundary and pins in the design are snapped during Level-1 editing

CUSTOM specifies that:

- The pin edges are snapped to the manufacturing grid
- The place-and-route boundary is snapped to the manufacturing grid

DIGITAL specifies that:

- The pin centers are snapped to the routing grid
- The place-and-route boundary is snapped to the placement grid

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Level-1 Editing: Place and Route Boundary and Pin Snapping

Block Type specifies the type of block to be generated, either CUSTOM or DIGITAL. This influences how the place and route boundary and pins in the design are snapped during level-1 editing.

CUSTOM specifies that the pin edges and the place and route boundary are snapped to the manufacturing grid.

DIGITAL specifies that the pin centers are snapped to the routing grid and the place and route boundary is snapped to the placement grid.

Using Level-1 Editing Commands: Move, Stretch, Reshape, and Chop



Level-1 Editing mode is the ability to edit the objects inside a soft block without using the *Descend* or *Edit In Place* commands.

You can access only the *prBoundary* and the pins.

The commands used to do *Level-1 Editing* are: *Move*, *Stretch*, *Reshape*, *Chop*.

Level-1 Editing allows:

- Editing the soft blocks without descending to hierarchy
- Editing the soft blocks by looking at adjacent blocks

Different methods available are: Moving Soft Blocks, Constant Area Stretch for Soft Blocks, Reshaping Soft Blocks, and Chopping Soft Blocks.



Using Level-1 Editing Commands: Move, Stretch, Reshape, and Chop

Level-1 Editing mode is the ability to edit the objects inside a soft block without using the *Descend* or *Edit In Place* commands.

You can access only the *prBoundary* and the pins.

The commands used to do *Level-1 Editing* are: *Move*, *Stretch*, *Reshape*, *Chop*.

Level-1 Editing allows:

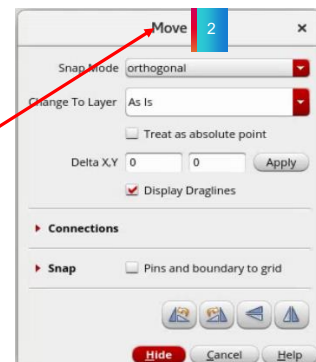
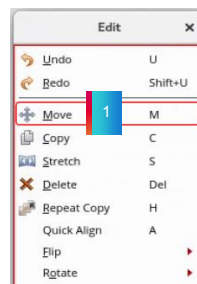
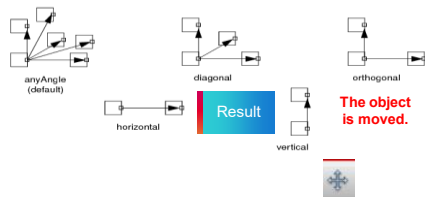
- Editing the soft blocks without descending to hierarchy
- Editing the soft blocks by looking at adjacent blocks

1. Moving Soft Blocks

1. In the layout canvas, choose **Edit – Move** or click the **Move** icon or press the bindkey **M**
2. Select one or more objects and move to the desired location.

Result: The selected objects move to the desired location.

- Select multiple objects to be manually relocated.
- You can move multiple pins from multiple blocks in a single move.
- Moving a Level-1 place-and-route boundary:
 - Digital block:
 - The place-and-route boundary is snapped to the placement grid.
 - Pins with placement status placed are snapped to the routing grid.
 - Analog block:
 - Both place-and-route boundary and placed pins are snapped to the manufacturing grid.



You can select the objects to be moved either before or after you start the *Move* command.

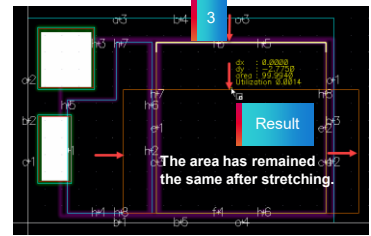
Moving an Object in Level-1 Editing Mode

Use the *Move* command in Level-1 Editing mode to move place-and-route boundaries, snap boundaries, and shape pins without entering Edit In Place mode. You can select the objects to be moved either before or after you start the *Move* command.

2. Constant Area Stretch for Soft Blocks

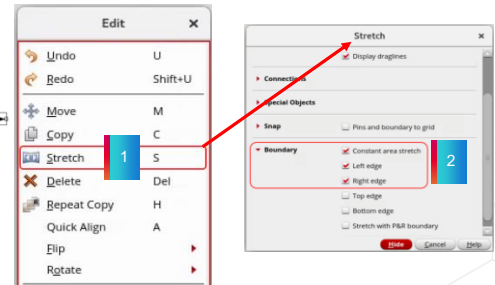
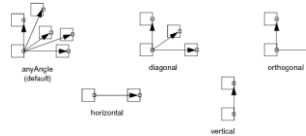
1. In the layout canvas, choose **Edit – Stretch** or click the **Stretch** icon or press the bindkey **S** to stretch objects in *Level-1 Editing*.
2. Enable **Constant area stretch** option in **Boundary** section and select **Left edge** and **Right edge** options.
3. Stretch the top edge of **block3** *prBoundary*. 

Result: The left and right edges move to keep the area constant.



Stretch

- Digital block:
 - Place-and-route boundary snaps to the placement grid.
 - Pin centers snap to the routing grid.
- Analog block:
 - Both place-and-route boundary and pin edges snap to the manufacturing grid.



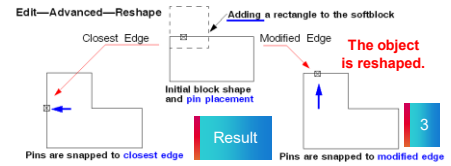
Stretching an Object in Level-1 Editing Mode

Use the *Stretch* command in Level-1 Editing mode to stretch place-and-route boundaries, snap boundaries, and other pins without entering Edit In Place mode. You can stretch multiple pins, including the place-and-route boundary and snap boundary both at level 1 and at the top level, in a single operation.

3. Reshaping Soft Blocks

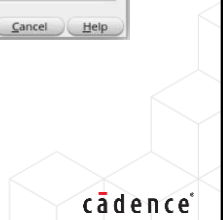
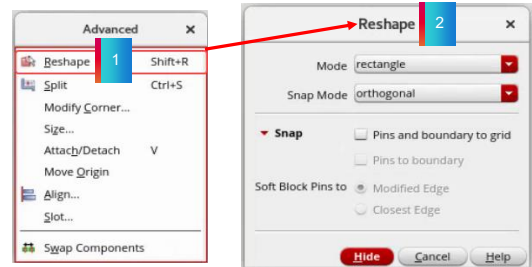
1. In the layout canvas, choose **Edit – Advanced – Reshape** or press the bindkey **Shift+R** to reshape objects in *Level-1 Editing*.
2. To create rectilinear soft blocks, use the **Reshape** command to add the additional area. (OR)
3. Select the place-and-route boundary and use the **Reshape** command to add the new coordinates.

Result: The selected object is reshaped.



Reshape

- Digital block:
 - Place-and-route boundary snaps to the placement grid.
 - Pin centers snap to the routing grid.
- Analog block:
 - Both place-and-route boundary and pin edges snap to the manufacturing grid.



Reshaping an Object in Level-1 Editing Mode

Use the *Reshape* command in Level-1 Editing mode to reshape a place-and-route boundary without entering Edit In Place mode.

4. Chopping Soft Blocks

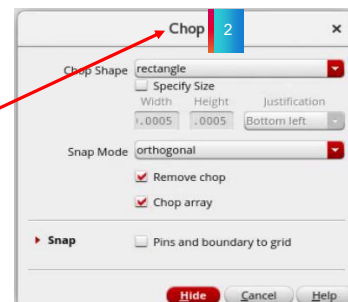
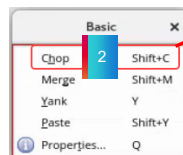
1. In the layout canvas, select the boundary of the desired soft block to chop (**block4**).
2. Choose **Edit – Basic – Chop** or press the bindkey **Shift+C** to chop objects in *Level-1 Editing*.
3. Chop the selected object.

Result: The selected object is chopped.



Chop

- Digital block:
 - Place-and-route boundary snaps to the placement grid.
 - Pin centers snap to the routing grid.
- Analog block:
 - Both place-and-route boundary and pin edges snap to the manufacturing grid.



Partial chopping of instances and vias is not supported.

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Chopping an Object in Level-1 Editing Mode

Use the *Chop* command in Level-1 Editing mode to cut away part of a place-and-route boundary without entering Edit In Place mode.

Use the *Chop* command to cut away part of an object or cut an object into pieces.

Updating Block Planning



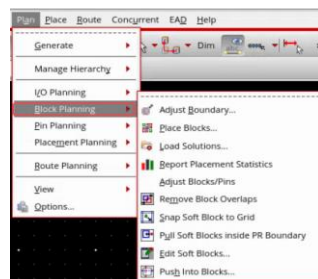
In the layout canvas, use the **Plan – Block Planning** commands to adjust soft blocks in the design.

After running the Block Placer, you may want to make minor manual adjustments or automated adjustments, such as using the **Block Planning** commands to remove overlaps or pulling blocks inside the boundary.

You can use Block Planning commands to:

- Resize or create area boundaries for the selected virtual hierarchy
- Automatically place all hard and soft blocks
- Minimize wire length and area
- Load alternative solutions
- Display a placement statistics report after block placement
- Remove overlaps between soft blocks

- Align soft blocks with the routing or manufacturing grid
- Pull soft blocks towards the core area
- Push top-level implementation of power structures (including metal shapes and poly layers) down to the block level



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Updating Soft Blocks

In the layout canvas, use the **Plan – Block Planning** commands to adjust soft blocks in the design.

After running the block placer, you may want to make minor manual adjustments or automated adjustments such as using the **Block Planning** commands to remove overlaps or pulling blocks inside boundary.

You can use **Block Planning** commands to Resize/create area boundaries for the selected virtual hierarchy, auto-place all hard and soft blocks, minimize wire length/area, load other solutions, Display the placement statistics report after block placement, to remove soft block overlaps, snap soft blocks to the routing grid or manufacturing grid, pull soft blocks to the core area, and to push the top-level implementation of power structures, including metal shapes and poly layers, into the block level.

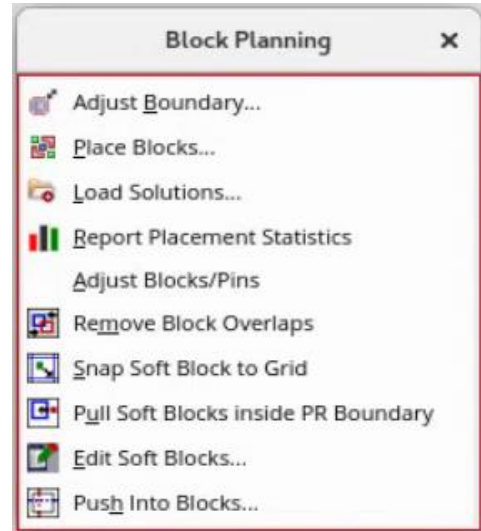
The *Plan–Block Planning* menu provides the following sub-options:

- **Adjust Boundary:** Resize or create the boundary for the selected virtual hierarchy area.
- **Place Blocks:** Automatically place all hard and soft blocks and minimize wire length and chip area.
- **Load Solutions:** Load other solutions that block placer generates.
- **Report Placement Statistics:** Display placement statistics report after block placement.
- **Adjust Blocks/Pins:** Adjust soft blocks by abutting and pushing the blocks.
- **Remove Block Overlaps:** Remove overlaps in blocks.
- **Snap Soft Block to Grid:** Pulls soft blocks to manufacturing or placement grid.
- **Pull Soft Blocks inside PR Boundary:** Pulls soft blocks inside the PR boundary.
- **Edit Soft Blocks:** Lets you modify the attributes of soft blocks in a design.
- **Push Into Blocks:** Pushes the top-level implementation of power structures to the block level.

Block Planning Commands

The **Plan – Block Planning** menu provides the following sub-options:

- **Adjust Boundary:** Resize or create the boundary for the selected virtual hierarchy area. 
- **Place Blocks:** Automatically place all hard and soft blocks and minimize wire length and chip area. 
- **Load Solutions:** Load other solutions that block placer generates. 
- **Report Placement Statistics:** Display placement statistics report after block placement. 
- **Adjust Blocks/Pins:** Adjust soft blocks by abutting and pushing the blocks.
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- **Snap Soft Block to Grid:** Pulls soft blocks to the manufacturing or placement grid. 
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Using the Pin Optimizer



You can do **Pin Optimization** to position the pins of the blocks in a manner that helps obtain the shortest possible net length at a particular level in the design.

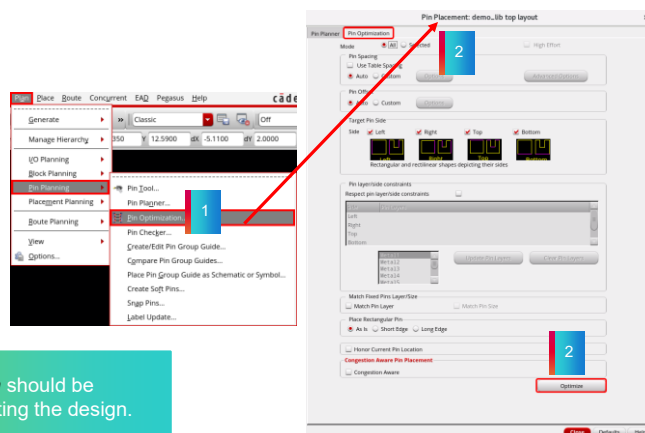
Pin Optimization must be done even for blocks that are placed by the Block Placer because the aggregate length of the nets connecting the pins of the blocks may not be optimal or the shortest.

To optimize the soft block pins and top-level pins:

1. In the layout window, choose **Plan – Pin Planning – Pin Optimization** or click the **Pin Optimizer for All Pins** icon or choose **Plan – Pin Planning – Pin Planner** to open the Pin Placement form.
2. In the Pin Placement form, in the **Pin Optimization** tab, select the following options:
 - i. Select *Mode* as **All**.
 - ii. With other options as default values, click **Optimize**.

Result: All the pins are optimized.

Pin Optimization should be done before routing the design.



Using the Pin Optimizer

You can do **Pin Optimization** to position the pins of the blocks in a manner that helps obtain the shortest possible net length at a particular level in the design.

Pin Optimization must be done even for blocks that are placed by the Block Placer because the aggregate length of the nets connecting the pins of the blocks may not be optimal or the shortest.

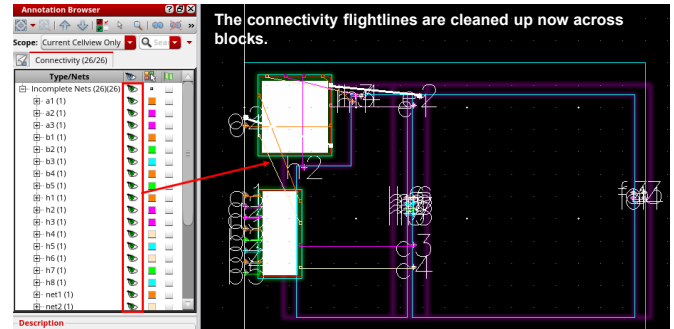
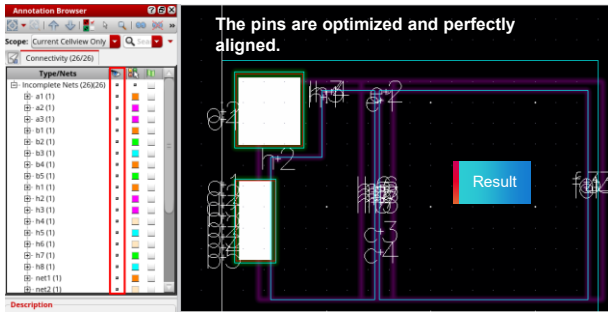
To optimize the soft block pins and top-level pins:

1. In the layout window, choose **Plan – Pin Planning – Pin Optimization** or click the **Pin Optimizer for All Pins** icon or choose **Plan – Pin Planning – Pin Planner** to open the Pin Placement form.
2. In the Pin Placement form, in the **Pin Optimization** tab, select the following options:
 - i. Select *Mode* as **All**.
 - ii. With other options as default values, click **Optimize**.

Result: All the pins are optimized.

Pin optimization should be done before routing the design.

Result: Using the Pin Optimizer



Aligning the Soft Block Pins with the Top-Level Pins

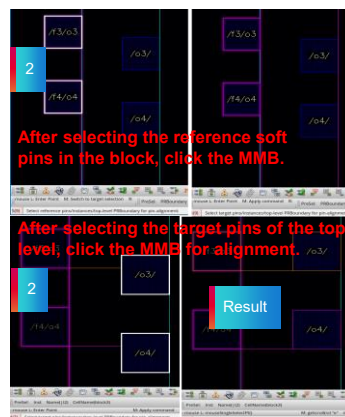
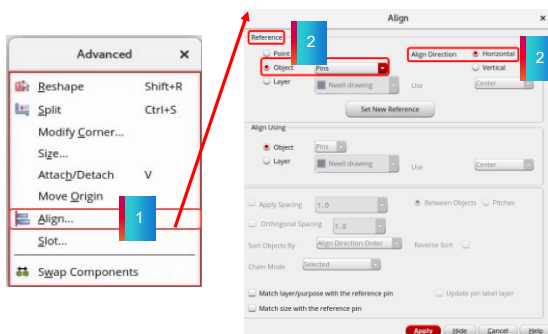


Use the **Align** command to align the objects in the design.

To align the objects in your design:

1. In the layout canvas, choose **Edit – Advanced – Align** or click the **Align** icon to invoke the Align form.
2. In the Align form, in the **Reference** section, select the following options and do the alignment:
 - i. Object: **Pins**
 - ii. Align Direction: **Horizontal**
 - iii. After selecting the reference soft pins in the block, click the **MMB**.
 - iv. After selecting the target pins of the top level on the RHS, click the **MMB**.

Result: The soft block pins are aligned with the top-level pins.



Aligning the Soft Block Pins with the Top-Level Pins

Use the **Align** command to align the objects in the design.

To align the objects in your design:

1. In the layout canvas, choose **Edit – Advanced – Align** or click the **Align** icon to invoke the Align form.
2. In the Align form, in the **Reference** section, select the following options and do the alignment:
 - i. Object: **Pins**
 - ii. Align Direction: **Horizontal**
 - iii. After selecting the reference soft pins in the block, click the **MMB**.
 - iv. After selecting the target pins of the top level on the RHS, click the **MMB**.

Result: The soft block pins are aligned with the top level pins.

Pin Alignment Report in the CIW

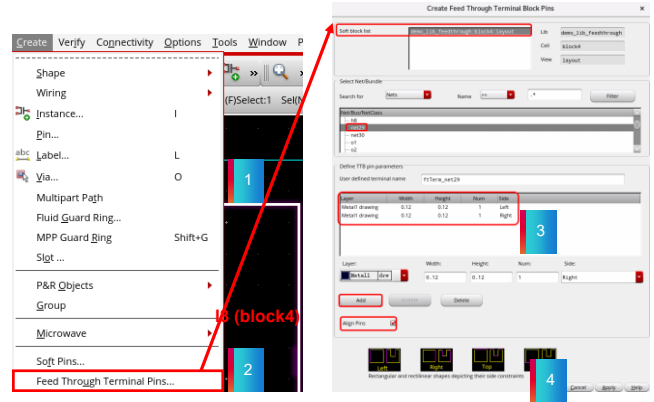
Observe the CIW for the *Pin Alignment Report*.



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To create feed through terminal pins:

- Result:** The feed through terminal pins are created on instance **I3 (block4)**.

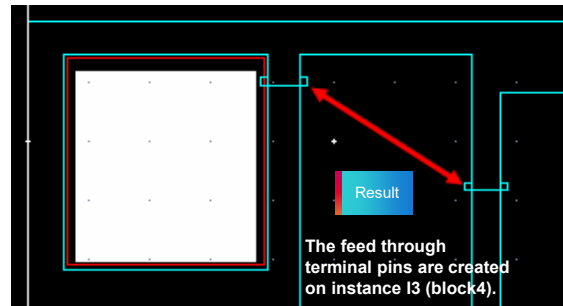
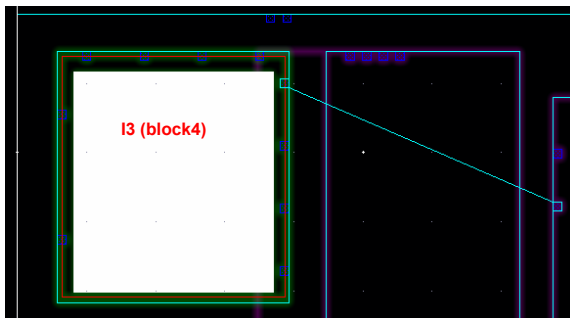


Refer to the videos below for the Virtuoso Floorplanner commands:

Channel: [Virtuoso Floorplanner](#)

Creating Feed Through Terminal Pins

Result



FAQ: Placing the Hard and Soft Blocks in a Design



Which command can you use to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area?

You can use the Block Placer command to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area.



Which command can you use to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area?

You can use the Block Placer command to automatically place all the hard and soft blocks in a design and minimize the wire length and chip area.

FAQ: Loading the Solutions in the Design



Which command can you use to load all the possible solutions that Block Placer generates?

You can use the Load Solutions command to load the solutions that Block Placer generates.



Which command can you use to load all the possible solutions that Block Placer generates?

You can use the Load Solutions command to load the solutions that Block Placer generates.

FAQ: Snapping the Top-Level and Level-1 Pins in the Design



Which command can you use to snap both the top-level and level-1 pins to the grid appropriate to the block type?

You can use the Snap Pins command to snap both the top-level and level-1 pins to the grid appropriate to the block type.



Which command can you use to snap both the top-level and level-1 pins to the grid appropriate to the block type?

You can use the Snap Pins command to snap both the top-level and level-1 pins to the grid appropriate to the block type.

FAQ: Retaining Constant Area in the Design



In the Stretch form, which option can you use to stretch the boundary keeping the area constant?

You can use the Constant area stretch option in Boundary section of the Stretch form to stretch the boundary keeping the area constant.



In the Stretch form, which option can you use to stretch the boundary keeping the area constant?

You can use the Constant area stretch option in Boundary section of the Stretch form to stretch the boundary keeping the area constant.

FAQ: Positioning the Pins of the Blocks in the Design



You can do ----- to position the pins of the blocks in a manner that helps in obtaining the shortest possible net length at a particular level in the design.

You can do Pin Optimization to position the pins of the blocks in a manner that helps in obtaining the shortest possible net length at a particular level in the design.



You can do --- ----- to position the pins of the blocks in a manner that helps obtain the shortest possible net length at a particular level in the design.

You can do Pin Optimization to position the pins of the blocks in a manner that helps obtain the shortest possible net length at a particular level in the design.

Quiz: Setting the Cell Type in the Design



What are the methods can you use to set the Cell Type in your design?

Select all that apply.

- A. Command Interpreter Window (CIW)
- B. Configure Physical Hierarchy (CPH)
- C. Navigator Assistant
- D. Property Editor Assistant
- E. All of the above

Answer: A and B

You can use either the Command Interpreter Window (CIW) or Configure Physical Hierarchy (CPH) to set the Cell Type in your design.

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Answer: A and B

You can use either the Command Interpreter Window (CIW) or Configure Physical Hierarchy (CPH) to set the Cell Type in your design.

Quiz: Generating the Physical Hierarchy in the Design



You can use the Generate Physical Hierarchy (GPH) command to generate the components in your design while maintaining the hierarchy levels you have defined in the Soft Block mode.

- A. True
- B. False

Answer: A

Yes. You can use the Generate Physical Hierarchy (GPH) command to generate the components in your design while maintaining the hierarchy levels you have defined in the Soft Block mode.



Answer: A

Yes. You can use the Generate Physical Hierarchy (GPH) command to generate the components in your design while maintaining the hierarchy levels you have defined in the Soft Block mode.

Quiz: Cutting the Objects in the Design



To cut away part of an object or cut an object into pieces, you can use the ---- command in the layout window.

Enter the answer in the blank field.

Answer: Chop

To cut away part of an object or cut an object into pieces, you can use the Chop command in the layout window.



Answer: Chop

To cut away part of an object or cut an object into pieces, you can use the Chop command in the layout window.

Quiz: Calculating the Number of Connections in the Design



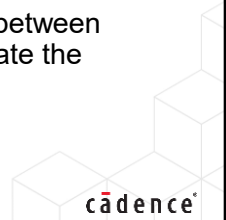
Which command can you use to calculate the number of connections between each pair of selected instances, draw flight lines between each connected pair, and indicate the number of connections by displaying a numerical value on the flight lines? Select all that apply.

- A. Analyze Connectivity
- B. Snap Pins
- C. Find/Replace
- D. Net Name Display
- E. None of the above

Answer: A

You can use the Analyze Connectivity command to calculate the number of connections between each pair of selected instances, draw flight lines between each connected pair, and indicate the number of connections by displaying a numerical value on the flight lines.

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Answer: A

You can use the Analyze command to calculate the number of connections between each pair of selected instances, draw flight lines between each connected pair, and indicate the number of connections by displaying a numerical value on the flight lines.

Quiz: Highlighting the Hard and Soft Blocks in the Design



Which form can you use to assign different highlight colors for the hard and soft blocks so that they are visually distinguishable in the layout canvas? Select all that apply.

- A. Configure Physical Hierarchy
- B. Design Planning and Analysis Options
- C. Generate Physical Hierarchy
- D. Highlight Options
- E. None of the above

Answer: B

You can use the Design Planning and Analysis Options form to assign different highlight colors for the hard and soft blocks so that they are visually distinguishable in the layout canvas.

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Answer: B

You can use the Design Planning and Analysis Options form to assign different highlight colors for the hard and soft blocks so that they are visually distinguishable in the layout canvas.

Module Summary

This module:

- Defines and applies the Cell Type attribute to enable correct floorplanning behavior for blocks and instances.
- Uses the Configure Physical Hierarchy (CPH) utility to control logical-to-physical mappings and floorplan configuration.
- Generates and places the hard and soft blocks using the Virtuoso Floorplanner and Block Placer.
- Performs Level-1 editing to modify soft blocks, boundaries, and pins without descending the hierarchy.
- Applies the Pin Optimizer to minimize the net length and improve the block-to-block connectivity.
- Aligns the soft block pins with top-level pins and creates feed-through terminal pins to enhance routability.



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Labs



- Lab 4-1 Defining the Cell Type Attribute
- Lab 4-2 Using the Configure Physical Hierarchy Utility
- Lab 4-3 Generating and Placing the Blocks with the Floorplanner
- Lab 4-4 Editing the Level-1 Objects
- Lab 4-5 Using the Pin Optimizer
- Lab 4-6 Aligning the Soft Block Pins with the Top-Level Pins
- Lab 4-7 Creating Feed Through Terminal Pins

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Module 5

Top-Down Floorplanning

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Module Objectives

In this module, you

- Analyze to accommodate a change in the die size
(Start fresh and determine the area and shapes of the top-level blocks)
- Set the Cell Type from the Configure Physical Hierarchy (CPH) utility
- Use the soft block estimation system to create soft blocks for all the top-level cells
- Choose how to edit the soft blocks to fit into the new die area
- Use the I/O Placer
- Generate, place, and prepare to route the top-level floorplan



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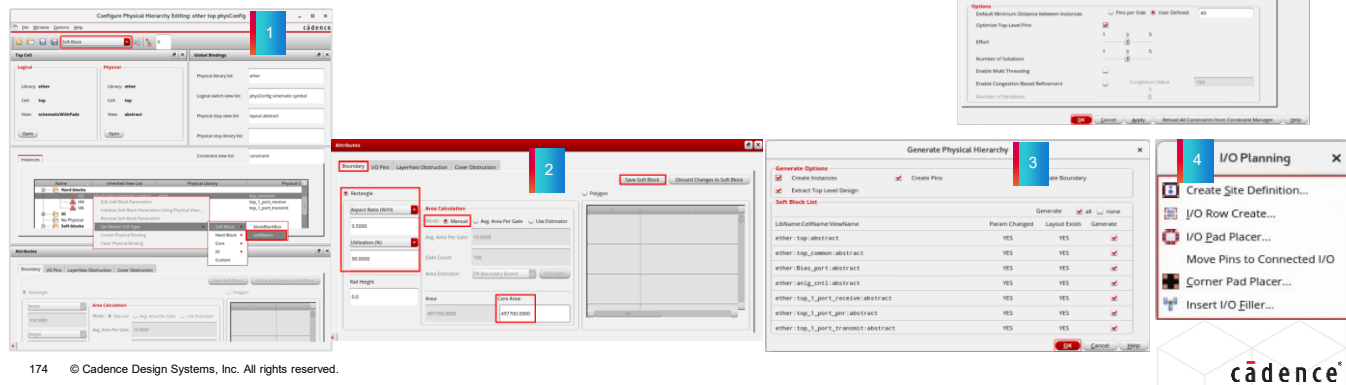
Accommodating a New Die Size



Use the top-down floorplanning to accommodate a new die size using the soft blocks in your design.

To accommodate a new die size in your design:

1. Set the Cell Type from the CPH.
2. Use Configure Physical Hierarchy (CPH) to generate soft blocks for the hard blocks.
3. Use Generate Physical Hierarchy (GPH) to estimate the block sizes.
4. Use the I/O Planning for your I/O planning and placement tasks.
5. Given a top-level area, use the Block Placer to place the blocks.



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Accommodating a New Die Size

Use the Top-Down Floorplanning to accommodate a new die size using the soft blocks in your design.

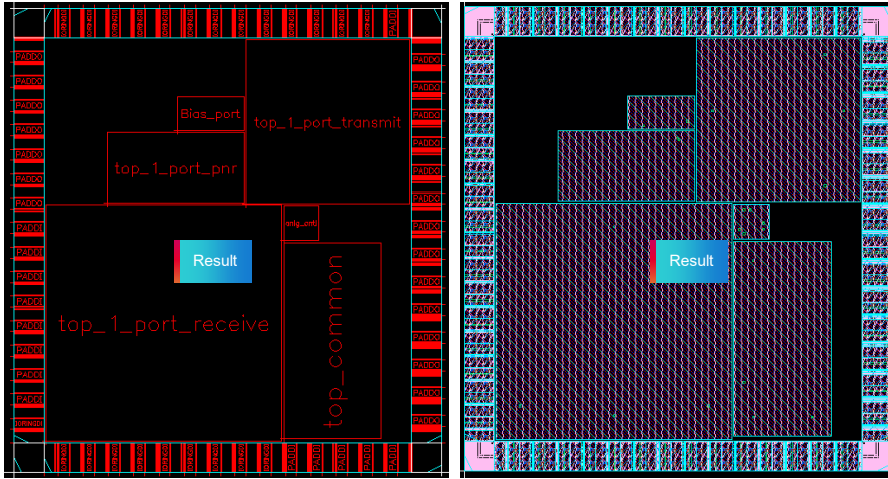
To accommodate a new die size in your design:

- Set the Cell Type from the CPH.
- Use Configure Physical Hierarchy (CPH) to generate soft blocks for the hard blocks.
- Use Generate Physical Hierarchy (GPH) to estimate the block sizes.
- Use the I/O Planning for your I/O planning and placement tasks.
- Given a top-level area, use the Block Placer to place the blocks.

Result: The selected blocks are fit into the space as specified.

Result: Accommodating a New Die Size

Result: The blocks are placed with routable solution.



Result: Accommodating a New Die Size (continued)

Result: The blocks are placed with routable solution.

Setting the Cell Type from the CPH



Floorplanner requires information about the cell type to distinguish the cells at the floorplanning level. Floorplanner behaves differently depending on the cell type.

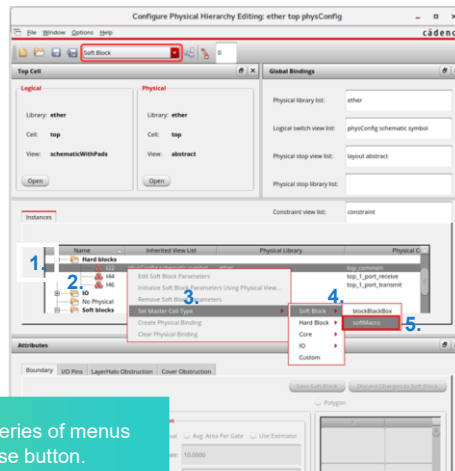
For example, **I/O Placer** will work on pads only and not on other cells.

You can use the **Instances** table in the **Soft Block** mode of the **CPH** form to set the Cell Type.

To set the Cell Type from the CPH in your design:

1. **Left-click** to select the block for which you want to set the Cell Type.
2. **Right-click** to pop up a menu from the cell for which you want to set the Cell Type.
3. Slide the pointer to **Set Master Cell Type**.
4. Slide the pointer to **Soft Block**.
5. Slide the pointer to **blockBlackBox** or **softMacro**.

Result: The selected block's Cell Type is set as **blockBlackBox** or **softMacro**.



You can set the Cell Type from a series of menus that you display with the right mouse button.

Setting the Cell Type from the CPH

Floorplanner requires information about the cell type to distinguish the cells at the floorplanning level.

Floorplanner behaves differently depending on the cell type.

For example, **I/O Placer** will work on pads only and not on other cells.

You can use the **Instances** table in the **Soft Block** mode of the **CPH** form to set the Cell Type.

To set the celltype from the CPH in your design:

1. **Left-click** to select the block for which you want to set the celltype.
2. **Right-click** to pop up a menu from the cell for which you want to set the Cell Type.
3. Slide the pointer to **Set Master Cell Type**.
4. Slide the pointer to **Soft Block**.
5. Slide the pointer to **blockBlackBox** or **softMacro**.

Result: The selected block's celltype is set as **blockBlackBox** or **softMacro**.

You can set the cell type from a series of menus that you display with the right mouse button.

I/O Placement Engine: Managing I/O Cells



In the layout window, choose **Plan – I/O Planning** to display the I/O Placement menu commands namely: *I/O Row Create*, *I/O Pad Placer*, *Corner Pad Placer*, and *Insert I/O Filler*.

Use the *I/O Planning* for your Floorplanner tasks.

The below commands have been provided for the I/O placement engine in the Virtuoso® platform:



- **I/O Row Create**

- Create rows to place the pads



- **I/O Pad Placer**

- Position pads in the rows



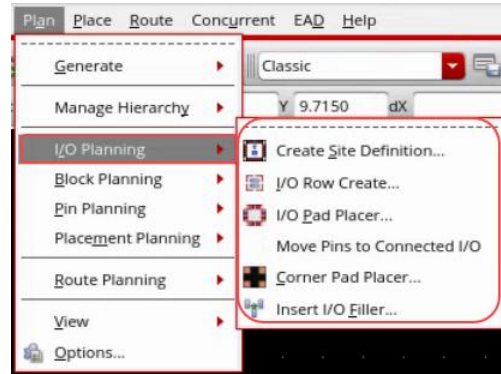
- **Corner Pad Placer**

- Corner pads must have Type = Corner



- **Insert I/O Filler**

- Add filler cells to the *Insert Filler Cells* form
- Filler cells must match the row
- Place filler cells in pad ring



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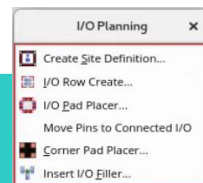
I/O Planning and Placement

The I/O placement engine consists of set of commands by which user can optimize I/O position. It is a row-based placement methodology.

- The first step in I/O planning and placement is to create I/O rows.
- Once the I/O rows are created, then place PADs inside the I/O rows.
- After PADs are placed inside the I/O rows, you can insert filler cells in the gaps between the pads and the rows.
- Thereafter, you can place corner cells on the 4 corners of the design.

I/O Placer Commands: I/O Row Create, I/O Pad Placer, Corner Pad Placer, and Insert I/O Filler.

- *Row Creation:* Creates I/O rows of PAD type *siteDefs* available in technology.
- *PAD Placement:* Places Cell Type of PAD instances in I/O rows.
- *Corner Cells:* Inserts corner cells not in the row but in between the rows to maintain signal continuity between I/O ring.
- *Filler Cells:* Inserts filler cells once Pads instances are placed to maintain signal continuity in the I/O row.



I/O Planning and Placement

The IO placement engine consists of set of commands by which user can optimize IO position. It is a row-based placement methodology. The first step in IO planning and placement is to create IO rows. Once the IO rows are created, then place PADs inside the IO rows. After PADs are placed inside the IO rows, you can insert filler cells in the gaps between the pads and the rows. Thereafter, you can place corner cells on the 4 corners of the design.

The following commands have been provided for IO placement engine in the Virtuoso platform:

Row Creation

- Creates IO rows of PAD type siteDefs available in technology.

PAD Placement

- Places celltype of PAD instances in IO rows.

Corner Cells

- Inserts corner cells not in the row but in between the rows to maintain signal continuity between IO ring.

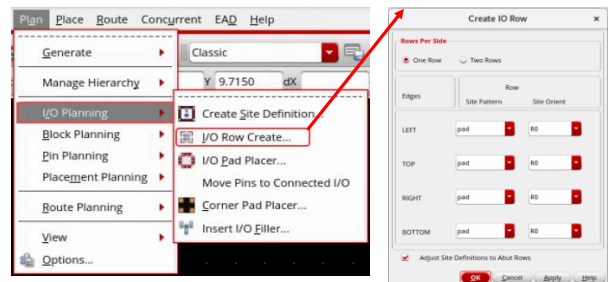
Filler Cells

- Inserts filler cells once Pads instances are placed to maintain signal continuity in the IO row.

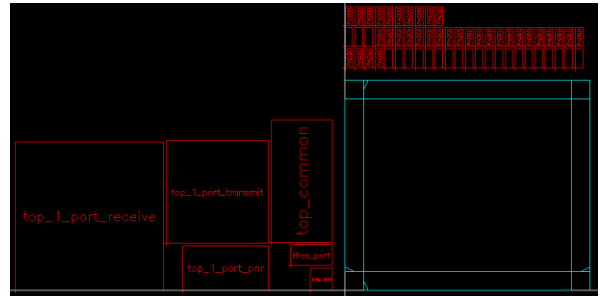
Using the I/O Planning to Create the Pad Rows



In the layout window, choose **Plan – I/O Planning – I/O Row Create** or click the **IO Row Create** icon to open the Create IO Row form.



- Use the *Create IO Row* form to create the rows
- Creates I/O rows of PAD type *siteDefs* available in technology
- Rows Per Side
 - One Row
 - Two Rows
- Edges
 - Site Pattern Site Orient
 - o pad R0
- Rows are created
- Pads are placed outside boundary during generation



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Row Creation

In Floorplanner, the first step that IO planning and placement has is to create IO rows. The rows determine the legal placement area for pads, where to place the PADs and the type of PADs that are to be placed. You can manually create rows and use IO Placer with manually created rows. However, the manually created rows need to have the right settings. There is no restriction on where the rows should be created. Also, it is not mandatory to create rows on the periphery.

To create rows, you can click the Plan – I/O Planning – I/O Row create command. The Create IO Row form is displayed.

Note: The site definition of the type 'PAD' will be read from the technology file and displayed in the *Site Pattern* fields. If there is no site definition of type PAD available, the row creation UI is not displayed.

In the Create IO Row form, the *Rows Per Side* section has two options, *One Row* and *Two Rows*. *One Row* is selected by default and the options for Row 1 are displayed. A maximum of two rows can be created on each side (i.e. TOP, LEFT, BOTTOM, and RIGHT) of the top level boundary. If you want to create two rows on each side, select *Two Rows*. This will insert the UI for the second row.

To create a row, you need to specify the row site patterns and their orientations for each side from the *Site Pattern* and *Site Orient* combo fields respectively and then click *OK*. This will create the row at the appropriate position. If the row already exists at the position where new row is to be created, the existing pad rows will be replaced by the new row. If you run this command again and some rows already exists, they will be deleted automatically.

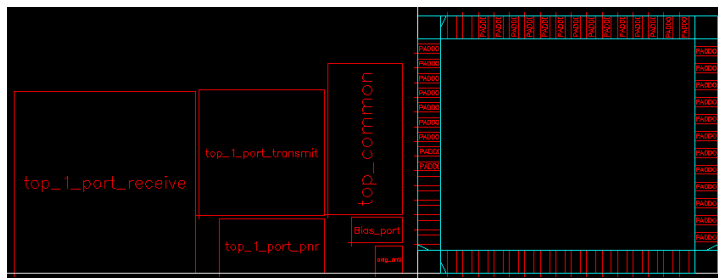
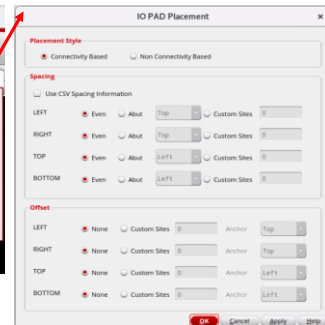
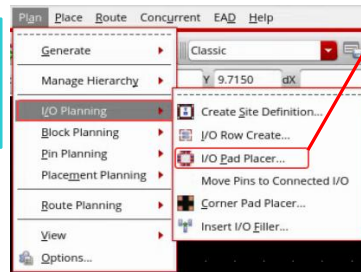
Note: The site pattern is extracted from the technology file. The site orientation is the relative orientation of the pad inside the row and not the final pad orientation.

Using the I/O Planning to Create the Pad Placement



In the layout window, choose **Plan – I/O Planning – I/O Pad Placer** or click the **IO Placer** icon to open the IO PAD Placement form.

- Use the *IO PAD Placement* form to place the pads
- Places Cell Type of PAD instances in I/O rows
- Style:
 - Connectivity Based
 - Non-Connectivity Based
- Spacing:
 - Even
 - Abut:
 - Left/Right/Center
 - Top/Bottom/Center
 - Custom:
 - Specify spacing distance per side
- Offset:
 - None
 - Custom:
 - Specify spacing distance per side
 - Anchor
 - Specifies the corner from which offset must be calculated when Offset is set to Custom Sites.



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PAD Placement

PAD placement is an essential part of the IO placement methodology. It is the next step after row creation in IO Placer. IO rows should be present before running PAD placement. The PADS are placed on the periphery of the design in the IO rows.

To place PADS on IO rows, click the **Plan – I/O Planning – I/O Pad Placer** command. The IO PAD Placer form is displayed.

The IO PAD Placement form has two sections – *Style* and *Spacing*. When the IO Placer form is invoked, you can select the placement style and put the placement constraints if required.

Style

There are three types of placement styles that are supported, Connectivity Based and Non-Connectivity Based. The Style section allows you to select either 'Connectivity Based' or 'Non-Connectivity Based' style of placement. By default, Connectivity Based style is selected.

▪ Connectivity Based

You can use this option to place the PADs in the rows minimizing the net length between the PADs and the placed blocks in the design.

Note: The assumption is that in Connectivity mode, the core instances are placed inside the prBoundary.

▪ Non Connectivity Based

This option enables you to distribute the PADs around the periphery of the design boundary without considering the connectivity between PADs and blocks placed in the design.

Spacing

Spacing constraints defines the inter spacing between the PADs in a given row. This is a post processing step for the PAD placement engine.

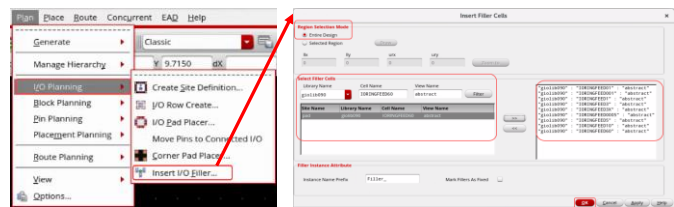
Offset

Specifies the position of the first IO PAD cell. Positions of the remaining PAD cells are calculated based on the specified Spacing values.

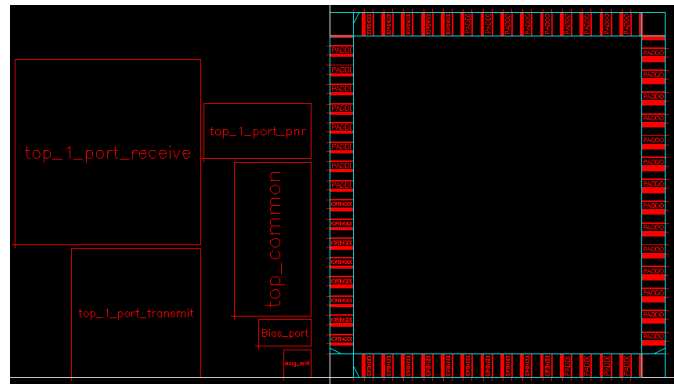
Using the I/O Planning to Insert the Filler Cells



In the layout window, choose **Plan – I/O Planning – Insert I/O Filler** to open the Insert Filler Cells form.



- Use the *Insert Filler Cells* form to insert the filler cells
- Places Cell Type of I/O RING instances
- Insert filler cells after PAD placement is done in the I/O rows of the design
- The filler cells are inserted in the empty space between PADs in an I/O row
- The *Insert Filler Cells* form has three sections:
 - *Region Selection Mode*
 - *Select Filler Cells*
 - *Filler Instance Attribute*



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Filler Cells Insertion

To open the Insert Filler Cells form, click **Plan – I/O Planning – Insert I/O Filler** command. The Insert Filler Cells form is displayed. The Insert Filler Cells form has three sections: *Region Selection Mode*, *Select Filler Cells*, and *Filler Instance Attribute*.

Region Selection Mode

In this section, you can select either of the two modes, i.e, work on entire design or work on a selected region.

Entire Design: Enables you to select the entire design for inserting filler cells. This means that filler cells will be inserted in all the IO rows of the design.

Selected Region: Enables you to select a region for inserting filler cells. This means that only few IO rows need to be used for filler insertions.

Select Filler Cells

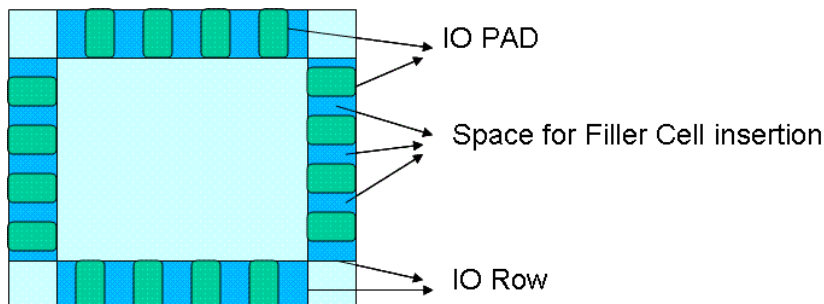
The *Select Filler Cells* section facilitates you in selection of few cells from the entire range of libraries/cells/views. The few selected cells will be used by the engine as filler cells.

Filler Instance Attribute

In this section, you can define the instance name prefix and the placement status of the filler cell using the *Instance Name Prefix* and *Mark Fillers As Fixed* options.

Example: Using the I/O Planning to Insert the Filler Cells

- The Filler cells insertion is needed:
 - To maintain signal continuity in the I/O ring for power and ground distribution
 - For Electro Statics Discharge (ESD) protection reasons



The Filler cells insertion is needed to maintain signal continuity in the I/O ring.

Filler Cells

The Filler cells insertion is needed to maintain signal continuity in the IO ring for power and ground distribution and for Electro Statics Discharge (ESD) protection reasons.

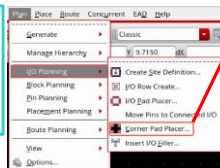
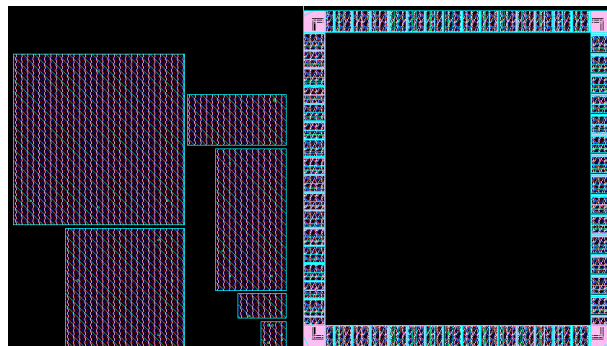
Filler cells insertion is done after PAD placement is done in the IO rows of the design. The filler cells are inserted in the empty space between PADs in an IO row. In order to fill the gap, the filler instances with the exact height matching the row site height are chosen. This is done to minimize the number of filler cells that are inserted by the insert filler engine. The filler cells that are to be inserted will be used from the Filler Cells table in the Insert Filler Cells form.

To insert filler cells, you first need to define the design area. It could either be an entire design or a selected region of a design. After defining the design area, select the filler cells from the libraries that are to be used. To select a filler cell from libraries, you can browse libraries and select few cells to be used as filler cells.

Using the I/O Planning to Place the Corner Cells



In the layout window, choose **Plan – I/O Planning – Corner Pad Placer** to open the Corner Pad Placement form.

- Use the *Corner Pad Placement* form to place the corner pads
- Places four corner cells on four corners of the design
- Corner cells insertion and deletion is done after PAD and filler cell placement in the I/O rows of the design
- In the *Corner Pad Placement* form, specify:
 - *Instance Name* from the drop-down list box
 - *Orientation* from the drop-down list box

Placing Corner Cells

The procedure to insert corner cells is as follows

Selecting corner instances

Ensure that the corner instances are already present in the design. The corner instances can be selected in the following ways:

- Navigator
- Search Engine
- Canvas Selection

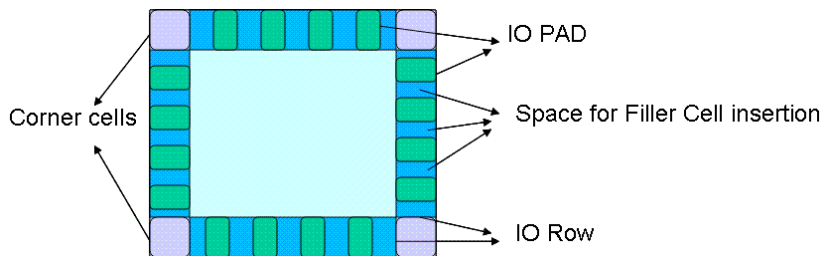
Once the corner instances are selected in the design, invoke the *Corner Pad Placer* command by clicking **Plan – I/O Planning – Corner Pad Placer**. The Corner Pad Placement form is displayed. The form is by default populated with the selected instances. However, if the corner instances are not selected, the form will not be displayed, and a corresponding message is displayed in the status bar.

Placing the corner

In the Corner Pad Placement form, you can specify another instance for each corner from the *Instance Name* drop-down list box. You can also set the orientation for each corner from the *Orientation* drop-down list box. Once the instance is selected and the orientation is set, click the OK button. A row will be created automatically. Simultaneously, a corner cell is placed in that row.

Example: Using the I/O Planning to Place the Corner Cells

- The Corner cells insertion is needed:
 - To maintain signal continuity in the I/O ring between the I/O row power and ground distribution
 - Four corner cells are placed on four corners of the design



The Corner cells insertion is needed to maintain signal continuity in the I/O ring.

Corner Cells

Corner cells insertion is needed to maintain signal continuity in the I/O ring between the I/O row power and ground distribution. The four corner cells are required to be placed on the four corners of the design.

Corner cells insertion and deletion are done after PAD, and filler cell placement is done in the IO rows of the design.

The high level algorithm for corner cells is as follows:

- The list of the selected instances is filled in the corner UI.
- Engine takes the input instance name and its orientation for each corner from the UI.
- The corner row is created of the appropriate site(s) in the design before placing the corner cells.
- Corner cells will be placed in the appropriate row as per the orientation specified.

Before inserting corner cells, ensure the following:

- The library has at least four instances with their *cellType* set to corner. If not, a suitable message is displayed in the CIW.
- The row in which the corner cells must be placed has the same *sitePattern* as the corner cells. Corner cells can be placed only in rows with matching *sitePattern*.
- The technology file contains the *sitePattern* under the *siteDefs* section. If not, a suitable message is displayed in the CIW.

If all the above conditions are met, a corner row of the given *sitePattern* is created, and cells with *cellType* corner are inserted in these rows.

Module Summary

This module covered:

- Applies top-down floorplanning to accommodate the changes in die size by re-evaluating top-level block areas and shapes.
- Set the Cell Type using the Configure Physical Hierarchy (CPH) utility to enable correct floorplanning behavior.
- Uses the soft block estimation system to generate soft blocks for all top-level cells.
- Modifies and edits the soft blocks to ensure they fit within the new die area constraints.
- Performs I/O Planning to create I/O rows, place pads, insert fillers, and add corner cells.
- Generates, places, and prepares the top-level floorplan for routing using a structured, routable solution.



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Lab



Lab 5-1 Creating a Top-Down Floorplan

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Module 6

Course Conclusions

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Summary

In this course, you learned to

- Set up and run Module Generator
- Analyze Modgen Enhancements
- Set up and run Floorplanner
- Set up and run Top-Down Floorplanning



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References

Refer to these links for more details on the topics discussed in this course.

- [Virtuoso Module Generator User Guide Product Version IC25.1](#)
- [Virtuoso Design Planning and Analysis User Guide Product Version IC25.1](#)



- [Virtuoso Module Generator User Guide Product Version IC25.1](#)
- [Virtuoso Design Planning and Analysis User Guide Product Version IC25.1](#)



Module 7

Next Steps

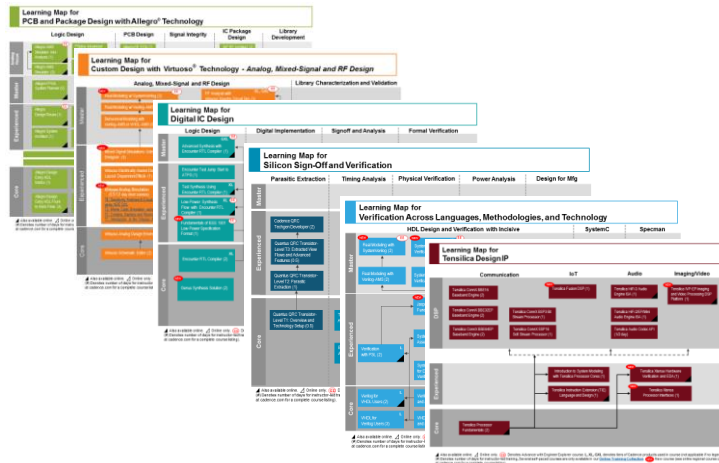
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Learning Maps

Cadence® Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers.

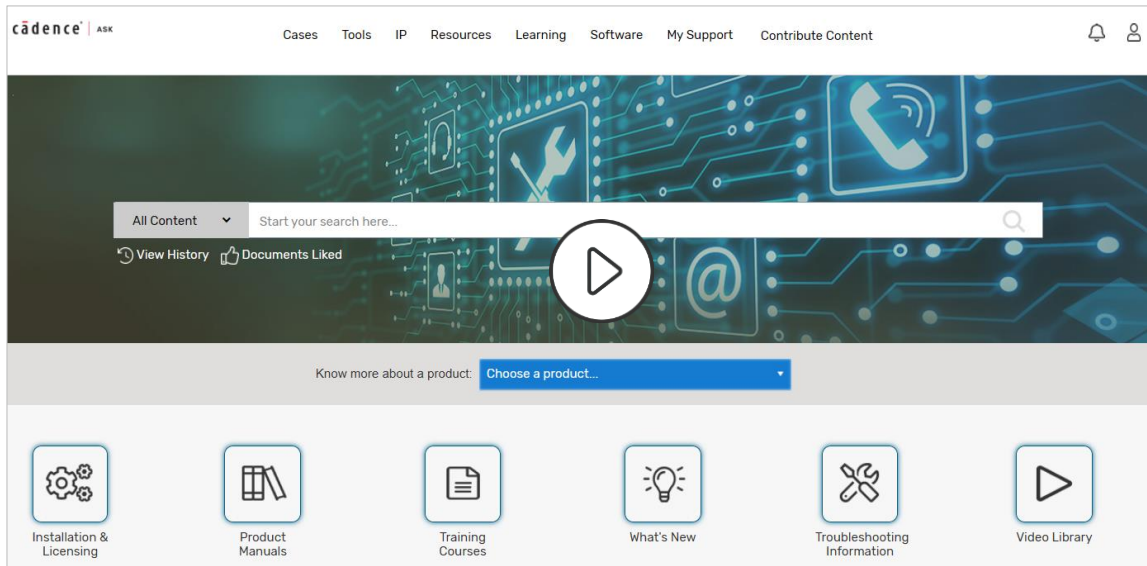
Click [here](#) to see all our courses in each technology area and the recommended order in which to take them.



Go here to view the learning maps:

http://www.cadence.com/Training/Pages/learning_maps.aspx

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Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion
- Complete the Course's Digital Badge exam to earn your Badge

Thank you!



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